

AccessMatrix Common Administration Guide - 6.0.1-GA (AI)

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1. Preface - AccessMatrix Common Administration Guide

This guide contains administration and management instructions on various aspects of the AccessMatrix system in general, e.g. security policies, users, groups, etc. that are configurable via the Admin Console.

Intended Reader

It is intended for the security administrators to perform the security administration tasks and for the auditors and the information security personnel to have a detailed understanding of the security administration procedures.

Document Scope

This document provides a general overview on the various aspects of the AccessMatrix system that are configurable via the Admin Console. For in-depth details and instructions on specific components of the AccessMatrix system, refer to the respective user or developer guides.

For feedback or questions regarding this document, contact doc@i-sprint.com.

Related Documents

For more information, see the following documents in the AccessMatrix Product Suite:

- *AccessMatrix Release Notes*
- *[Universal Authentication Server \(UAS\) Administration Guide](#)*
- *[Universal Sign-On \(USO\) Administration Guide](#)*
- *[Universal Access Management \(UAM\) Administration Guide](#)*
- *[Universal Credential Manager \(UCM\) Administration Guide](#)*
- *[AccessMatrix Common Deployment Guide](#)*

2. Introduction to AccessMatrix Administration

This page provides an overview of the AccessMatrix product suite.

AccessMatrix System

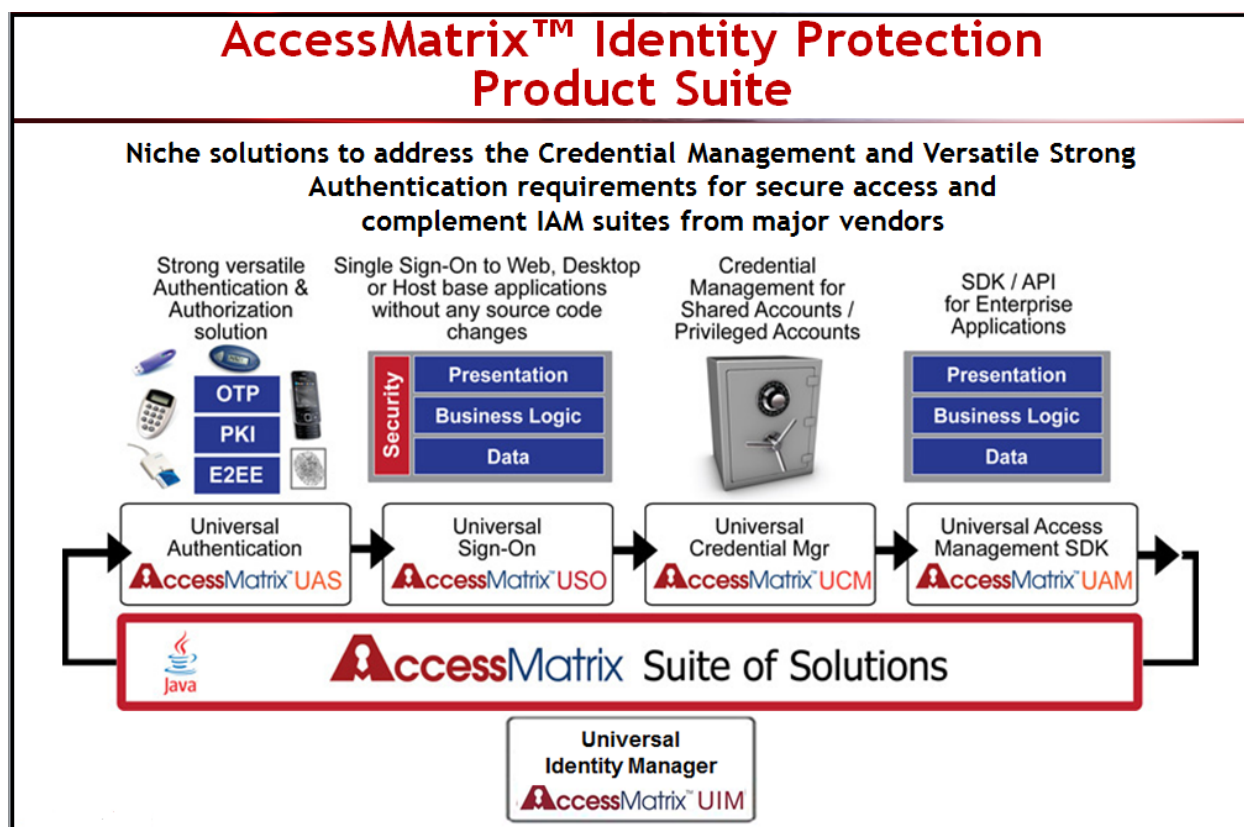
AccessMatrix is a security framework that protects enterprise applications. It is highly configurable and provides practical and powerful features to manage the security needs of large organizations.

Generally, the AccessMatrix security framework provides the following features:

- Centralized administration services for security administrators to easily and effectively manage security policies, application permissions, and user privileges throughout the entire organization.
- Centralized authentication, authorization, and audit services for users to securely access different enterprise and e-business applications and resources.

AccessMatrix Product Suite

AccessMatrix product suite is an integrated suite of security products or solutions built upon the common AccessMatrix security framework (a.k.a. AccessMatrix security infrastructure).



3. AccessMatrix Admin Console

This page introduces the AccessMatrix Admin Console.

About Admin Console

Upon successful setup of a new AccessMatrix database, three initial administrators will be automatically created in the AccessMatrix system. Below table shows the created administration IDs and passwords, its purposes and the admin roles assigned:

Initially Created Administrators				
Administrator Id	Purpose	Can it be deleted if it is not in use?	Default Password	Assigned Admin Role
amadmin	Performs initial administrative tasks	No	password	InitialAdminRole (includes all admin rights)
amadmin1	Performs initial administrative tasks and serve as a complementary administrator in situations where dual control is required, such as approving administrative requests or secondary login to Token Import Utility.			
backupadmin	Serves as a backup administrator account to perform contingency or emergency administrator operations such as resetting passwords for the above initial administrator accounts in the case both administrator account passwords were lost and no other administrator is available or created in the system.	Yes		

Log in to AccessMatrix Admin Console:

-
1. In the web browser, inside the URL box, enter
"http://<AM_Server_Name_or_IP_Address>:<Port_Number>/<Admin_Console_Context_Root>"
(e.g. "http://127.0.0.1:8080/am5").
 2. On the Admin Console **Login** page, select the desired language from the **Language** drop-down menu and log in with the administrator Id and password provided above.

**Notes:**

- While logging in to AccessMatrix Admin Console, entering a certain number of incorrect passwords will lock the account. The number of incorrect passwords (login attempts) allowed depends on the account lockout threshold set by the system administrator. By default, this value is set to "3".
- During the first login, the **Change Password** window will be displayed to force the user to change the default password. This window will also be displayed upon logging in if the user's password has expired.
- The administrator may refer to [Appendix A. Admin Role and Admin Right Definitions](#) to find out the relevant admin role and admin rights required to perform respective administrative tasks.

Enable Reset Password via CLP Feature

The reset password feature allows users to reset their own password using a **Forgot Password** button displayed on the Admin Console **Login** page. It is disabled by default, and must first be enabled.

This feature makes use of the AccessMatrix Common Login Page (CLP), a separate web-based client authentication component bundled with AccessMatrix. Before you can enable this feature, you must ensure the following:

- CLP should already be deployed. You can refer to [AccessMatrix Common Login Page Deployment Guide](#) for more information on CLP and how to deploy, configure, and use it.
- The self-service password reset feature for CLP must also be enabled. Refer to [AccessMatrix Common Login Page Deployment Guide - Configure Self-Service Password Reset](#) for instructions on this.



Note: You should skip step 5 of the **Enable SSPR** section, which is an optional step to enable the force password change feature. This step is not applicable for the password reset configurations described here.

When ready, perform the following steps to enable the reset password feature:

-
1. Navigate to <AM_ROOT>\tomcat\webapps\am5\WEB-INF\classes\ and open the **wpe_config.properties** file.
 2. Search for the line forgot.password.enable=false and update the values accordingly:

Properties (.properties files)

```
#AM-11328: You can enable self-service password reset
from Admin Console by setting forgot.password.enable to
true.
#AM-11328: You can also configure the redirect link to
one of your choosing.
forgot.password.redirect=http://localhost:8086/clp/sspr
forgot.password.enable=true
```

- forgot.password.redirect - Ensure that the server name and port are accurate. Update them if necessary (e.g. "https://localhost:8334/clp/sspr" for a default secure deployment of CLP).
 - forgot.password.enable - Set the value to "true" to enable the reset password feature.
3. Save the changes and restart the server.
 4. Access the Admin Console **Login** page and verify that the **Forgot Password** button is now displayed.

About Admin Console Dashboards

The AccessMatrix Admin Console dashboards' user interfaces are designed to help each user achieve specific goals with effectiveness and efficiency in all contexts of usage. Select the desired dashboard from the drop-down list at the top-right corner of the Admin Console.

All pages in the Admin Console are grouped under these three different dashboards for easy navigation:

- **Operation Dashboard**
- **Administration Dashboard**
- **Report Dashboard**

Each page of the Admin Console dashboards' user interface is divided into left and right panes. The left pane consists of a navigation tree that enables the user to navigate through various nodes, and the right pane has a **Search For** field to search records in the Admin Console. Depending on the navigation options selected, the contents displayed on the left and right panes may vary.

Upon selecting a navigation option, the left pane will show breakdown nodes of the selected option or display a segment tree if the content of the selected option is assignable to a segment. The right pane will always display a complete listing of the selected option at the left pane where the auto search is performed to retrieve all its values.

While performing create/edit/search actions in Admin Console, all attributes marked with (*) indicate mandatory and the administrator must enter a value to be able to proceed.

Perform searches in Admin Console:

- The search can be based on a wildcard (*) or exact phrase match and is case-sensitive.
e.g. If "DEMO*" is entered for a search field, the entry with an Id of "DEMO123" will be retrieved, however, not "123DEMO". Enter "*DEMO" to retrieve "123DEMO", or leave the search field blank to search for all available results.
- The search criterion is also based on 'AND' condition.
e.g. If the administrator enters values for both **Id** and **Name** fields, then only the results matching both values will be retrieved.

Adding custom attribute when you create or edit a user/group/segment:

1. Select **Add Custom Attribute**. The **Add New Attribute** dialog box is displayed.
2. Enter the attribute name, data type and its attribute value, and then click **Apply**.



Notes:

- If the attribute value is to be stored in an encrypted form in the security registry, check the **Encryption** option. This action is not reversible once the entry is saved.
- After adding the attribute, the administrator is allowed to edit its value.

About AccessMatrix Server License

The server license is used to enable general features (e.g. core features) and product-specific features (e.g. UAS, USO, UCM, UAM, and UIM features).



Important Notes:

- If AccessMatrix system is set up in a distributed environment, where the administrator is requested to set up more than one

AccessMatrix system (i.e. having more than one AccessMatrix server) where these AccessMatrix systems will be accessing to a centralized repository, the administrator is required to create a new server entry for each AccessMatrix system. This can be done via one of the AccessMatrix systems that have been granted access to the security registry.

- The administrator will not be able to edit/delete the server settings if there is a pending/rejected admin request pertaining to that server entry.

Load AccessMatrix Server License

1. Navigate to **Administration Dashboard > System > Server & Cache > Server** on the left pane to view the server details.
2. Select a Server-Id as listed on the **Search Server Result** page.
3. Click **Edit**.
4. Select the **Licensed Features** tab on the right pane. See below the three options to load the server license.

- **Upload License File**

- Select the first radio button, click **Choose File** or **Browse** to select a license file.



Note: On Windows Explorer, use the **Browse** button. On the Chrome browser, use the **Choose File** button.

- Click **Upload Licence File** to upload the license file to the server. An information window with the message "Licence file has been uploaded" appears. Click **OK** to proceed.

- **Copy & Paste License Key**

- Select the second radio button and click > . The **Licence Key** dialog box is displayed.
- Copy the license key, paste it into the dialog box, and click **OK**. If successful, the license will be updated according to the provided license key.

- **Use Evaluation License**

- Select the third radio button and click **Use Evaluation Licence**. The **Choose features** dialog box is displayed.
 - Select the product(s) to be included as part of the evaluation license and click **OK**.
-



Note: Only those selected product(s) will be ticked together with **CORE FEATURE**.

5. Click **Save** at the bottom of the page to accept the changes.
6. Log out and log in again for the new license to take effect.

4. Using AccessMatrix Admin Console

The following sections provide details and instructions on using the various dashboards and features in the AccessMatrix Admin Console:

- [Operation Dashboard](#)
- [Administration Dashboard](#)
- [Report Dashboard](#)
- [Advanced Configurations](#)
- [Utility Tools](#)

4.1. Operation Dashboard

This dashboard provides functions that facilitate the operational requirements of an organization. It helps with performing and executing day-to-day operational tasks such as monitoring and approving admin requests, assigning login realm to the user and changing user's information to serve the needs of the organization users.

The **Operation Dashboard** of the Admin Console user interface contains the first-level navigation bar with the **Home**, **Request**, **User & Segment**, **Token**, and **About** options.

Selecting this dashboard will also display a **Summary** table on the right pane. This behavior is subject to the `load.workflowRequest.for.all.OpDashBoard` setting in the `wpe_config.properties` file. When set to "false", the **Summary** table will only be displayed in the **Request** menu (**Operation Dashboard > Request**). When set to "true", it will be displayed in all menus under the **Operation Dashboard** (except **About**). Refer to [Common Deployment Guide - AccessMatrix Admin Console Properties](#) for more details.

Summary

Request	Description
PIN mailer request	
Pending my approval	Displays the number of pending PIN mailer requests to be approved by the logged in user.
Pending others approval	Displays the number of pending PIN mailer requests to be approved by other approvers.
Admin Request	
Pending my approval	Displays the number of pending admin requests to be approved by the logged in user. The displayed value will depend on the <code>workflowRequest.count.limit.OpDashBoard</code> setting in <code>wpe_config.properties</code>. Refer to Common Deployment Guide - AccessMatrix Admin Console Properties for the detail. For example, if the limit value is set to "50" and the count of Pending my approval requests exceeds 50, then the UI will display "50+". Otherwise, the actual count will be displayed without the suffix "+".

Request	Description
Pending others approval	Displays the number of pending admin requests to be approved by other approvers.

4.1.1. User Management

This page provides details on managing users as individuals, in a user group or as part of a segment, and user request management.

Manage Users

In the AccessMatrix system, a user is an entity that accesses the system. It belongs to a default store (a.k.a local repository) or an external user store. Users can be categorized as interactive users (i.e. administrators or normal users) or non-interactive users (a.k.a. agents), as described below:

- **Interactive User (Administrator or Normal User)**

- An Administrator is a user who is assigned with certain administrative roles to manage security policies, groups, users, applications, etc., in the AccessMatrix system. At least one initial administrator is created during the initial setup of AccessMatrix.
- A normal user is a user who is not assigned with any administrative role but allowed access to AccessMatrix system and entitled applications to utilize the USO functions provided by the system for individual use.

- **Non-Interactive User (Agent)**

The non-interactive user, also known as an agent, accesses the system to retrieve the represented applications defined in the system on behalf of the interactive user. It acts as a gateway between the interactive user and the security server. Usually, this kind of administrator is not created explicitly for the daily operations of the AccessMatrix system. If there are multiple applications with different functions, it is recommended to create a separate non-interactive user for each application. The non-interactive user is authenticated based on the TLS client's certificate DN used as part of client-authenticated TLS. A non-interactive user, `usoagent`, is created by default, and it is part of "InitialAdminGroup", which has all privileges. This is to ease API testing by application and integration with other AccessMatrix components. For a more secure deployment, especially in a production environment, it is highly recommended to reduce the privileges or create other non-interactive users with only the required privileges.

You manage users on the **User** tab. Navigate to **Operation Dashboard > User & Segment > User**. The following tasks can be performed on the **User** tab:

- **Create a new user**

Click **Create** to display the **Create User** page, enter the user's information in each empty field and click **Save**.

i **Note:** Default store users can be created or deleted from the AccessMatrix System, while external users, e.g. Active Directory (AD) users, cannot be created or deleted from AccessMatrix System.

- **Search an existing user or users**

You can find an existing user by entering their user Id, username, user store or segment on the **Search User** list, and then click **Search** to display the **Search User Result** list.

i **Note:** When all the input fields in the **Search User** list are empty, clicking **Search** will show all existing users on the **Search User Result** list.

- **View an existing user**

Select a user in the **Search User Result** list to view that user's details.

- **Edit User Information**

Click **Edit** on the **View User** page to modify its information.

i **Note:** You cannot edit or delete the current login user Id information.

- **Delete a user**

Click **Delete** on the **View User** page and click **OK** in the confirmation pop-up box.

- **Assign a user to a segment**

On the **Create/Edit User** page, select a segment on the left pane and click **Choose Segment** to auto-fill the **Segment** attribute on the right pane.

i **Note:** The **Segment Hierarchy** licensed feature is required to assign local users to internal segments. They cannot be assigned to external segments.

- **Assign a user to a group**

On the **Edit User** page, select the **Membership** tab, check the **User Group** to be assigned and click **Save**.

- **Change Password of a user**

On the **View User** page, click **Reset Password**. Enter a new password in both the **New Password** and **Confirm New Password** fields. Click **Save**.

i **Notes:**

- The **Reset Password** button is only applicable for a user whose login accounts contain the SystemPasswordBasic login module, and the user is not the current login user.

- To change the password for the current login user, select **Change Password** at the top-right corner of Admin Console. Enter the old password, new password, the confirmed new password, and then click **OK**.



Important Note:

- The administrator cannot edit/delete a user if there is a pending/rejected admin request of that user entry.

Having administered the desired users, the next task may be to generate the security and audit reports for monitoring and auditing the users' daily activities accessing the system. This is discussed in details in **Report Dashboard**.

View User Page Attribute Description

Attribute Name	Description
*User Id	Unique user Id. Notes: • Both the maximum character length and allowed character set may vary depending on the user Id format restrictions configured in Global Setting. • The maximum possible character length, if permitted in the Global Setting configurations, is 64 characters. • It is strongly recommended that you only use ASCII characters.
*User Name	Name of user. Note: The maximum character length is 64 characters. It is strongly recommended that you only use ASCII characters.
Description	Description of user.
Version	User entry version.
*Interactive	Defines if the user is an interactive or non-interactive user.
User Store	Name of the user store the user belongs to.
Distinguished Name	The administrator may specify the user's Distinguished Name (DN) manually. This setting is required if the user would log in to the AccessMatrix system using the digital certificate authentication method where the user's DN will be retrieved automatically based on the digital certificate assigned to the user.
Account Status	To activate or disable the user account. • "Activated" means the user is universally enabled. This is the default setting.

Attribute Name	Description
	"Disabled" means the user is universally disabled either by the administrator or by the system due to a certain policy, e.g. the user has been inactive for a certain period.
Segment (if Segment license is applied)	Name of the segment the user belongs to.
Login	Last successful login time.
Logout	Last logout time.
Token Sharing Group ID	Unique group Id if token sharing is applicable. This defines the list of group Ids that supports multiple Token Sharing Groups (TSGs).
Attribute Tab	
This tab contains system predefined as well as custom (user-defined) attributes. The predefined attributes include the email address and the mobile number. The system can use this stored information to deliver a message to a user. For example, for SMS-OTP to send OTP to the mobile number or send activation/authorization code for the Vasco Multi-Device. The administrator is allowed to add, edit or delete any custom attributes. Notes: - For string attributes: - The maximum character length is 96 characters. - There is no restriction on character sets that can be used. Do note that AccessMatrix does not check the validity of the values (e.g. it does not check if the value specified for the Email attribute is a valid email address). As such, the user/administrator must ensure that the value specified for each attribute is valid. - For integer attributes: - The maximum value is 2147483647.	
Membership Tab	
This tab contains the group information for membership assignment as follows: group Id, group name, user store, and membership. - To assign a group to the user, check the Membership checkbox of the user group. - To unassign the group from the user, uncheck the Membership checkbox of the user group. Note: If there are more than 1000 records in the Membership tab display list, only 1000 records will be displayed. Therefore, you need to refine your search list by entering your search filter using the wildcard widget (i.e. "group2*") or the exact value in the empty field of Group Id at the end of the list. - Uses the wildcard asterisk "*" to search for all words beginning with, for example, group2 (i.e. "group2*"). The asterisk stands in for any number of letters following the "group2" in this example. Do not place the asterisk before or in between the word "group2".	

Attribute Name	Description
USO Entitlements Tab	
This tab will be displayed if the server license contains the USO feature. This tab indicates whether a user (Application Id) can access a USO application (Application Name). The entitlement can be a user to the USO application directly, or it can also be assigned via the user group (Edit Group - USO Entitlements tab); as long as the user is part of the user group, the user has the entitlement.	
Login Accounts Tab	
This tab contains the authentication realm and login module assigned to a user. Authentication realms that are automatically assigned can be identified based on the corresponding value displayed under the Assigned Automatically column. There are three types of automatically assigned authentication realms: • Authentication realms that are automatically assigned on a global level have the "Yes" value. Global-level (i.e. all users) realm auto-assignment is configured within the respective authentication realm's configuration page. • Authentication realms that are automatically assigned on a segment level have the "Yes (Segment)" value. Segment-level (i.e. all users in a segment and any sub-segments further down its segment hierarchy) realm auto-assignment is configured within the respective segment's configuration page. • Authentication realms that are automatically assigned on a user group level have the "Yes (Group)" value. Group-level (i.e. all users in a user group) realm auto-assignment is configured within the respective user group's configuration page. The administrator may manage the user's authentication realm by performing the following tasks: • Assign Authentication Realm Click the Add button to choose the authentication realm. • Edit Authentication Realm The authentication realm panel allows the modification of fields like realm Status, Start Time, and End Time. • Remove Authentication Realm Click the Delete button to remove the realm from the user. • Edit login module of the Authentication Realm The Login Module panel allows the modification of fields like Status, Force Change Password, Start Time, and End Time.	
Admin Privilege Tab	
This tab contains the admin roles and group admin roles assigned to the user. The group admin role panel (located beneath the admin role panel) is a read-only panel that displays the admin roles inherited from the assigned group membership. The administrator may manage the user's admin roles by performing the following tasks: • Assign user's admin role Click the Add button to choose the admin role. • Edit user's admin role The user admin role panel allows the modification of the Start Time, End time, Delegation Type, and Restriction fields.	

Attribute Name	Description
Remove admin role Click the Delete button to remove admin role from the user.	
Application Privilege Tab	
This tab contains the application roles that can be assigned to the user. This tab will be displayed if the server license contains UAM feature.	
SAML Application Privilege Tab	
This tab contains the list of SAML applications that the user can access. This tab will be displayed if the server license contains UAM feature.	
OAuth Client Privilege Tab	
This tab contains the list of OAuth Clients that the user can access. This tab will be displayed if the server license contains UAM feature.	
OAuth Consented Scopes Tab	
This tab lists all the scopes the user has granted OAuth Clients access to. Individual scopes are separated by spaces in the list. Do note that only OAuth Clients with scopes granted by a consenting user will be displayed. Click Reload to refresh the list as required. This tab will be displayed if the server license contains UAM feature.	
OAuth Session Persistence Tab	
This tab lists all valid access tokens/refresh tokens of the active OAuth Clients' sessions. You can invalidate the selected access tokens or check the Select to Invalidate All checkbox, and then click Invalidate for all the access tokens in the list. Click Reload to list all valid access tokens as required. This tab will be displayed if the server license contains UAM feature.	
Session Persistence Tab	
This tab will display the login sessions if Session Persistence is enabled.	
Assigned Token Tab	
This tab contains the list of assigned tokens to a user. It will display the token serial number, token vendor, token model, token status and token sharing group Id/members. This tab will be displayed if the user has View token rights.	
Access Token Tab	
Note: This tab will only be displayed for non-interactive users; it will not be displayed for interactive users. Verify that you are viewing the details of a non-interactive user by ensuring that the *Interactive field has the value "No".	

Attribute Name	Description
	This tab displays the access token associated with a non-interactive user. If there is an existing token, the following information is displayed: • Current Status - Indicates what the status of the token is. • Token - Contains the masked value of the access token. • Version - Indicates what version the current token is (e.g. "5" indicates that the current token is the 5th generated token). The administrator may manage the non-interactive user's access token by performing the following tasks: • Revoke the current access token Click the Revoke Token button to change a token from the "Activated" status to the "Revoked" status. The current token is now no longer usable. • Generate a new access token Perform the following steps: 1. Specify a value for Access Token Expiry (in seconds) to manually define the expiry period, or leave it blank for no expiry. 2. Click the Generate button to generate a new access token. All previous versions of the access token are invalidated. 3. Copy the value of the access token displayed in the Access Token field and store it in a secure location. Important Note: You MUST save the access token value at this point. It will only be displayed once, and you will be unable to retrieve this unmasked value of the token ever again. 4. You may use the access token as a Bearer token when calling the AccessMatrix REST API - do this by setting the value you saved to the HTTP Authorization header. • Verify an access token Perform the following steps: 1. Enter the value of the access token to be verified into the Access Token text box. This could be the value of the access token that was previously generated (as per the steps above), or any valid access token. 2. Click the Verify button to verify the specified access token. If successful, a pop-up dialog is displayed with the message "Token is successfully verified".

Note that there are additional buttons on the **View User** page: the **Token Assignment**, **Biometric Data**, **Social Network Login**, and **Secondary Account** buttons. These buttons are independent of the tabs.

- **Token Assignment**

Select this button to display the assigned tokens with their token serial numbers, vendors and models. You can reassign token(s) using the following methods:

- **System Assigned (Token Group)**

Let the system perform the token assignment to the user via token group. Select **System Assigned**, add a **Token Group** using the selector, then click **OK**.

- **System Assigned (Issuance Token Group)**

Let the system perform the token assignment to the user via [issuance token group](#). Select **Issuance Token Group for System Token Assignment**, add a **Token Group** using the selector, then click **OK**.

- **Enter Serial Number/ID**

Manually assign a token to the user based on the provided token serial number and the vendor. Select **Enter Serial Number/ID**, specify a **Token Serial Number**, choose the appropriate **Token Vendor** from the drop-down list, then click **OK**.



Note: A token cannot be manually assigned to the user using this method if it meets any of the following conditions:

- The token belongs to a token group that has the **Redis Pool Enabled** setting set to "true", and the AccessMatrix server's **amsystem.properties** file has been configured to use the [Redis cache](#) (i.e. am.cache.vendor=redis in the properties file).
- The token belongs to an issuance token group. You can verify this by checking whether the **Issuance Token Group for System Token Assignment Id** field of the token is empty (either in the **Search Token Result** page or the **View Token** page).

Finally, you can also unassign some or all of the previously assigned tokens.

- **Biometric Data**

Select this button to display biometric data such as the biometric vendor, type, label and status of a specific user for authentication purposes. You can delete the biometric data on this page.

- **Social Network Login**

Select this button to view the social network type (i.e. Facebook, Google, LINE, and WeChat) application, Id type, user identity and name, and its time of linking for a specific user. You can unlink all the social network links of a user on this page.



Note: This button will only be displayed if the server license for the UAM module is enabled and there's at least one entry for the **Social Network Login** feature.

- **Secondary Account**

Select this button to display any secondary accounts linked to the user. You can link secondary accounts or unlink secondary account links of a user on this page.

Manage Groups

A group consists of multiple interactive users who have common access privileges to the authorized applications or consists of multiple administrators who have common administrative roles for a particular administrative group.

There are two categories of groups, i.e. default store groups (a.k.a. internal groups) and external groups (from external user stores). When a user becomes a member of a group, the user will automatically inherit/obtain the administrative roles/rights assigned to the group.

You manage groups on the **Group** tab. Navigate to **Operation Dashboard > User & Segment > Group**. The following tasks can be performed on the **Group** tab:

- **Create a group**

Click **Create** to display the **Create Group** page, enter the group's information on each empty field, and click **Save**.



Note: Default store groups can be created in or deleted from the AccessMatrix System, but external groups, e.g. Active Directory (AD) groups, cannot be created or deleted from AccessMatrix System. External groups can only be edited for performing administrative tasks.

- **Search an existing group or groups**

You can find existing group(s) by entering the group Id, group name, user store or segment on the **Search Group** list, and then clicking **Search** to display the **Search Group Result** list.



Note: When all the input fields on the **Search Group** list are empty, click **Search** will show all existing groups on the **Search Group Result** list.

- **View an existing group**

Select a group on the **Search Group Result** list to view that group's details.

- **Edit Group information**

Click **Edit** on the **View Group** page to modify its information.

- **Delete a group**

Click **Delete** on the **View Group** page and click **OK** in the confirmation pop-up box.

- **Assign user group to a segment**

On the **Create/Edit Group** page, select the segment on the left pane and click **Choose Segment** to auto-fill the **Segment** attribute on the right pane.



Note: The **Segment Hierarchy** licensed feature is required to assign local groups to internal segments. They cannot be assigned to external segments.



Important Note: The administrator cannot edit/delete the group if there is a pending/rejected admin request of the same group entry.

Having administered the desired groups, the administrator may proceed to administer users. This is discussed in detail in [Manage Users](#).

View Group Page Attribute Description

Attribute Name	Description
*Group Id	Unique group Id.
*Group Name	Name of group.
Description	Description of group.
Version	Group entry version.
Distinguished Name	Administrator may specify the group's Distinguished Name (DN) manually. This setting is required if the user would log in to the AccessMatrix system using digital certificate authentication method where user group's DN will be retrieved automatically based on the digital certificate assigned.
User Store	Name of user store the group belongs to.
Segment (if Segment license is applied)	Name of the segment the group belongs to.
Attributes Tab	
The administrator can add, edit or delete any custom attributes.	
Membership Tab	
This tab contains the users that are a member of the selected group. The following user information is displayed: user Id, user name and user store.	
Notes:	
Administrator is only able to view but not able to modify the group membership. Group membership can only be done from the user's page.	

Attribute Name	Description
	If there are more than 1000 records in the Membership tab display list. Only 1000 records will be displayed. You can select the pagination arrows (HOME, LEFT, RIGHT or END arrow) at the top of the list to view the next page.
USO Entitlements Tab	
This tab will be displayed if the server license contains the USO feature. This tab indicates whether a user (Application Id) can access the USO application (Application Name). The entitlement can be a user to the USO application directly, or it can also be assigned using the user group (Edit Group - USO Entitlements tab). As long as the user is part of a user group, the user has the entitlement. It also allows password sharing in a user group. USO helps to perform single-sign-on (SSO) to target applications. It does that by filling in the target application's user Id and password automatically. At the user level, every member of a group has his/her password. If stored at the group level, all group members will use the same user Id and password.	
Admin Privilege Tab	
This tab contains the admin roles assigned to the user group. The administrator may manage the group's admin roles by performing the following tasks: Assigning group's admin role Click the Add button to choose the admin role. Editing user's admin role The group admin role panel allows the modification of the Start Time, End Time and Delegation Type fields. Removing admin role Click the Delete button to remove admin role from the group.	
Application Privilege Tab	
This tab contains the application roles that can be assigned to the group. This tab will be displayed if the server license contains UAM feature.	
SAML Application Privilege Tab	
This tab contains the list of SAML applications that the group can access. This tab will be displayed if the server license contains UAM feature.	
OAuth Client Privilege Tab	
This tab contains the list of OAuth Clients that the group can access. This tab will be displayed if the server license contains UAM feature.	
Authentication Policy Tab	
This tab contains the Authentication Realm attribute. Authentication realms that are added under this attribute are automatically assigned to all users belonging to this user group. These	

Attribute Name	Description
	authentication realms are assigned to group users in addition to any globally, automatically assigned authentication realms. Only realms that are not already globally, automatically assigned can be added here.

Manage Segments

A segment is a basic administration unit in AccessMatrix which represents an entity within an organization such as business division, subsidiary, department or unit. A segment (a.k.a. parent segment) may contain sub-segments (a.k.a. child segments), users, user groups, applications and token groups. User(s) can be granted administration roles and assigned as security administrator(s) to manage the restricted segment(s), the sub-segment(s) and their corresponding groups, users and applications according to the delegation type.

There are two categories of segments; internal and external segments as described below:

- **Internal Segments**

Internal segments are segments created under the Root Segment of the Default Store. The Root Segment is created (during system installation and initialization setup), and it can be added with no limit of segments (or sub-segments). Thus, the administrator may create groupings and manage groups and users and applications according to the organizational structure and have it represented in a segmented hierarchical model where the users and the groups are created and saved in the AccessMatrix system's default store.

- **External Segments**

External segments are associated with a user store that points to an external user registry (e.g. Active Directory) which is not editable and only viewable. However, groups and users residing in external segments can be granted administrative privileges to perform certain administrative tasks, e.g. assign admin privilege, view the report, etc.

You manage segments on the **Segment** tab. Navigate to **Operation Dashboard > User & Segment > Segment**. The following tasks can be performed on the **Segment** tab.

- **Add Internal Segment/Sub-Segment**

On the **Segment** page, choose a segment from the left pane as the parent segment for the new segment to be created and select **View**. Next, click **Add Segment** on the **View Segment** page to

create a child segment. Notice the previously selected segment Id is shown as the **Parent Segment Id**. Next, enter the segment Id, segment name, its description and click **Save**.

- **Edit Internal Segment Attributes**

On the **View Segment** page, click **Edit**. Make the required changes and click **Save**.

- **View Internal/External Segment/Sub-Segment**

You can view the internal segments, external segments and their sub-segments on the left **Segment Hierarchy** pane of the **Segment** tab. Select a segment at the left pane and select **View**.



Notes:

- To view external segment(s), make sure segment configuration to the external user store has been done accordingly.
- To map segments to the existing external stores, refer to [the mapping on the Segment or OU Tab](#) of the respective user store.
- To integrate the user registry as a user store, read [Manage User Stores](#).

- **Remove Internal Segment/Sub-Segment**

Click **Delete** on the **View Segment** page and click **OK** in the confirmation pop-up box.

- **Search Segment's Group for Administration Tasks**

Select the **Group** tab. On the **Search Group** page, select a desired segment on the left pane and select **Choose Segment**. Click **Search** on the right pane to find all the user groups that belong to the selected segment.

- **Search Segment's User for Administration Tasks**

Select the **User** tab. On the **Search User** page, select a desired segment on the left pane and select **Choose Segment**. Click **Search** on the right pane to find all the users that belong to the selected segment.



Notes:

- **Segment** and **Segment Hierarchy** are licensed features. Hence, the appropriate licensed feature must be subscribed to in order to be able to use segments. Otherwise, this feature will not be available in the Admin Console.
 - The administrator cannot edit/delete a segment if there is a pending/rejected admin request of that segment entry.
 - The administrator is only allowed to create the user(s)/group(s) and assign to internal segment via the Admin Console. This is not allowed for external segments via the Admin Console. To assign a user or user group to a segment, refer to [Assign User Group to a Segment](#).
-

View Segment Page Attribute Description

Attribute Name	Description
*Segment Id	Unique segment Id.
*Segment Name	Name of segment.
Description	Description of segment.
Version	Segment entry version.
Attribute Tab	
This tab contains custom (user-defined) attributes. The administrator is allowed to add, edit or delete any custom attributes.	
Authentication Policy Tab	
This tab contains the Authentication Realm attribute. Authentication realms that are added under this attribute are automatically assigned to all users belonging to this segment. Realm auto-assignment is inherited down the segment hierarchy, meaning all users in any of its sub-segments (a.k.a child segments) further down the segment hierarchy will also be auto-assigned the realm. These authentication realms are assigned to segment users in addition to any globally, automatically assigned authentication realms. Only realms that are not already globally, automatically assigned can be added here.	

4.1.2. Request Management

An administrator can request to have his or her role to be able to create/edit/delete a segment, group, token group, actor's group, and client access, or be able to perform all operations (create, edit, and delete) when carrying out his or her administration works. In this case, an admin requester is a Maker, and a Checker is another administrator tasked to approve, reject or delete the requests.

Refer to [Configure Admin Workflow](#) for more information. The administrator can retrieve the below request lists:

- Requests created by the currently logged-in administrator (grouped as **My Request**) where the requests can be either processed or yet to process.
- Requests pending for approval by a current logged-in administrator or other administrators (grouped as **My Task**).



Note: In order for an administrator (maker) to create a request, refer to [Create a Request](#) under the Configure Admin Workflow section.

To retrieve the request list:

1. Locate the **Admin Request** page by navigating to **Operation Dashboard > Request > Admin**.
2. On the right pane, select the **My Request** tab for the pending request list or the **My Task** tab for the pending approval list.
 - a. **My Request** - By default, the **My Request** tab is selected and this will display the list of requests raised by the currently logged-in administrator if any. Some of the requests may have been processed e.g. rejected while others are still pending for approval by other administrators. Request details can be viewed by clicking on the desired request Id.
 - b. **My Task** - This tab will display the list of requests pending for approval. The requests in this list are raised by other administrators and have not been processed i.e. pending for approval by a current logged-in administrator or other administrators. Click on the desired request Id to view the details, in order to perform further action on the request.

View Request Page Attribute Description

Attribute Name	Description
*Request Id	Unique request Id number.

Attribute Name	Description
Version	Version of request entry. Each time an amendment is made to the request, the version number increases by 1.
Type	Type of request.
Maker	Requester Id.
Action	Action performed on the request.
Checker	Approver Id.
Status	Current status of the request.
Content Summary	Request summary.
Details Tab	This tab contains the details of the request.
Histories Tab	This tab contains the histories of the administrator's actions on the request as well as the comments specified for the request.
Changes Tab	This tab contains the changes to the fields and displays the old value and new value, if applicable.



Note: Once a request is made, the administrator (maker) can edit or withdraw the request as required. The administrator (checker) can use the **Approve**, **Reject**, or **Delete** buttons on the **Details** tab to take action after checking. The Administrator will be prompted for confirmation. Upon successful execution of an action, a screen will appear allowing the administrator to specify the comment for the request.

4.1.3. Login Account Management

A user may be assigned various login modules, each associated with a specific login method. The login modules assigned to a user are also referred to as login accounts.

An administrator can manage an individual user's login accounts via that user's **View User** page.

However, AccessMatrix also supports the management of multiple login accounts at once. Each batch update must be made to login accounts of the same type.

This feature is provided as a convenient method to view and update the statuses of batches of multiple login accounts. It may be useful in situations where there is a need to enable a number of disabled login accounts at once.

Manage Login Accounts

Administrators with the relevant admin privileges can manage login accounts on the **Login Account** tab. Navigate to **Operation Dashboard > Login Account > Login Account**. The following tasks can be performed on this tab:

- **Search for login account(s)**

You can find login accounts assigned to users by selecting a login module type, status, and the user store on the **Search Login Account** search form. Click the **Search** button to begin the search.

- **Modify login account status**

On the **Login Account** page, after searching for login accounts, you can update the statuses of the login accounts returned in the search query. Tick the checkboxes of the login accounts to be updated, select a **New Status**, and click the **Modify Account** button. A confirmation message is displayed after a successful update.



Note: Refer to [Manage User Authentication Service - Usable Login Account Status](#) for details on the different login account statuses.

4.2. Administration Dashboard

This dashboard aggregates all system and AccessMatrix products related settings, administrations, policies, configurations, and functions. Administration dashboard of Admin Console user interface contains the first-level navigation bar with the **Home, System, Security Policy, Configuration, UCM,** and **About** options.

Selecting this dashboard will display the user's last successful login date and time based on the user's most recent login events or history.

4.2.1. Manage Admin Role Delegation

The following terms are used in admin role delegation:

- **Administration Right**

An administration right is operation permission of a system resource. For example, for the "Edit User" administration right, "Edit" is the operation permission of the "User" resource type. All administration rights in the AccessMatrix system are grouped according to their resource types to form many sets of administration rights, e.g. **User Admin Rights, System Configuration Rights**, etc.

- **Administration Role**

An admin role is a named collection of administration rights that can be granted to users or user groups. Each user or user group can be granted with one or multiple admin roles where each assigned role can be applied with a start time restriction, end time restriction, user store restriction or segment restriction to further restrict user scopes of access. In addition to that, each assigned admin role has to be associated with a delegation type to extend or restrict the delegation power of the given admin role.

- **Delegation Type**

Delegation types of admin roles are defined in the table below:

Delegation Type	Description
Own Only	User will own the admin role, however, not allowed to delegate it to other users.
Own + Delegate	User will own the admin role and also able to delegate it to other users.
Delegate Only	User will not own the admin role, though able to delegate it to other users.
Own + Delegate One Level	User will own the admin role and delegate to other users with the delegation type "Own Only" to restrict the target users from further delegating.
Delegate One Level Only	User will not own the admin role, however, able to delegate to other users with delegation type "Own Only" to restrict the target users from further delegating.

AccessMatrix System provides these predefined system roles:

Admin Role Name	Description
AuditorRole	Allows user to view all the dashboards, as well as reports.
InitialAdminRole	User will have all admin rights.
PolicyAdminRole	User will have system configuration rights.
SelfServiceRole	Allows user to view USO application password in Self-Service Facility.
TokenAdminRole	User will have token management rights.
UCMAdminRole	User will have UCM admin rights.
UCMPasswordOwner	Allows user to reset UCM credential's password.
UserAdminRole	User will have user admin rights.
ViewReportOnlyRole	Allows user to view the Report Dashboard only. However, user must also have the "Report - View" right and be assigned with the "View" admin right for the specific resource/object to view, e.g the "Token - View" right.
WorkflowAdminRole	User will have workflow admin rights.


Note: The administrator will not be able to edit/delete the admin role if there is a pending/rejected admin request pertaining to that admin role entry.

To locate the **Admin Role** page, navigate to **Administration Dashboard > Security Policy > Admin & Audit > Admin Role** on the left pane. The following tasks can be performed on the admin roles:

- Create Admin Role**
Click **Create** to display the **Create Admin Role** page, enter the details and click **Save**.
- Search for Admin Role**
Can be searched by the admin role Id or the admin role name. All available admin roles are displayed by default.
- View Admin Role**
Click an admin role Id under the **Search Admin Role Result** list to view the details.
- Edit Admin Role**
Click **Edit** on the **View Admin Role** page to modify the details and click **Save**.
- Delete Admin Role**
Click **Delete** on the **View Admin Role** page and click **OK** in the confirmation pop-up box.

View Admin Role Page Attribute Description

Attribute Name	Description
*Admin Role Id	Unique Id of admin role.
*Admin Role Name	Name of admin role.
Description	Description of admin role.
Version	Admin role entry version.
Admin Role Tab	
Resource column	Defines the resource type of an operation, e.g. "Policy"
Operation column	Lists all operations of the corresponding resource type, e.g. "Create", "Edit", "Delete", "View", etc.



Note: The **Admin Role** tab displays the administration rights of all the system resources. To include/remove admin right(s) to the admin role, check/uncheck the individual checkbox(es) of the admin right(s) respectively. Tick the **Select/Unselect All** checkbox found on top of the **Operation** column to select/deselect all admin rights. And tick the respective **Select/Unselect All** checkbox to select/deselect admin rights for that specific resource group only.

4.2.2. Manage User Authentication Service

AccessMatrix User Authentication Service enables organizations to rapidly deploy a wide variety of authentication methods to address the business requirements for evolving and robust authentication mechanisms through a single, unified framework. This authentication infrastructure supports multiple authentication mechanisms such as Ids/Passwords, dynamic passwords, certificates, biometrics, and security tokens using the Pluggable Authentication Module (PAM) approach.

In AccessMatrix, the authentication realm is used to associate users' credentials for authentication based on a lookup module and the login module(s) where user entries can be created into a local repository of AccessMatrix or integrated from Lightweight Directory Access Protocol (LDAP)-compliant directory. Authentication realm(s) would then be assigned to users for authentication purposes.

Configure User Lookup Module

To configure the user lookup module for authentication, first, locate the user entries into the respective user store. The user entries can be in a local user registry which is created into the local repository (a.k.a default store) of AccessMatrix or in an external user registry which is either:

- Integrated into AccessMatrix from the existing LDAP-compliant directory (this is discussed in [Manage User Stores](#)).
- Modeled as a Dummy User Store in AccessMatrix.
- Connected to AccessMatrix with its fields mapping using Java Database Connectivity (JDBC) connection.

To locate the **User Lookup Module** page, navigate to **Administration Dashboard > Configuration > Authentication > User Lookup Module** on the left pane. The following tasks can be performed on the user lookup modules:

- **Create User Lookup Module**
Click **Create** to display the **Choose User Lookup Module Implementation** page. Select an implementation and click **Next** to open the **Create User Lookup Module** page. Enter the user lookup module's details and click **Save**.
 - **Search for User Lookup Module**
Can be searched by using the user lookup module Id or the user lookup module name. All available user lookup modules are displayed by default.
-

- **View User Lookup Module**

Click on a user lookup module Id under the **Search User Lookup Module Result** list to view the details.

- **Edit User Lookup Module**

Click **Edit** on the **View User Lookup Module** page to modify the details and click **Save**.

- **Delete User Lookup Module**

Click **Delete** on the **View User Lookup Module** page and select **OK** in the confirmation pop-up box.

Federated/Authenticated Lookup Mechanisms

- **SAML**
- **FIDO UAF** (*Deprecated*)
- **Social Network Login**
 - Facebook
 - Google
 - LINE
 - WeChat

Supported Types of User Lookup Modules Implementation

Lookup Module	Description
Attribute Based Lookup	This user lookup module performs the lookup of the user based on the user attribute (for example, email).
Auto-Provisioning Lookup	This user lookup module ensures the existence of the asserted user and grant asserted admin roles.
Microsoft Entra ID Lookup	This Microsoft Entra ID Lookup module performs the lookup of the user entries from Microsoft Entra ID User Store for authentication.
Certificate DN Lookup	This user lookup module provides a search for the requested user in either an internal or external user store after a successful client certificate authentication done by AccessMatrix server. This lookup is based on certificate filter that uses certificate DN lookup implementation. A predefined Certificate DN Lookup module called "CertificateDnLookup" will be created upon installation. The administrator may make use of the

Lookup Module	Description
	predefined entry with necessary changes or create a new entry that caters for the user requirements.
Dummy Lookup	This user lookup module provides a dummy search for users through a trusted user lookup method. It is used when AccessMatrix is not the master user store and to reduce the efforts of user replication. A dummy user store is used so that no actual user replication is required. For example: - When the AccessMatrix server is used as a proxy authenticating other authentication servers like Remote Authentication Dial-In User Service (RADIUS). - When AccessMatrix is served as an authentication server for user tokens, and yet it is not the master user store. A predefined Dummy Lookup module called "DummyLookup" will be created upon installation. The administrator is not required to create another entry of the Dummy Lookup module unless the predefined entry could not fulfil the user requirements. The administrator is also not advised to delete the existing entry of the Dummy Lookup module unless it is necessary to do so.
Facebook Login Lookup	This user lookup module uses Facebook Login to look up a user and perform single sign-on to AccessMatrix. This lookup module must be used together with the Social Network Login module. Note: This lookup module will not be displayed from the User Lookup Module list box if the UAM feature is disabled.
FIDO UAF Lookup (Deprecated)	This user lookup module parses FIDO UAF authentication response and obtains user identity from the given response.
FIDO2 Discoverable Credentials Lookup	This user lookup module is used for FIDO2 usernameless (or ID-less) authentication, where the user is not known during login. It must be used together with another user lookup module (eg. "SystemLookup"). The lookup modules are linked using the multiple choice lookup module.
Google Sign-In Lookup	This user lookup module uses Google Sign-In to look up a user and perform single sign-on to AccessMatrix. This lookup module must be used together with the Social Network Login module. Note: This lookup module will not be displayed from the User Lookup Module list box if the UAM feature is disabled.
JDBC Lookup	This user lookup module provides a search for users in a JDBC-based external registry through JDBC connections. In other words, the user and group information is stored in an external JDBC-based database.

Lookup Module	Description
LDAP Lookup	This user lookup module provides a search for the requested user in an external LDAP compliant user store, i.e. an LDAP user registry, e.g. AD registry. To find out more about LDAP user store integration, refer to Manage User Stores. Predefined LDAP Lookup modules called "DefaultActiveDirectoryLookup" and "DefaultOpenLDAPLookup" will be created upon installation. The administrator may make use of the predefined entries with necessary changes or create new entries that cater to user requirements.
LINE Sign-In Lookup	This user lookup module uses LINE Sign-In to lookup user and single sign-on to AccessMatrix. This lookup module must be used together with the Social Network Login module. Note: This lookup module will not be displayed from the User Lookup Module list box if the UAM feature is disabled.
Multiple Choice Lookup	This user lookup module supports user identity lookup on the chaining of JDBC/LDAP lookup. The lookup module supports only three modules in one chain and will search the store in the order specified.
OIDC Relying Party Lookup Module (For Authorization Code flow)	This user lookup module is used to exchange authorization code for OIDC tokens to get user identity.
Out-of-Band Lookup Module	(For CLP use only) This user lookup module is used for scan-and-login token verification to both identify and authenticate the user by providing a challenge image for the token to scan. Currently, it is supported for OneSpan/Vasco Multi-Device token.
SAML Artifact Resolution Lookup and Login	This user lookup module is used for SAML Idp initiated Artifact binding Authentication where AM5 serves as a SAML Service Provider. It will parse and validate a user's artifact response to obtain user identity and other user-related assertion attributes (if any).
SAML Web Browser SSO Lookup and Login	This user lookup module is used for SAML Web Browser SSO where AccessMatrix server serves as a SAML Service Provider. It will parse and validate the SAML response to obtain user identity and other user's related assertion attributes (if any).
System Lookup	This user lookup module provides a search for the requested user in the system user registry (a.k.a default store). A predefined system user lookup module called "SystemLookup" will be created upon installation. Generally, the administrator should not create a new entry of the System Lookup module as there should be only one entry of the System Lookup module throughout the whole AccessMatrix System. The administrator is also not advised to delete the existing entry of the System Lookup module unless necessary to do so.

Lookup Module	Description
WeChat Sign-In Lookup	This user lookup module uses WeChat Sign-In to lookup user and single sign-on to AccessMatrix. This lookup module must be used together with the Social Network Login module. Note: This lookup module will not be displayed from the User Lookup Module list box if the UAM feature is disabled.
 Important Note: - Administrator will not be able to edit/delete a user lookup module if there is a pending/rejected admin request of that user lookup module entry. - Having administered the desired user lookup modules, the administrator may proceed to administer the associated login modules to allow the users to log in to the system using the desired login/authentication method supported.	

View User Lookup Module Page Attribute Description

Attribute Name	Description
*User Lookup Module Id	Unique Id of user lookup module .
*User Lookup Module Name	Name of user lookup module .
Description	Description of user lookup module .
Version	User lookup module entry version.
Implementation	Type of user lookup module implementation.
User Store	User store associated with the user lookup module .

 **Note:** Detailed configuration differs depending on the lookup module.

Attribute Based Lookup

Attribute Name	Description
Attributes Tab	
Attribute	Either one of these fields (Attribute, Field) are set but not both. Custom attribute(s) may be added by following the instructions described in Adding Custom Attribute .

Attribute Name	Description
Field	Either one of these fields (Attribute , Field) are set but not both. Possible selection value(s): "User UUID"

Auto-Provisioning Lookup

Attributes Tab
There are no available attributes, however, custom attribute(s) may be added by following the instructions described in Adding Custom Attribute .
Agent User Id Tab
Assign an agent to be assigned to the module.

Microsoft Entra ID Lookup

Attributes Tab
There are no available attributes, however, custom attribute(s) may be added by following the instructions described in Adding Custom Attribute .

Certificate DN Lookup

Attribute Name	Description
Configuration Tab	
Certificate Filter	This filter is used to map an attribute in the certificate with an attribute in the user store for user lookup. The value should be specified in the form of an equation, where the left side of the equation represents the user store attribute and right side of it represents the certificate attribute (or the certificate attribute with the Java Regular Expression group). Format for the value: "<field name of user store>=<field/field's component name of client certificate OR field/field's component name of client certificate plus the Java Expression Group>" To apply a Java Regular Expression group to the field/field's component name of the certificate, specify the field/field's component name of certificate followed by [1].

Attribute Name	Description
Java Regular Expression group	This is used to extract the value from the certificate attribute specified in Certificate Filter (e.g. Certificate Subject, Issuer or etc.) based on the Java Regular Expression defined here. For example: "^CN=.*\([([A-Z0-9]{7}+)\)\),.*" If this is used, ensure that in the Certificate Filter value, the certificate's attribute (e.g. Subject for Subject, or Issuer for Issuer) is followed by [1] as the regular expression group name of this field, e.g. "Subject[1]".
Allow Trusted App PKI User Authentication	When enabled, this allows trusted applications (e.g. CLP) to use PKI user authentication.
Agent-based Access Restriction	This is used to whitelist the agent (which must be a non-interactive user/agent) of the trusted application. If the trusted application's agent is not whitelisted, PKI authentication for its application's end users will fail. This attribute must be configured if Allow Trusted App PKI User Authentication is set to "true". Default: false

Dummy Lookup

Attributes Tab
There are no available attributes, however, custom attribute(s) may be added by following the instructions described in Adding Custom Attribute .

Facebook Login Lookup

Attribute Name	Description
Attributes Tab	
App Id	Unique Id of application. This value is generated from Facebook Developer Console.
App Secret	Secret code of an application. This value is generated from Facebook Developer Console.
Use Authorization Code Flow	Indicate whether an input is an authorization code or access token. - Setting to "Yes" causes the authorization code flow to be used. Authorization code flow is used for application with server-side component and when the access token is not exposed to the client-side. - Setting to "No" causes the authorization code flow to not be used; therefore, the input must be an access token. Default: false

Attribute Name	Description
User fields to Query	Returns the queried user field, e.g. "Id", "name", or "email". Default: id, name, email
Return Access Token in Response	Determines whether to return access token to the AccessMatrix client-side that calls API to login via Facebook. Default: No
HTTP Proxy	
Enabled	Setting to "true" will enable the HTTP Proxy. Default: false
Proxy Host Name	Refers to the URL of the proxy server.
Port	Refers to the port of the proxy server.
User Name	Refers to the username used to authenticate the proxy server.
User Password	Refers to the user's password used to authenticate the proxy server.
NTLM Authentication	Indicates whether NTLM authentication is used. If NTLM Authentication is enabled, Domain Name must be specified. In addition to that, it might be necessary to configure the AM5 server host domain name or IP address in the amsystem.properties file with the property key "am.social.ntlm.host" for each instance. Default: false
Domain Name	Refers to the domain name of AD used in NTLM authentication.
HTTP Reverse Proxy	
Replace Facebook URL's host name	Several Facebook URLs will be triggered during runtime. By default, the hostname of those URLs is "https://graph.facebook.com". If a different hostname is specified (e.g. "http://myforwardProxyhost"), then it will replace the hostname of the corresponding Facebook URLs. Default: https://graph.facebook.com
Auto Create New User	
Enabled	Setting this to "true" will map: - email (user profile attribute from Facebook) to user Id and user's email - name (user profile attribute from Facebook) to the username Default: false
User ID Mapping	If "Facebook Email" is selected, the user's Facebook account is expected to be registered with email as his or her primary Facebook Contact or else an exception will be thrown to request users to do so.

Attribute Name	Description
	If "Facebook ID" is selected, the user's Facebook Email, which is retrieved from the user's Facebook Profile, will be mapped as the User ID during the auto user creation process; however, if the Facebook Email is not available, the Facebook ID will be used instead. Options: Facebook Email, Facebook ID Default: Facebook Email
User Store	Refers to the target JDBC user store. Default: DefaultStore

FIDO UAF Lookup (Deprecated)

Attribute Name	Description
Attributes Tab	
Token Batch	The token batch of the UAF token (Deprecated) or Google Authentication OATH TAP token.

FIDO2 Discoverable Credentials Lookup

Attributes Tab	
There are no available attributes. However, custom attribute(s) may be added by following the instructions described in Adding Custom Attribute .	

Google Sign-In Lookup

Attribute Name	Description
Attributes Tab	
Client Id	Unique Id of a client. This value is generated from Google Cloud Platform Console.
Client Secret	Secret code of the client. This value is generated from Google Cloud Platform Console.
Use Authorization Code Flow	Indicate whether an input is an Authorization Code or ID Token. - Setting to "Yes" causes the authorization code flow to be used. Authorization code flow is used for application with server-side component and when the access token is not exposed to the client-side. - Setting to "No" causes the authorization code flow to not be used; therefore input must be an ID token.

Attribute Name	Description
	Default: false
Return Access Token in Response	Determine whether to return an access token. Default: No
HTTP Proxy	
Enabled	Setting to "true" will enable the HTTP Proxy. Default: false
Proxy Host Name	Refers to the URL of the proxy server.
Port	Refers to the port of the proxy server.
User Name	Refers to the username used to authenticate the proxy server.
User Password	Refers to the user's password used to authenticate the proxy server.
NTLM Authentication	Indicates whether NTLM authentication is used. If NTLM Authentication is enabled, Domain Name must be specified. In addition to that, it might be necessary to configure the AM5 server host domain name or IP address in the amsystem.properties file with the property key "am.social.ntlm.host" for each instance. Default: false
Domain Name	Refers to the domain name of AD used in NTLM authentication.
HTTP Reverse Proxy	
Replace Google Account's URL hostname	Several Google Account URLs will be triggered during runtime. By default, the hostname of the URLs is "https://accounts.google.com". If a different hostname is specified (e.g. "http://myforwardProxyhost"), then it will replace the hostname of the corresponding Google URLs. Default: https://accounts.google.com
Replace Google API's URL hostname	Several Google API URLs will be triggered during runtime. By default, the hostname of the URLs is "https://www.googleapis.com". If a different hostname is specified (e.g. "http://myforwardProxyhost"), then it will replace the hostname of the corresponding Google URLs. Default: https://www.googleapis.com

Jdbc Lookup

Attributes Tab
There are no available attributes, however, custom attribute(s) may be added by following the instructions described in Adding Custom Attribute .

LINE Sign-In Lookup

Attribute Name	Description
Attributes Tab	
Client Id	Unique Id of client. This value is generated from LINE Developer Console.
Client Secret	Secret code of the client. This value is generated from the LINE Developer Console.
User Fields to Query	Returns the queried user field. sub represents the user ID generated by LINE. Default: sub,name
Return Access Token in Response	Determine whether to return an access token. Default: No
HTTP Proxy	
Enabled	Setting to "true" will enable the HTTP Proxy. Default: false
Proxy Host Name	Refers to the URL of the proxy server.
Port	Refers to the port of the proxy server.
User Name	Refers to the username used to authenticate the proxy server.
User Password	Refers to the user's password used to authenticate the proxy server.
NTLM Authentication	Indicates whether NTLM authentication is used. If NTLM Authentication is enabled, Domain Name must be specified. In addition to that, it might be necessary to configure the AM5 server host domain name or IP address in the amsystem.properties file with the property key "am.social.ntlm.host" for each instance. Default: false
Domain Name	Refers to the domain name of AD used in NTLM authentication.

Attribute Name	Description
Auto Create New User	
Enabled	Setting this to "Yes" will map: - email (user profile attribute from LINE) to user Id and user's email - name (user profile attribute from LINE) to the username Default: No
User Store	Refers to the target JDBC user store. Default: DefaultStore

LDAP Lookup

Attribute Name	Description
Attributes Tab	
Find UserStore based on NT Domain value	To support for Windows/AD domain user lookup especially for support of multiple Windows/AD domains, LDAP Lookup will expect the user Id to be in the format of "DOMAIN\USERID". - If set to "true", LDAP Lookup will extract the "DOMAIN" value and search for the user in an AD user store where the NT Domain value specified in the AD user store matches the "DOMAIN" value. - If set to "false", LDAP Lookup will search for the user in the user store specified in the Lookup module itself. Default: true

Multiple Choice Lookup

Configuration Tab
This tab consists of three lookup module attributes to support identity lookup chaining.

Out-of-Band Lookup

Attribute Name	Description
Attributes Tab	
*Verification Login Module	Select the required login module from the drop-down list for token verification.

Attribute Name	Description
Challenge Image Type	Select the type of challenge image that needs to be scanned. Default: VascoMultiDevice

SAML Artifact Resolution Lookup and Login

Attribute Name	Description
Attributes Tab	
SAML Service Provider	A configuration of SAML Service Provider, which will be used to resolve artifact and parse artifact response.
SAML Identity Provider	A configuration of SAML Identity Provider, which will be used to generate Artifact Resolve. The Artifact Resolve will be forwarded to the Artifact Resolution URL of the SAML Identity Provider.
Response Authentication Context Mapping	The authentication context(s) required by the lookup module for mapping with the authentication context(s) of the SAML Response.
Auto Create New User	This attribute indicates whether a new user would be auto-created if it is not found in the user store of the user lookup module. Default: true
User Identity Mapping	This value is used as a user store's attribute, which is mapped with the resolved user identity of SAML Response. Default: UUID
Custom Handler	
Short Class Name for Create New User	The custom class name that implements ICreateNewUser.java .
Short Class Name for User Lookup Data Binding	The custom class name that implements IUserLookupDataBinding.java .

SAML Web Browser SSO Lookup and Login

Attribute Name	Description
Configuration Tab	
SAML Service Provider	A configuration of SAML Service Provider, which will be used to parse SAML response. (mandatory)

Attribute Name	Description
SAML Identity Provider	A configuration of SAML Identity Provider, which will be used to parse SAML response. (mandatory)
Is IDP Initiated Authentication Response	This is to specify if the SAML response is generated based on the Identity Provider-initiated (without SAML request) Web Browser SSO Profile or the Service Provider-initiated (with SAML request) Web Browser SSO Profile. Default: true
User Identity Mapping	The attribute that will be used to resolve the user identity based on the response. Default: UUID
Response Authentication Context Mapping	The context value to use in the validation of the SAML response to ensure that the user is authenticated using the same authentication context value.
Short Class Name for User Lookup Data Binding	The name of the custom class that implements IUserLookupDataBinding.java , which produces the lookup result used for user lookup based on the validated SAML Response.
User Auto Provisioning Tab	
Auto Create New User	If set to "true", either the User Auto Provisioning Module or Short Class Name for Create New User attribute must be configured. This option is only supported for the Default User Store and JDBC User Store. Default: false
User Auto Provisioning Module	The user auto-provisioning module to be used for the auto-provisioning of users in the AccessMatrix server.
External ID Type	The ID type of the external user identity that will be linked with an AccessMatrix user when User Identity Mapping is set to "Secondary Account".
Short Class Name for Create New User	The name of the custom class that implements ICreateNewUser.java , a custom plugin that provisions users after the successful parsing of the SAML Response. This custom plugin can be used for auto-provisioning instead of the user auto provisioning module. If User Identity Mapping is set to "Secondary Account", this class must also handle the identity mapping provisioning that links the SAML SP user with an AccessMatrix user.

System Lookup

Configuration Tab
There are no available attributes, however, custom attribute(s) may be added by following the instructions described in Adding Custom Attribute .

WeChat Sign-In Lookup

Attribute Name	Description
Attributes Tab	
Client Id	Unique Id of client. This value is generated from WeChat Developer Console.
Client Secret	Secret code of the client. This value is generated from WeChat Developer Console.
Return Access Token in Response	Determines whether to return an access token. Default: No
HTTP Proxy	
Enabled	Setting to "true" will enable the HTTP Proxy. Default: false
Proxy Host Name	Refers to the URL of the proxy server.
Port	Refers to the port of the proxy server.
User Name	Refers to the username used to authenticate the proxy server.
User Password	Refers to the user's password used to authenticate the proxy server.
NTLM Authentication	Indicates whether NTLM authentication is used. If NTLM Authentication is enabled, Domain Name must be specified. In addition to that, it might be necessary to configure the AM5 server host domain name or IP address in the amsystem.properties file with the property key "am.social.ntlm.host" for each instance. Default: false
Domain Name	Refers to the domain name of AD used in NTLM authentication.

Configure Login Module

Login module provides a login or authentication method to verify user login credentials. Each login or authentication method could be optionally configured with a login policy and password policies where applicable.

To locate the **Login Module** page, navigate to **Administration Dashboard > Configuration > Authentication > Login Module** on the left pane. The following tasks can be performed on login modules:

- **Create Login Module**

Click **Create** to display the **Choose Login Module Implementation** page. Select an implementation and click **Next** to open the **Create Login Module** page. Enter the login module's details and click **Save**.

- **Search for Login Module**

Can be searched by login module Id or login module name. All available login modules are displayed by default.

- **View Login Module**

Click on a login module Id under the **Search Login Module Result** list to view the details.

- **Edit Login Module**

Click **Edit** on the **View Login Module** page to modify the details and click **Save**.

- **Delete Login Module**

Click **Delete** on the **View Login Module** page and select **OK** in the confirmation pop-up box.

Supported Authentication Mechanisms (Embedded)

- **System Password Basic (1FA)**

- **End-to-End Encryption Authentication**

- Gemalto Luna Payment HSM
- Gemalto Protect Server External
- Pseudo (verification in AccessMatrix Server)

- **Knowledge Based Authentication**

- **Out-of-Band One-Time Password via SMS or Email**

- AccessMatrix UAS SMS OTP
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- Vasco Digipass
 - **One-Time Password Hardware Tokens**
 - Vasco (Go3, Go6, DP260, DP27X)
 - OATH (Gemalto, Vasco, Oberthur, Yubico) with RFC 6030 PSKC for token import
 - Dynamicode
 - **Mobile Tokens**
 - Vasco Digipass for Mobile
 - V-Key
 - YesSafe
 - **Face Biometrics**
 - Sensetime
 - **Public Key Infrastructure (PKI) Signature Verification**

External Authentication Mechanisms

- **AD/LDAP**
- **SGFIndex Identity Login**
- **Singapore Assurity via RTAP**
- **Social Network Login**

Social Network Login is configured as part of the lookup module where its Single Sign-On token is verified and the user is identified at the same time

Types of Login Module Implementation

Login Module	Description
Auto-Provisioning Login	(Unused)
Microsoft Entra ID Login	This login module provides LDAP-based password basic login methods for the Microsoft Entra ID users with administrator rights in an external user store (Microsoft Entra ID User Store). It is assumed that the administrator has already set up the Microsoft Entra ID account in a read-only operation mode that also means that the administrator cannot change or reset the password.

Login Module	Description
Biometric Login	This login module is used for Biometric login (i.e. NEC fingerprint) where its users may reside in either an internal or external user store.
China OTP Token Login	This login module is used to provide OTP verification for the China_OTP token. Note: This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution.
Composite Login	This login module provides composite (a.k.a. combined) login methods for the user who may reside in either an internal or external user store. Currently, the composite login methods can support a combination of a dynamic password (a.k.a. OTP) and static password in the form of "<OTP><Static Password>". Predefined composite login modules called "DefaultADVascoCompositeLogin" and "DefaultOpenLDAPVascoCompositeLogin" will be created upon installation. Note: Administrator may make use of the predefined entries with necessary changes or create new entries that cater to the user requirements.
Device-based Password Login Module	This login module allows you to set the maximum number of registered devices per user for seamless authentication using device Ids. It also allows you to specify a login module to handle the authentication request. The login module supports multiple passwords based on a device Id. Note: You should use different device Ids if you use various applications on a physical device. For example, a mobile phone with multiple device-based password login modules.
E2EE via PKCS#11 Login Module	This login module provides Safenet's PSG-based E2EE login method for the user using PKCS#11 API who may reside in either an internal or external user store. Note: This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution.
E2EEA Migrator Login Module	This login module is used to facilitate the migration of an existing user's password from one E2EEA Login Module to another E2EEA Login Module. It can be used to easily migrate user passwords when switching from one HSM to another. Note: This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution.
FIDO UAF Login (Deprecated)	This login module is used to trigger FIDO authentication (Deprecated) requests and parse FIDO authentication responses. Note: This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution.
FIDO2 Token Login	This login module is used for the registration of FIDO2 tokens and authentication using registered FIDO2 tokens. Note: This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution.

Login Module	Description
Generic Biometric Login	This login module is used for any biometric vendor as long as it's integrated into AccessMatrix through plugin, e.g. SenseTime. Note: This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution.
i-Sprint SMS OTP Login	This login module is the i-Sprint™ SMS OTP-specific login module that provides an i-Sprint™ SMS OTP login method for the user who may reside in either an internal or external user store. Predefined i-Sprint™ SMS OTP Login modules called "SMSOTPLLogin" and "SSFOTPLLogin" will be created upon installation. "SSFOTPLLogin" will be used as a default reauthentication login module for Self Service configuration. Note: The administrator may make use of the predefined entry with necessary changes or create new entries that cater to the user requirements. This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution.
Ldap Login	This login module provides LDAP-based password basic login methods for the user in an external user store, e.g. Active Directory registry. Predefined LDAP Login modules called "DefaultActiveDirectoryLogin" and "DefaultOpenLDAPLogin" will be created upon installation. Notes: • The administrator may make use of the predefined entries with necessary changes or create new entries that cater to the user requirements. • To support change/reset the password, the secure SSL/TLS connection in AD/LDAP user store should be enabled. The AD/LDAP user with administrator rights should be configured in the user store.
Lookup Authenticated Login	This login module is used with lookup modules that do authentication, e.g. FIDO UAF Lookup (<i>Deprecated</i>). Note: This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution.
Luna SA E2EE Login	This login module provides SafeNet's Luna SA-based E2EE login method for the user who may reside in either an internal or external user store. Note: This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution.
Luna SA E2EEA Login Module	This login module provides SafeNet's Luna SA-based E2EEA login method for the user who may reside in either an internal or external user store. Note: This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution.
Multiple Choice Login	This login module verifies user credentials using multiple login modules. Verification is considered successful if the user can be verified by one of the configured login modules. This login module can be useful for a transitional period or in supporting two types of tokens where the user may be assigned with one of the tokens or even both tokens.

Login Module	Description
OATH Token Login	This login module is OATH token-specific login module, which provides an OATH token-based login method for the user who may reside in either an internal or external user store. A predefined OATH Token Login module called "DefaultOATHTokenLogin" will be created upon installation. Note: Administrator may make use of the predefined entry with necessary changes or create new entries that cater to the user requirements. This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution.
Push to Login Module	This login module is used to provide a push notification-based login method via OneSpan's Mobile Authenticator Studio mobile app. It may be used for users who reside in either an internal or external user store. Note: This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution.
Pseudo E2EEA Login Module	This login module provides a secure end-to-end encrypted authentication method, where the password is encrypted at the client's side using a public key and decrypted at AccessMatrix server with a private key. The password is verified in the AccessMatrix server. The storage key used to encrypt password hash when at rest can optionally be stored within a Hardware Security Module for stronger security using AccessMatrix server key management. Note: This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution.
Q&A KBA Login	This login module provides a Q&A KBA-based login method for the user who may reside in either an internal or external user store. A predefined Q&A KBA Login module called "Q&AKBALogin" will be created upon installation. The administrator shall not require to create another entry of the Q&A KBA Login module unless the predefined entry could not fulfil the user requirements. The administrator may make use of the predefined entry with necessary changes to cater to the user requirements.
Q&A KBA Login (one step)	This login module provides a Q&A KBA-based login method for the user who may reside in either an internal or external user store. It allows the user to configure their own Q&As (up to 10 questions). Users can log in to answer the defined questions in one submission. This module will then verify multiple answers all at once (one step/in one call). The administrator can reset the users' questions and answers to cater to the users' requirements.
Radius Login	This login module provides the RADIUS-based login method for the user who may reside in either an internal or external user store. A predefined RADIUS Login module called "RADIUSLogin" will be created upon installation. Note: Administrator may make use of the predefined entry with necessary changes or create new entries that cater to the user requirements.
Remote UAS Password Login	This login module provides the system's password basic login method for user authentication. It resets/changes the password of credentials residing in an external user store, i.e. UIM user store.

Login Module	Description
	Note: This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution.
Response Only Token Agnostic Login	This login module is token-specific login module, which responds only token agnostic login method for the user who may reside in either an internal or external user store. Note: Administrator may create new entries that cater to the user requirements.
SafeNet PHEFT Login	This login module provides SafeNet ProtectHost EFT (PHEFT) login method for the user who may reside in either an internal or external user store. A predefined SafeNet PHEFT – E2EE login module called "SafeNetPHEFTE2EE" will be created upon installation. Note: This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution. The administrator is advised not to delete the existing entry of SafeNet PHEFT login module unless necessary to do so.
SafeNet ProtectServer E2EEA Login Module	This login module provides SafeNet ProtectServer-based E2EEA login method for the user who may reside in either an internal or external user store. Note: This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution.
SAML Artifact Resolution Lookup and Login	This login module provides SAML's login method where it validates a user's artifact response to obtain user identity and other user-related assertion attributes if any. Note: This login module is specific to AccessMatrix Universal Access Management (UAM) solution.
SAML Web Browser SSO Lookup and Login	This login module provides SAML's login method where it obtains the user's identity and other users- related assertion attributes if any. Note: This login module is specific to AccessMatrix Universal Access Management (UAM) solution.
Server User Impersonation Login (For server process/tasks only)	This login module is provided for the server task to log in as another interactive user. Note: This login module is specific to the AccessMatrix Universal Identity Manager (UIM) solution.
SGFindex Identity Login	This login module is an SGFinDex-specific login module that will call the Govtech endpoint to obtain end-user identity in SingPass. The obtained user identity is used to cross-check and validate users who may reside in either an internal or external user store of Financial Information Service Provider (FISP) IDP. Note: This login module is specific to AccessMatrix Universal Access Management (UAM) solution. This login module will not be displayed from the User Login Module list box if UAM feature is disabled.

Login Module	Description
SHDLDC (Shanghai Donglian Dynamic Code) Token Login	This login module is used to provide OTP verification for SHDL K8 token. Note: This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution.
Social Network Login	A special login module for social network login usage. It must be used together with social network lookup modules, such as Facebook Login and Google Sign-In. Note: This login module is specific to AccessMatrix Universal Access Management (UAM) solution. This login module will not be displayed from the User Login Module list box if UAM feature is disabled.
SSL Client Certificate Login	This login module relies on the AccessMatrix application server to perform client-to-server certificate authentication for users (who may reside in either an internal or external user store). The Certificate DN Lookup must be used as its lookup module. A predefined SSL Certificate Login module called "ClientCertificateOverSSL" will be created upon installation. The administrator may make use of the predefined entries with necessary changes or create new entries that cater to the user requirements.
System Password Basic	This login module provides the system's password basic login method for the user, particularly in the System user registry (a.k.a default store). A predefined system login module called "SystemPasswordBasic" will be created upon installation. Note: Generally, the administrator should not create another entry of the System Password Basic login module as there should be only one entry of the System Password Basic login module throughout the whole AccessMatrix System. The administrator is also strongly advised not to delete the existing entry of the System Password Basic login module unless necessary to do so.
Thales payShield Login Module	This login module is used to generate public\private key pairs as part of the pin translation process with Thales payShield 9000 or 10000 HSM. It requires the Thales HSM host and port, the number of connections and the message header information. Thales payShield Login Module supports TLS mutual authentication. You may specify the keystore path and password where the client certificate is stored in the Thales payShield Login Module configuration. Note: This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution.
Trust Associated Login	This login module provides a trusted login method (i.e. USO working with Kerberos) for the user who may reside in an external user store, such as Active Directory registry. A predefined Trust Associated Login module called "DefaultNTLMLogin" will be created upon installation.

Login Module	Description
	Note: Administrator may make use of the predefined entry with necessary changes or create new entries that cater to the user requirements.
Utimaco Crypto Server E2EEA Login Module	This login module provides a secure end-to-end encrypted authentication (E2EEA) login for the user by using Utimaco crypto server HSM. You configure the login policy and HSM information in it. The user's sensitive data like the password is verified in the HSM. Note: This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution.
Vasco Digipass for Web Login	(Unused) This login module is used to support software-based token with OTP generation and Data Signing functionality based upon the knowledge of a password and the possession of a (Digipass) secret stored in the user's PC using Java applet. A predefined Vasco Digipass for Web Login module called "DP4WebLogin" will be created upon installation.
Vasco DP110 Login	(Unused) This login module is used to provide OTP verification for DP110 provided by Vasco. It is a hybrid Digipass with a hardware component, and a software component, which leveraged on a Java applet running on the user's internet browser, and a USB stick plugged into the same machine. These two components together will generate an OTP on demand. A predefined Vasco DP110 Login module called "DP110Login" will be created upon installation.
VASCO Token Login	This login module is VASCO token-specific login module that provides VASCO token-based login method for the user who may reside in either an internal or external user store. A predefined VASCO Token Login module called "DefaultVascoTokenLogin" will be created upon installation. Note: Administrator may make use of the predefined entry with necessary changes or create new entries that cater to the user requirements. This login module is specific to the AccessMatrix Universal Authentication Server (UAS) solution.



Notes:

- Some other third-party tokens are supported by AccessMatrix.
- An administrator will not be able to edit/delete a login module if there is a pending/rejected admin request of that login module entry.
- Having administered the desired user lookup modules and login modules, the administrator may proceed to administer the associated authentication realms to allow the users to log in to the system using the desired login/authentication method supported.

View Login Module Page Attribute Description

Attribute Name	Description
*Login Module Id	Unique Id of login module.
*Login Module Name	Name of login module.
Description	Description of login module.
Version	Login module entry version.
Implementation	Type of login module implementation.
Handler	Handler of the login module.

The attributes of the **Configuration** tab differ depending on the login module. Refer below for some of the login module configurations available.

Microsoft Entra ID Login

Attribute Name	Description
Login Configuration	
Login Policy	The login policy specifies if the bad login count should be incremented and if the login account should be locked upon exceeding the maximum bad login count, and other policy controls. If no login policy is attached, then the corresponding policy controls like the incrementing bad login count will not take effect.
Authentication Level	The strength of authentication ranges from 1 (lowest) to 10 (highest). Default: 1
Scope	A space-separated list of scopes, or permissions, that required by the AccessMatrix server. Default: User. Read Directory.AccessAsUser.All

Default Active Directory Login via LDAP

Attribute Name	Description
Login Configuration	
Login Policy	Refer to the description found in the System Password Basic attribute table.

Attribute Name	Description
Authentication Level	Refer to the description found in the System Password Basic attribute table. Default: 1
Use User's User Store	Set to "true" if LDAP Lookup and NT Domain - User Store mapping is used. It will use the user store of the user instead of the user store configured in this login module. Default: false
Password Expiry Policy	Refers to the AD password expiry policy of this login module. Refer to Configuring Password Expiry Policy section for more details.
Reset Password Options	
Allow Manual Password Reset	Setting this to "true" will enable manual reset of AD user's password. Default: true
Allow Reset By System Assigned	Setting this to "true" will enable AD user's password generation. Default: false
Allow Unlock On Reset Password	Set this to "true" to unlock the user on password reset (it works only if the user store of the current login module supports the "Unlock" feature). Unlocking a user will not affect the password reset. Default: false
Reset Password Specific Configuration	
Event Notification Policy	Refer to the description found in the System Password Basic attribute table.
Password Generation Policy	
Password Length	Length of generated AD user's password. Value should be from 6 to 30. Default: 8
Force Password Change	
Behavior	This attribute determines whether a user will be prompted to do a forced password change if the AD returns a force password change code (i.e. error 1907 - ERROR_PASSWORD_MUST_CHANGE) when the user attempts to log in. The following options are available: "Successful login with forced password change" - The user is successfully logged in and then prompted to change their password. "Successful login without forced password change" - The user is successfully logged in. The force password change flag is ignored and the user is not prompted to change their password.

Attribute Name	Description
	"Unsuccessful login" - The user is unable to login and an error is returned. The event is treated as an unsuccessful login. Default: Successful login with forced password change
i	Note: An additional Scripted Rules tab is hidden by default. It is an advanced feature to customize additional actions during login. To display this tab, add loginModule.scriptedRules = true in wpe_config.properties file. (Location of wpe_config.properties file: <AccessMatrix_Installation_Directory>/tomcat/webapps/am5/WEB-INF/classes/)

Device-based Password Login Module

Attribute Name	Description
Login Configuration	
Login Policy	Refer to the description found in the System Password Basic attribute table.
Authentication Level	Refer to the description found in the System Password Basic attribute table. Default: 1
Login Module	Specifies the login module used to handle the authentication request from the device that stores the password. The login module can be an E2EEA login module or a SystemPasswordBasic login module. Note: For security reasons, it is recommended that you use an E2EEA login module.
Max Registered Device Per User	Maximum number of registered devices per user. The oldest registered devices will automatically be deregistered until the number of registered devices is within the limit. "0" means no limit. Default: 0

E2EEA Migrator Login Module

Attribute Name	Description
Login Configuration	
Login Policy	Refer to the description found in the System Password Basic attribute table.
Authentication Level	Refer to the description found in the System Password Basic attribute table. Default: 1

Attribute Name	Description
Activated	Setting this to "true" will enable the migration of the user password from the source login module to the target login module. Upon the completion of user password migration, set this to "false" to disable further migration. Default: true
Login Module	
*From	The source E2EEA login module that the user password will be migrated from.
*To	The target E2EEA login module that the user password will be migrated to.

Multiple Choice Login

Attribute Name	Description
Login Configuration	
Login Policy	Refer to the description found in the System Password Basic attribute table.
Authentication Level	Refer to the description found in the System Password Basic attribute table. Default: 1
First(1st) Login Module	The first login module to verify credentials. Supports 1-step login (no challenge generation) modules only.
Second(2nd) Login Module	If the first login module cannot verify a credential, verify using this login module. Supports 1-step login (no challenge generation) modules only.
Third(3rd) Login Module	If the second login module cannot verify a credential, verify using this login module. Supports 1-step login (no challenge generation) modules only.

Push to Login Module

Attribute Name	Description
Login Configuration	
Login Policy	Refer to the description found in the System Password Basic attribute table.
Authentication Level	Refer to the description found in the System Password Basic attribute table. Default: 1

Attribute Name	Description
Token Vendor	
Vendor	The token vendor used. Default: VASCO Note: The list of available options depends on the token vendors enabled in your AM license.
Configure Message Template	
Title	The title of the push notification prompting the user to approve/reject the login request in the OneSpan Mobile Authenticator Studio mobile app. Value should not be greater than 80 characters. Default: Push Login
Message	The message of the push notification prompting the user to approve/reject the login request in the OneSpan Mobile Authenticator Studio mobile app. Default: Do you want to accept this login?
Session Validity (in minutes)	The validity period of the push to login session. After the configured time period, the transaction is marked as expired. Default: 5
Push Login Configuration	
Notification Action Identifier	This value needs to match the Action Identifier setting in the DIGIPASS.xml file used to generate the OneSpan Mobile Authenticator Studio mobile app. Default: notificationsTds
Version Tag	The version tag of the push notification. This should be left as the default value. Default: 01

Radius Login

Attribute Name	Description
Login Configuration	
Login Policy	The login policy specifies if bad login count should be incremented and if login account should be locked upon exceeding maximum bad login count, as well as other policy controls. If no login policy is attached, then the corresponding policy controls like incrementing bad login count will not take effect. For more details, refer to the description found in the System Password Basic attribute table.
Authentication Level	Strength of authentication from 1 (lowest) to 10 (highest). Unused. For more details, refer to the description found in the System Password Basic attribute table. Default: 1 Note: This is an attribute of the login module used for UAM product.
IDP Session Authentication Context	If specified, this value is the SAML authentication context that will be set by the SAML IDP into the SAML response.
Primary Radius Configuration	
*Primary Radius shared secret	Specifies the primary Radius shared secret.
Primary Radius server IP	Specifies the primary Radius server IP. Default: 127.0.0.1

Attribute Name	Description
Primary Radius server port	Specifies the primary Radius server port. Default: 1812
Secondary Radius Configuration	
Secondary Radius shared secret	Specifies the secondary Radius shared secret.
Secondary Radius server IP	Specifies the secondary Radius server IP.
Secondary Radius server port	Specifies the secondary Radius server port.
Radius Additional Options	
Enforce Message-Authenticator	If set to "true", AccessMatrix will enforce the sending and receiving of the Message-Authenticator attribute in all RADIUS packets. The Message-Authenticator is a checksum that is used to ensure the integrity of the RADIUS packet. This configuration causes all packets sent by the RADIUS server to include this attribute, and for any incoming packets lacking this attribute to be rejected. This ensures that only packets with the Message-Authenticator attribute are processed, which mitigates the Blast-RADIUS vulnerability. Default: false Note: This attribute is set to "false" by default for backwards compatibility.

SAML Web Browser SSO Lookup and Login

Attribute Name	Description
Login Configuration	
Login Policy	Refer to the description found in the System Password Basic attribute table..
Authentication Level	Refer to the description found in the System Password Basic attribute table. Default: 1
User Lookup Module	Specify the SAML Web SSO lookup module. Note: Referred by the Service Provider when AccessMatrix server functions as the Service Provider. (SP-initiated)

SGIndex Identity Login

Attribute Name	Description
Attributes Tab	
Login Configuration	
Login Policy	The login policy specifies if bad login count should be incremented and if login account should be locked upon exceeding maximum bad login count, as well as other policy controls. If no login policy is attached, then the corresponding policy controls like incrementing bad login count will not take effect. Default: DefaultBasicLoginPolicy
Authentication Level	(Unused) Strength of authentication ranges from 1 (lowest) to 10 (highest). Note: This is an attribute of the login module used for UAM product. Default: 1
Identity Checking	
Environment	This setting determines the default SGIndex user identification URL to use if the SGIndex User Identification URL attribute is not set. Possible values: "Staging" "Production" Default: Staging
SGIndex User Identification URL	This setting would override the default behavior implied by the Environment attribute (by overriding the default SGIndex User Identification URL determined by Environment).
SGIndex APP ID	SGIndex APP ID credential supplied upon SGIndex onboarding.
Request Signing Key Store	

Attribute Name	Description
Keystore	Path to FISP insurer's signing keystore so that the parameters of the SGFindex User Identification request can be digitally signed. The keystore and its corresponding settings (as listed below) should be the same as the one specified under fisp - signingKeys of the sgf.yml file in SGFindex OAuth proxy .war file.
Keystore Passphrase	This refers to the keystore passphrase.
Keystore Alias	Key alias/label as listed in the keystore.
Identity Mapping Tab	
User's Custom Attribute Name	This setting is used to retrieve the AM5 user identity for SGFindex cross identity checking.
HTTP Proxy Tab	
Enabled	Setting to "true" will enable the HTTP Proxy. Default: false
Proxy Host Name	Refers to the URL of the proxy server.
Port	Refers to the port of the proxy server.
User Name	Refers to the username used to authenticate the proxy server.
User Password	Refers to the user's password used to authenticate the proxy server.
NTLM Authentication	Indicates whether NTLM authentication is used. If NTLM Authentication is enabled, Domain Name must be specified. In addition to that, it might be necessary to configure the AM5 server host domain name or IP address in the amsystem.properties file with the property key "am.social.ntlm.host" for each instance. Default: false
Domain Name	Refers to the domain name of AD used in NTLM authentication.

System Password Basic

Attribute Name	Description
Login Configuration	
Login Policy	The login policy specifies if bad login count should be incremented and if login account should be locked upon exceeding maximum bad login count, as well as other policy controls. If no login policy is attached, then the corresponding policy

Attribute Name	Description
	controls like incrementing bad login count will not take effect. Default: DefaultBasicLoginPolicy
Authentication Level	This is an attribute of login module used for UAM product. During authorization, a resource may require permission with a higher authentication level. For example, given a URL which requires authentication level 2; so if a user is logged in with authentication level 1 and tries to access it, WSA will challenge the user to re-authenticate with login module of higher authentication level (at least authentication level 2 and above). Value should be from 1 (lowest) to 10 (highest). Default: 1
Others	
Hash scheme	The hash algorithm for System Password Basic. Possible values: "AM5 compatible" "Salted SHA-256" Default: Salted SHA-256
Password Policies	
Password Quality Policy	The password quality policy of this login module. Refer to Configure Password Quality Policy section for more details. Default: DefaultPasswordQualityPolicy
Password Expiry Policy	The password expiry policy of this login module. Refer to Configure Password Quality Policy section for more details.
Reset Password Options	
Allow Manual Password Reset	Setting this to "true" will enable the manual reset of the user password (i.e. the user creates a new password upon password reset). Default: true
Allow Reset By System Assigned	Setting this to "true" will enable new password generation by the system (i.e. the system generates a new password for the user upon password reset). Default: false
Reset Password Specific Configuration	
Event Notification Policy for System Assigned	This is an event-based trigger for sending an email or SMS. For example, this can be configured to send an OTP for user login when the Generate OTP event is triggered. Note: This is only the policy that will be invoked when resetting the password via the Allow Reset By System Assigned option.

Attribute Name	Description
Event Notification Policy for Manual Reset	This is an event-based trigger for sending an email or SMS. For example, this can be configured to send an OTP for user login when the Generate OTP event is triggered. Note: This is only the policy that will be invoked when resetting the password via the Allow Manual Password Reset option.
Password Generation Policy	
Password Length	Length of the generated password. Value should be from 6 to 30. Default: 8
Password Type	Type of characters in a generated password. Default: Uppercase Alphanumeric

Thales payShield Login Module

Attribute Name	Description
Login Configuration	
Login Policy	The login policy specifies if bad login count should be incremented and if login account should be locked upon exceeding maximum bad login count, as well as other policy controls. If no login policy is attached, then the corresponding policy controls like incrementing bad login count will not take effect.
Authentication Level	Value should be from 1 (lowest) to 10 (highest). Default: 1
Connection	
*Thales payShield HSM Host, Port and Number of Connections	Defines the connection setting to the HSM. Specify the value in "host:port:number of connections" format, e.g. "192.168.1.44:80:2". For a number of connections, consult Thales payShield HSM administrator. Unless set otherwise by the HSM administrator, the default number of connections should be set to "2".
*Message Header	The message header of Thales payShield 9000 or 10000 HSM. For example, "myName". Value can be any length from 1 to 255 characters and it is configured during HSM installation.
Enable TLS	Enable TLS for HSM host connection. Default: false
Keystore	Path to the keystore file containing the client certificate.

Attribute Name	Description
Keystore Passphrase	This refers to the keystore passphrase.
Private Key Passphrase	This refers to the client private key passphrase in the keystore.
Connection Timeout (seconds)	Socket connection timeout value in seconds. "0" means an infinite timeout. Value should not be less than 0. Default: 15
Heartbeat Interval (minutes)	Interval of heartbeat connection in minutes to HSM to keep the connection alive. The value must be lower than the idle timeout value configured in HSM. If the socket connection is unhealthy (temporarily disconnected), this value will also be used as the interval attempt to reconnect the socket. Value should not be less than 1. Default: 5
Maximum Connection Retry	Specify the maximum number of retry attempts for HSM connection. "0" means no retry. Value should not be less than 0. Default: 3
Asymmetric Cipher (affects generated key pairs in future)	
Algorithm	Default: RSA
Key Size	Default: 2048
RPIN OAEP Hash Algorithm	The hash algorithm of RPIN (RSA encrypted PIN) OAEP (Optimal Algorithm Encryption Padding). Default: SHA256
Number of Key Pairs	Value should be from 4 to 16. Default: 4

Trust Associated Login

Attribute Name	Description
Login Configuration	
Login Policy	The login policy specifies if bad login count should be incremented and if login account should be locked upon exceeding maximum bad login count, as well as other policy controls. If no login policy is attached, then the corresponding policy controls like incrementing bad login count will not take effect.

Attribute Name	Description
Authentication Level	Value should be from 1 (lowest) to 10 (highest) Default: 1
IDP Session Authentication Context	Specifies the SAML authentication context that will be configured by the SAML IDP into the SAML Response. Note: To select multiple values, press Ctrl + click on the relevant values.
Others	
Non-Interactive User and Client Type Restrictions	Allows access to the login module only for specified users with corresponding client types. If none are specified, no restrictions will be applied. Client type values: "2" XML RPC "7" SOAP "12" REST Note: The specified users must be available in the Default Store, and not belong to any segment.

Response Only Token Agnostic Login

Attribute Name	Description
Login Configuration	
Login Policy	The login policy specifies if bad login count should be incremented and if login account should be locked upon exceeding maximum bad login count, as well as other policy controls. If no login policy is attached, then the corresponding policy controls like incrementing bad login count will not take effect.
Authentication Level	Value should be from 1 (lowest) to 10 (highest) Default: 1
Retry Configuration	
Auto Retry	Set to "true" to enable Auto Retry (will retry every token(s) assigned to user). Notes: If set to "true": Only 'Response-Only(RO)' mode will be functional for this login module.

Attribute Name	Description
	- Bad login count checking in all corresponding token policies must be disabled to allow Auto Retry. If set to "false": - A user with multiple tokens can log in for each token only once based on its vendor and/or serial number being pass to API - "authn/reauthenticate". AccessMatrix system will select the correct token to use. At any instant when both parameters (vendor and/or serial number) match for more than one token, AccessMatrix system will prompt an error message.
Default Token Vendor	Enter a default token vendor to be used for authentication. Note: This setting is only applicable if Auto Retry is set to "false".

Configure Authentication Realm

The authentication realm associates the lookup module with one or more login modules (where lookup module enables the system to know where to look for the user, and the specified login methods are used to authenticate users). Thus, having administered the lookup modules and login modules as described in the previous sections, the next task will be to administer the necessary authentication realms.

To locate the **Authentication Realm** page, navigate to **Administration Dashboard > Configuration > Authentication > Authentication Realm** on the left pane. The following tasks can be performed on the authentication realms:

- **Create Authentication Realm**

Click **Create** to display the **Create Authentication Realm** page, enter its details and click **Save**.

- **Search for Authentication Realm**

Can be searched by authentication realm Id or authentication realm name. All available authentication realms are displayed by default.

- **View Authentication Realm**

Click an authentication realm Id under the **Search Authentication Realm Result** list to view the details.

- **Edit Authentication Realm**

Click **Edit** on the **View Authentication Realm** page to modify the details and click **Save**.

- **Delete Authentication Realm**

Click **Delete** on the **View Authentication Realm** page and select **OK** in the confirmation pop-up box.

Pre-defined Authentication Realms

Authentication Realm	Description
40151	This authentication realm associates the "SystemLookup" lookup module with the "SystemPasswordBasic" login module. It is the default realm that provides basic password authentication to log in to the AccessMatrix server.
40153	This authentication realm associates the "CertificateDnLookup" lookup module with the "ClientCertificateOverSSL" login module.
40170	This authentication realm associates the "DefaultOpenLDAPLookup" lookup module with the "DefaultOpenLDAPLogin" login module. This realm provides LDAP password authentication against the OpenLDAP server.
40171	This authentication realm associates the "DefaultActiveDirectoryLookup" lookup module with the "DefaultNTLMLogin" login module. This realm provides NTLM password authentication against the AD server.
40172	This authentication realm associates the "DefaultActiveDirectoryLookup" lookup module with the "DefaultActiveDirectoryLogin" login module. This realm provides AD password authentication against the AD server.
40192	This authentication realm associates the "DefaultActiveDirectoryLookup" lookup module with the "DefaultVascoTokenLogin" login module. This realm provides VASCO token-based password authentication.
ADRadius	This authentication realm associates the "DefaultActiveDirectoryLookup" lookup module with the "DefaultActiveDirectoryLogin" as 1st login module and RADIUSLogin as 2nd login module. This realm provides AD password authentication against an AD server, followed by a Radius server.
ADSMS	This authentication realm provides 2-factor authentication that associates the "DefaultActiveDirectoryLookup" lookup module with the "DefaultActiveDirectoryLogin" login module being the 1st factor and "SMSOTPLogin" login module being the 2nd factor in the chained authentication. This realm provides AD password authentication that is chained to i-Sprint SMS OTP authentication.

Authentication Realm	Description
ADVascoComposite	This authentication realm associates the "DefaultActiveDirectoryLookup" lookup module with the "DefaultADVascoCompositeLogin" login module. This realm provides VASCO token-based password concatenated with AD password authentication.
DefaultOATHToken	This authentication realm associates the "SystemLookup" lookup module with the "DefaultOATHTokenLogin" login module. This realm provides OATH token-based password concatenated with system password basic authentication.
DP110AuthRealm	This authentication realm associates the "SystemLookup" lookup module with the "SystemPasswordBasic" as the 1st login module and "DP110Login" as the 2nd login module. This realm provides system password basic authentication, followed by DP110 token authentication.
OpenLdapVascoComposite	This authentication realm associates the "DefaultOpenLDAPLookup" lookup module with the "DefaultOpenLDAPVascoCompositeLogin" login module. This realm provides VASCO token-based password concatenated with OpenLDAP password authentication.
SafeNetE2EE	This authentication realm associates the "SystemLookup" lookup module with the "SafeNetPHEFTE2EE" login module. This realm provides SafeNet PHEFT-based E2EE password concatenated with system password basic authentication.
SystemRealm	This authentication realm associates the "SystemLookup" lookup module with the "SystemPasswordBasic" login module. It is the default realm that provides basic password authentication to log in to the AccessMatrix Administration Console. Note: Generally, Administrator should not create another entry of "SystemRealm" as there should be only one entry of "SystemRealm" throughout the whole AccessMatrix system. The administrator should not delete the existing entry of "SystemRealm".



Notes:

- In AccessMatrix, an authentication realm can be configured to provide a chained authentication known as multiple-factor authentication of up to five factors of authentication.
- The administrator will not be able to edit/delete an authentication realm if there is a pending/rejected admin request of that authentication realm entry.
- Upon completion of AccessMatrix installation, there are several predefined authentication realms created in the system. Under typical user requirements, these entries should be good enough to

use with some or no changes required. Nevertheless, the administrator may create new entries to suit user requirements.

View Authentication Realm Page Attribute Description

Attribute Name	Description
*Authentication Realm Id	Unique Id of authentication realm.
*Authentication Realm Name	Name of authentication realm.
Description	Description of authentication realm.
Version	Authentication realm entry version.
Configuration Tab	
Automatically assigned to users	Allow this authentication realm to be automatically assigned to all users.
User Lookup Module	Lookup module is used to validate user identity (i.e. user search in the user store(s) or other validation methods).
First(1st) Login Module	The first login method to use for user authentication.
Second(2nd) Login Module	Optional subsequent user authentication method.
Third(3rd) Login Module	Optional subsequent user authentication method required.
Fourth(4th) Login Module	Optional subsequent user authentication method required.
Fifth(5th) Login Module	Optional subsequent user authentication method required.
Enable Login/Logout History Saving	Saving of login and logout history.
User Response	Return name-value pair responses used upon successful authentication.
Realm-Based Session Policy	The session policy of this authentication realm. If no session policy is assigned, then the session policy defined in Global Setting will be used.
Contextual Policy	(Optional) It contains the risk score settings depending on the country, geolocation, time-of-day and rooted status.



Note: The administrator can define other custom attributes. This is described in [Adding Custom Attribute](#).

Configure Login Account

The administrator can configure the usable login account statuses (i.e. activated, disabled, suspended, unassigned or others) for all types of login modules. During the administration of the users' login account, the login modules will use the selected login account statuses.

To locate the **Login Account** page, navigate to the **Administration Dashboard > Configuration > Authentication > Login Account** on the left pane.

- **Edit Login Account**

Click **Edit** on the **View Configuration** page to modify the details and click **Save**.

View Configuration Page Attribute Description

Attribute Name	Description
*Configuration Id	Unique login account Id.
*Configuration Name	Configuration name of login account.
Description	Description of login account.
Version	Login account entry version.
Implementation	Type of login account implementation.
Attributes Tab	
Usable Login Account Status	Refers to the login account statuses that are usable for all login modules. Default: Activated, Disabled, Suspended
Skip suspended accounts' login modules in multi-factor authentication (at least one module cannot be skipped)	If enabled, all login modules after the first will be skipped if the user's login account is suspended. Default: false

Usable Login Account Status

Attribute Value	Description
(default)	This is the default status of the login account configuration
Activated	If the login module is activated, the login module is ready to be used.

Attribute Value	Description
Disabled	If the login module is disabled, the login module is not ready to be used. The login module will be disabled under 2 conditions, i.e. the administrator manually disables the login module, or the system automatically disables the login module after x times of unsuccessful login trial.
Suspended	If the login module is suspended, the login module is not ready to be used. An administrator can suspend the login module if it is not used for a certain period.
Attribute value for Token	Description
Unassigned	This status indicates that a token has been assigned to a user before, but the user no longer requires the token. This means that the token is currently unassigned from the user, but the user has not returned the token. In this case, the token status should stay unassigned until the token is returned. A returned token can then be marked as active and can be reassigned to another user.
Locked	The status when the token is locked and cannot be used for authentication. Note: Token status will be set to "Locked" if token reaches the bad login count set in the token policy.
Frozen	This temporarily disables the token authentication. Note: Token is frozen when it is temporarily suspended. The user will not be able to use it until it goes back to the assigned state.
Lost	The status when the token has been reported as lost.
Damaged	The status when the token has been reported as damaged.
In_Transit	This is an intermediate state where the token is currently being transited.
Expired	The status when the token has expired and cannot be used for authentication. It is used for SMS OTP where OTP is expired when used after the defined expiry time in token policy.
Invalidated	The status when the token has been invalidated and cannot be used anymore.
PreActivation	The status when the token needs to be activated for assignment to a user.
Returned	The status when the token has been returned to the token issuer.
Faulty	The status when the token is faulty and cannot be used anymore.
Deleted	The status when the token has been deleted.
Unsuspending	The status when the token has been changed from the "Suspended" state to the "Unsuspending" state.

Attribute Value	Description
BatteryExpired	The status when the token's battery is already expired. This is for hardware token where the battery life is defined in the token batch.
Elevated	Currently, it's unused unless defined by the client application.
Revoked	The status shows the token is officially cancelled.
Terminated	The status shows the token is being terminated.
Converted	The status when the token has been converted.
New	The status shows the token is new.
Pending Print	The status when the token is pending for print.
Running	The status when the token is running.
Completed	The status when the token has been completed processing.
Deactivated	The status when the token instance is lost. Used for VASCO Multi-Device tokens. Note: Any token instance that has been flagged as deactivated will not be used for any token related operations.

Configure Login Policy

This policy defines behaviour settings, precondition checks, post-event actions of login, reset password and change password operations. This policy is optional during login module administration. By default, a predefined login policy called "DefaultBasicLoginPolicy" will be created upon successful installation of AccessMatrix. This predefined policy is applied on the SystemPasswordBasic login module used by the system users for logging in to AccessMatrix.

Types of login policy

Login Policy	Description
Basic Login Policy	This login policy can be used for all login modules.
Biometric Login Policy	This login policy is similar to Basic Login Policy with the addition of biometric configurations for Biometric Login Module use.
 Note: Administrator will not be able to edit/delete a policy if there is a pending/rejected admin request of that policy entry.	

To locate the **Login Policy** page, navigate to **Administration Dashboard > Security Policy > Authentication > Login Policy** on the left pane. The following tasks can be performed on the login policies:

- **Create Login Policy**

Click **Create** to display the **Choose Login Policy Implementation** page. Select an implementation and click **Next** to open the **Create Login Policy** page. Enter the details and click **Save**.

- **Search for Login Policy**

Can be searched by login policy Id or login policy name. The default basic login policy is displayed by default.

- **View Login Policy**

Click a login policy Id under the **Search Login Policy Result** list to view the details.

- **Edit Login Policy**

Click **Edit** on the **View Login Policy** page to modify the details and click **Save**.

- **Delete Login Policy**

Click **Delete** on the **View Login Policy** page and select **OK** in the confirmation pop-up box.

View Login Policy Page Attribute Description

Attribute Name	Description
*Login Policy Id	Unique login policy Id.
*Login Policy Name	Name of login policy.
Description	Description of login policy.
Version	Login policy entry version.
Implementation	Type of login policy implementation.
Attributes Tab	
Login Account	
Check login account status	If set to "true", then the login account status will be checked and will display corresponding error message if counter already reached the set bad login count. Otherwise, user will be allowed to login if correct password is entered regardless if the login account status was already

Attribute Name	Description
	disabled. Default: true
Disable login account if bad login count reaches	Sets the maximum bad login count. If bad login count reaches the specified number, then login account will be disabled. Default: 3
Allow resetting password without force password change	Allows password reset without forcing a password change. The behaviour will depend on the selected option: "Disabled" - Disallow password reset. "Enabled in API only" - Allows password reset in API only. "Enabled in API and Admin Console" - Allows password reset in both API and Admin Console. Default: Disabled
Allow enabling of login account without resetting password	Allows login account activation without resetting the password. Default: false Note: This attribute is applicable only for the Modify Account button, which is hidden by default. This button can change the login account's status. Uncomment the Modify Account button tag in the field-definitions.xml file to display the button.
Increment bad login count for the wrong password used during change password	Specifies the action to be done if change password failed due to an incorrect password. "Do not increment" - No action to be taken. "Increment same bad login counter as login" - Increment the bad login count if change password failed due to an incorrect password. "Increment separate bad login counter" - This will create or increment a separate counter if a change password failed due to an incorrect password. Default: Do not increment
Auto Unlock	
Enable Auto Unlock By Time	If enabled, then locked accounts will be unlocked after the specified time in the Auto Unlock Time Duration attribute.
Auto Unlock Time Duration (in minutes)	Sets the time duration when auto-unlock will be executed.
Others	
Include new password in change and reset password event	Setting this to "true" will include the new password in event object for successful change and reset password events.

Configure Password Policies

Password Policies can be divided into two groups, i.e. Password Expiry Policy and Password Quality Policy.

Configure Password Expiry Policy

This policy governs the user's login password expiration. This policy is optional during login module administration. By default, predefined password expiry policies called "DefaultADPasswordExpiryPolicy" and "DefaultPasswordExpiryPolicy" will be created upon successful installation of AccessMatrix.



Note: Administrator will not be able to edit/delete a policy if there is a pending/rejected admin request of that policy entry.

To locate the **Password Expiry Policy** page, navigate to **Administration Dashboard > Security Policy > Authentication > Password Expiry Policy** on the left pane. The following tasks can be performed on the password expiry policies.

- **Create Password Expiry Policy**

Click **Create** to display the **Choose Password Expiry Policy Implementation** page. Select an implementation and click **Next** to open the **Create Password Expiry Policy** page. Enter the details and click **Save**.

- **Search for Password Expiry Policy**

Can be searched by password expiry policy Id or password expiry policy name. All available password expiry policies are displayed by default.

- **View Password Expiry Policy**

Click a password expiry policy Id under the **Search Password Expiry Policy Result** list to view the details.

- **Edit Password Expiry Policy**

Click **Edit** on the **View Password Expiry Policy** page to modify the details and click **Save**.

- **Delete Password Expiry Policy**

Click **Delete** on the **View Password Expiry Policy** page and select **OK** in the confirmation pop-up box.

View Password Expiry Policy Page Attribute Description

Attribute Name	Description
*Password Expiry Policy Id	Unique Id of password expiry policy.
*Password Expiry Policy Name	Name of password expiry policy.
Description	Description of password expiry policy.
Version	Password expiry policy entry version.
Implementation	Type of password expiry policy implementation.

The lower section of the page contains the **Attributes** and **Segment-Specific Rules** tabs.

 **Note:** The details of the **Attributes** tab, and **Segment-Specific Rules** tab differ depending on the policy. This **Segment-Specific Rules** tab will only be displayed if the **Segment-Specific Rules** checkbox is ticked and the rules are set up for a segment during the password expiry policy creation.

For AD Password Expiry Policy Handler Implementation

Attribute Name	Description
Attributes Tab	
This tab contains system predefined as well as custom (user-defined) attributes.	
Active Directory Type	Specifies the Active Directory type to use - either the standard Active Directory (AD) or Active Directory Lightweight Directory Services (AD LDS). Default: AD
Use AD Expiry Policy	Specifies whether to enforce the AD password expiry policy to perform password expiry checking. If set to "true", the Number of days before the password expires attribute value will be ignored. Default: true
Number of days before the password expires (0 = no expiry)	Specifies the number of days before the password expires. Default: 30 Note: This attribute will be ignored if Use AD Expiry Policy attribute is set to "true".
Number of days before password expiry warning	Specifies the number of days before password expiry warning. Default: 5

Attribute Name	Description
Segment-Specific Rules Tab	
This tab allows setting password expiry rules to specific segment(s). Number of days before the password expires (0 = no expiry) and Number of days before password expiry warning values can be specified for each segment. Note: During the creation of the policy, click the checkbox for Segment-Specific Rules to display the segments hierarchy at the left pane. Select a segment on the left pane and then click Add to set the password expiry rules for this segment.	

For Password Expiry Policy Handler Implementation

Attribute Name	Description
Attributes Tab	
Number of days before password expires (0 = no expiry)	Specifies the number of days before password expires. Value should be from 0 to 999 Default: 30
Number of days before password expiry warning	Specifies the number of days before password expiry warning. Value should be from 0 to 999. Default: 5
Segment-Specific Rules Tab	
Same as the Segment-Specific Rules tab of the AD Password Expiry Policy Handler .	

Configure Password Quality Policy

This policy defines the user's login password quality based on its password length, password characters, dictionary checking, password history, password age, policy enforcement options, and more. This policy is optional during login module administration. By default, predefined password quality policies called "DefaultPasswordQualityPolicy" and "E2EEPHEFTPasswordQualityPolicy" will be created upon successful installation of AccessMatrix.

"DefaultPasswordQualityPolicy" is applied on the "SystemPasswordBasic" login module while "E2EEPHEFTPasswordQualityPolicy" is applied on the "SafeNetPHEFTE2EE" login module, where both are used by the system users for logging in to AccessMatrix.

**Notes:**

- If you do not see the password quality policy "E2EEPHEFTPasswordQualityPolicy" after installing AccessMatrix, you need to restart the server and then log in to Admin Console.
- Administrator will not be able to edit/delete a policy if there is a pending/rejected admin request of that policy entry.

To locate the **Password Quality Policy** page, select **Administration Dashboard > Security Policy > Authentication tab > Password Quality Policy** on the left pane. The following tasks can be performed on password quality policies:

- **Create Password Quality Policy**

Click **Create** to display the **Create Password Quality Policy** page, enter the details and click **Save**.

- **Search for Password Quality Policy**

Can be searched by password quality policy Id or password quality policy Name. All available password quality policies are displayed by default.

- **View Password Quality Policy**

Click a password quality policy Id under the **Search Password Quality Policy Result** list to view the details.

- **Edit Password Quality Policy**

Click **Edit** on the **View Password Quality Policy** page to modify the details and click **Save**.

- **Delete Password Quality Policy**

Click **Delete** on the **View Password Quality Policy** page and click **OK** in the confirmation pop-up box.

View Password Quality Policy Page Attribute Description

Attribute Name	Description
*Password Quality Policy Id	Unique Id of password quality policy.
*Password Quality Policy Name	Name of password quality policy.
Description	Description of password quality policy.
Version	Password quality policy entry version.

Attribute Name	Description
Attributes Tab	
Password Length	
Minimum length	Specifies the minimum password length. Value should be from 1 to 16. Default: 4
Maximum length	Specifies the maximum password length. Value should be from 2 to 64. Default: 20
Characters In Password	
Password should not contain characters within this Unicode range	Specifies the hexadecimal range of Unicode characters not allowed in the password. Unicode characters are supported and can be used in passwords. Use this setting to restrict specific Unicode characters from being used. For example, if the value specified is "4E00-9FFF", the Chinese, Japanese, Korean (CJK) Unified Ideographs are not allowed. Note: You can refer to https://www.unicode.org/charts/ for the full list of Unicode character hexadecimal ranges.
Minimum numeric characters	Specifies the minimum number of numeric characters that the password must have. Value should be from 0 to 8. Default: 1
Minimum alphabetic characters	Specifies the minimum number of alphabetic characters that the password must-have. Value should be from 0 to 8. Default: 0
Minimum uppercase characters	Specifies the minimum number of uppercase characters that the password must-have. Value should be from 0 to 8. Default: 0
Minimum lowercase characters	Specifies the minimum number of lowercase characters that the password must-have. Value should be from 0 to 8. Default: 0
Minimum non-alphanumeric characters	Specifies the minimum number of non-alphanumeric characters that the password must-have. Value should be from 0 to 8. Default: 0

Attribute Name	Description
Allow repeat of characters	Setting this to "true" will allow repeating of characters in a password. Default: true
Number of consecutive repeating characters allowed	When Allow repeat of characters is "true", this is not applicable. When it is "false", then repeat of characters is not allowed when the number of repeat characters exceeds this value. Specifies the number of consecutive repeating characters that the password can have. Value should be from 0 to 3. Default: 0
Allow sequence of alphabetic characters	Setting this to "true" will allow a sequence of alphabetic characters in a password. From version 5.6.2.6214-GA-E03 onwards, a backward sequence like "bca" is checked in addition to forwarding sequence like "abc". Default: true
Allow sequence of numeric characters	Setting this to "true" will allow a sequence of numeric characters in a password. From version 5.6.2.6214-GA-E03 onwards, a backward sequence like "321" is checked in addition to forwarding sequence like "123". Default: true
Maximum number of sequence of character	- When Allow sequence of alphabetic characters and/or Allow sequence of numeric characters is/are set to "true", then this setting is not applicable. - When Allow sequence of alphabetic characters and/or Allow sequence of numeric characters is/are set to "false", then this setting can be used to prohibit the number of a sequence of characters exceeding the specified value. For example, if this attribute is set to "2", then "ab" and "12" are allowed but not "abc" and "123" are not allowed.
Allow password same as User Id	A condition whether to allow the password to be the same as the user Id or contain user Id. Possible choices: "Exact Match" - does not allow the password to be the same as the user Id. Matching is case-insensitive. "Contain" - does not allow the password to contain user Id. For example, if the user id is "user1", then "user1pwd" will not be allowed. "Allow" - allow the password to be any variation of user Id. Default: Contain
Password initial character	Specifies the initial character of a password. Options: "0" - alphabetic character, e.g. password1

Attribute Name	Description
	"1" - numeric character, e.g. 1password "2" - non-alphanumeric character, e.g. @password "3" - any character Value should be from 0 to 3. Default: 3
Password should not contain these characters	Specifies character(s) that should not be in the password. To specify multiple characters that should not be in the password, list them in a string without spaces. For example, to prohibit the characters "a", "b", and "c", enter the value "abc".
Password should not begin with any of these characters	Specifies character(s) that should not be used as the first character in the password. To specify multiple characters that should not be the first character, list them in a string without spaces. For example, to prohibit the characters "a", "b", and "c", enter the value "abc".
Password should not end with any of these characters	Specifies character(s) that should not be used as the last character in the password. To specify multiple characters that should not be the last character, list them in a string without spaces. For example, to prohibit the characters "a", "b", and "c", enter the value "abc".
Password must contain at least one of these characters	Specifies character(s) that must be in the password. To specify multiple characters that should be used in the password, list them in a string without spaces. For example, to require the characters "a", "b", and "c", enter the value "abc".
Dictionary Checking	
Enabled	A flag of whether to enable or disable dictionary checking. Default: false
Case sensitive	Setting this to "true" will check case sensitivity between the password and dictionary words. Default: true
Combination checking	Setting this to "true" will check if there are more than one dictionary words in the password. For example, "new123" is allowed while "newpassword" or "dictionaryword" is not allowed. Default: true
File used as password dictionary	Specifies the password dictionary file. By default, this file is included in am-res.jar with dictionary.txt as the filename. To override or use user's dictionary, place the file in the <i>WEB-INF/classes/</i> directory. Default: dictionary.txt

Attribute Name	Description
Password History	
Number of previous passwords to check for reuse	Refers to the number of previous passwords to check if it can be reused or not during password change operation. Value should be from 2 to 10. Default: 3
Number of previous passwords to store in the database	The number of previous passwords that will be stored in the database. The value must be equal or greater than the number of previous passwords to check for reuse. "0" means the same as the number of previous passwords to check for reuse. Value should not be greater than 10.
Password Age	
Password age days	Number of days (after password last changed date) before the password can be changed again. Default: 0
Policy Enforcement Options	
On Reset Password Functionalities	Setting this attribute to "true" will enforce password checking against the password quality policy during a password reset. Default: false When enabled, the extent of checking will depend on the capabilities and features of each login module. Password age In general, password age will not be checked during a password reset to allow users to reset the password at all times. Safenet PHEFT E2EE Login Module* - Supported from versions 5.6.0.6013-E08, 5.6.2.6214-E02, 5.6.3.6300 - Password quality is configured on the HSM itself and not checkable by AccessMatrix server - Password history will be checked
	Note: The administrator can add custom (user-defined) attributes where this window is described in Adding Custom Attribute .

Applicability of password policy to various login module passwords

Not all attributes apply equally to the various login module passwords.

	Notes:
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- For Safenet PHEFT E2EE Login Module, the password quality policy is configured on the HSM itself.
- For SafeNet Protect Server Login Module, the attribute **Check password quality in HSM (if supported by HSM)** must be set to "true". For more information, refer to [Configure SafeNet ProtectServer E2EEA Login Module](#).
- "Y" denotes supported and the empty cells denote not supported.

Login Module Attribute	System Password Basic	SafeNet PHEFT E2EE Login Module	SafeNet Protect Server Login Module	Safenet Pseudo E2EE Login Module
Password Length				
Minimum length	Y		Y	Y
Maximum length	Y		Y	Y
Characters In Password				
Minimum numeric characters	Y		Y	Y
Minimum alphabetic characters	Y		Y	Y
Minimum uppercase characters	Y		Y	Y
Minimum lowercase characters	Y		Y	Y
Minimum non-alphanumeric characters	Y		Y	Y
Allow repeat of characters	Y		Y	Y
Number of consecutive repeating characters allowed	Y		Y	Y
Allow sequence of alphabetic characters	Y		Y	Y
Maximum number of sequence of characters	Y		Y	Y
Allow password same as User Id	Y		Y	Y

Login Module Attribute	System Password Basic	SafeNet PHEFT E2EE Login Module	SafeNet Protect Server Login Module	Safenet Pseudo E2EE Login Module
Password initial character	Y		Y	Y
Dictionary Checking				
Enabled	Y			Y
Case sensitive	Y			Y
Combination checking	Y			Y
File used as password dictionary	Y			Y
Password History				
Number of previous passwords to check for reuse	Y	Y	Y	Y
Number of previous passwords to store in the database	Y	Y	Y	Y
Password Age				
Password age days	Y	Y	Y	Y
Policy Enforcement Options				
On Reset Password Functionalities	Y	Y	Y	Y

Configure Realm-Based Session Policy

This policy defines the maximum session idle time and maximum session lifetime per authentication realm. This policy can be set during authentication realm administration, but setting it is optional. If this policy is not assigned to the authentication realm, then the **Session Idle Time** and **Session Lifetime** values in **Global Setting** will be used.



Note: Administrator will not be able to edit/delete a policy if there is a pending/rejected admin request of that policy entry.

To locate the **Realm-Based Session Policy** page, navigate to **Administration Dashboard > Security Policy > Authentication > Realm-Based Session Policy** on the left pane. The following tasks can be performed on realm-based session policies.

- **Create Realm-Based Session Policy**

Click **Create** to display the **Create Realm-Based Session Policy** page, enter the details and click **Save**.

- **Search for Realm-Based Session Policy**

It can be searched by realm-based session policy Id or realm-based session policy name. All available realm-based session policies are displayed by default.

- **View Realm-Based Session Policy**

Click a realm-based session policy Id under the **Search Realm-Based Session Policy Result** list to view the details.

- **Edit Realm-Based Session Policy**

Click **Edit** on the **View Realm-Based Session Policy** page to modify the details and click **Save**.

- **Delete Realm-Based Session Policy**

Click **Delete** on the **View Realm-Based Session Policy** page and select **OK** in the confirmation pop-up box.

View Realm-Based Session Policy Page Attribute Description

Attribute Name	Description
*Realm-Based Session Policy Id	Unique Id of realm-based session policy.
*Realm-Based Session Policy Name	Name of realm-based session policy.
Description	Description of realm-based session policy.
Version	Realm-based session policy entry version.
Attributes Tab	
Session Idle Time (in seconds)	Specifies the maximum session idle time per authentication realm. It is the validity of the user logged-in session when the user is idle or has no activity. In terms of API, this is the validity of the sessionToken. Session token has the last access time, and idle timeout is calculated based on this last access time. A

Attribute Name	Description
	refresh session API can be called to generate a new session token to extend the session (It has an updated last access time). If no value is specified, then the Session Idle Time value defined in Global Setting will be used. Value should be from 120 to 864000.
Session Lifetime (in seconds)	Specifies the maximum session lifetime per authentication realm. It is the maximum validity of the user logged-in session regardless of idleness/activity. In terms of API, this is the maximum validity of the session token regardless of the last access time. The validity checking is calculated based on the logged-in time, instead of the last access time. If no value is specified, then the Session Lifetime value defined in Global Setting will be used. Value should be from 120 to 864000.
Realm-Based Concurrent Session Policy	
Max. number of concurrent logins	Default: 1
Session Concurrency	Set different concurrent session per authentication realm. It contains different options of the concurrent session for a realm-based session policy. To allow more than one session of concurrent logins, and restrict the maximum number of concurrent logins per user, select a required option of the concurrent sessions, and then select the maximum number of concurrent logins in this list box. Default: "Allow concurrent logins" Refer to Configure Global Setting for more information.

4.2.3. Manage User Stores

Other than the built-in Default User Store, AccessMatrix Server supports external User Stores where the user is stored in a separate JDBC database table or in LDAP-based user registry. This feature simplifies integration effort of organizations and allows AccessMatrix Server to have read access to their external user store(s) directory without changing or writing to the LDAP(s) schema.

The LDAP-based user registries supported are:

- **Active Directory**
- **JDBC**
- **LDAP**
- **Novell eDirectory**
- **OpenLDAP**
- **Sun Directory Server**
- **Tivoli Access Manager (TAM)**

To integrate the LDAP-based user registry, a new user store must be created to configure:

- The LDAP connection to the required user registry.
- Attributes mapping of users, groups, and segment for the required user registry.
- Query cache of the required user registry.

For Microsoft Active Directory, there are three options on Microsoft Azure Cloud:

- **Microsoft Entra ID**
Communicates by using [Microsoft Graph API](#). See below for more information about [Microsoft Entra ID](#).
- **Microsoft Entra ID Domain Service (Microsoft Entra ID DS)**
Communicates by using LDAP API. From external applications like AccessMatrix, Microsoft Entra ID DS is almost as good as connecting to a local Active Directory server. See below for more information about the [Microsoft Entra ID Domain Service \(Microsoft Entra ID DS\)](#) and [Microsoft Entra ID \(Microsoft Entra ID\) User Store](#) implementations.
- **Active Directory on a Windows VM on Azure.**
This is as good as an Active Directory on a local Windows server.

To locate the **User Store** page, navigate to **Administration Dashboard > Configuration > Authentication > User Store** on the left pane. The following tasks can be performed on user stores:

- **Create User Store**

Click **Create** and select an implementation from the drop-down list that matches the required LDAP-based user registry.



Notes:

- If the **Implementation** drop-down list does not contain the desired LDAP server model, choose "LDAP User Store", which can be used for any model of LDAP-based server software. The administrator can test the configuration by clicking the **Test Connection** button on the **Create User Store** page.
- For Microsoft Entra ID Domain Service, choose "Active Directory User Store".

- **Search for User Store**

Can be searched by user store Id or user store name. All available user stores are displayed by default.

- **View User Store**

Click a user store Id under the **Search User Store Result** list to view the details.

- **Edit User Store**

Click **Edit** on the **View User Store** page to modify the details and click **Save**.

- **Delete User Store**

Click **Delete** on the **View User Store** page and click **OK** in the confirmation pop-up box.



Important Notes:

- The administrator will not be able to edit/delete the user store if there is a pending/rejected admin request pertaining to that user store entry.
- Once user store administration is completed, the administrator may proceed to associate user lookup modules and login modules to enable user login to the system using the supported authentication methods. This will be discussed in details in [Manage User Authentication Service](#).

Create Active Directory User Store

Active Directory (AD) User Store is a directory service that Microsoft developed for Windows domain networks.



Note: Refer below for configuration references of applicable attributes for the following user stores:

- LDAP User Store
 - Novell eDirectory Server User Store
-

- OpenLDAP User Store
- Sun Directory Server User Store
- TAM User Store

Create User Store Page Attribute Description

Attribute Name	Description
*User Store Id	Unique Id of AD user store.
*User Store Name	Name of AD user store.
Description	Description of AD user store.
Version	AD user store entry version.
Implementation	Type of AD user store implementation.



Note: For Active Directory user store, an NT Domain field to support Windows/AD domain user lookup (which is used for LDAP user authentication) will be displayed. Click [Manage User Authentication Service](#) to understand more about user authentication and user lookup.

Attribute Name	Description
Connection Tab	
Connection	
*Urls	Refers to the Active Directory server name/IP Address and port number to connect to, e.g. "LDAP://192.168.1.1:389". Note: Failover connections of LDAP servers can be set up by adding multiple URLs. For Java 11 and above, FQDN(Full Qualified Domain Name) must be used when SSL is enabled.
Use SSL	A flag to enable/disable Secure Sockets Layer (SSL) connection. Default: false
Authentication mechanism	Refers to the authentication mechanism/level used for authentication to the AD server. Default: Simple (Login Id + Password)
Login Id	Refers to the user Id of the AD credential with the host name, e.g. "WIN2003\demouser".
Password	Refers to the AD user's password.
Connection (Advanced)	

Attribute Name	Description
Use connection pool	Setting this to "true" will use the values configured in Global Setting > LDAP Connection Pool . Default: true
Connection timeout (seconds)	Specifies the maximum time (in seconds) to wait for connection results. "0" means there is no timeout. Value should be from 0 to 3600.
Read Timeout	Specifies the maximum time (in seconds) to wait for results on a read operation. "0" means there is no timeout. Value should be from 0 to 3600.
Use LDAP paging feature	Setting this to "true" will enable paging when searching (RFC 2696). LDAP paging allows retrieval of search result more than the query limit. The report may not work properly when this feature is disabled. Default: true Note: For older version of LDAP server, set this attribute to false (turn off) as this feature is not available.
LDAP Referral	Option specifying whether referrals to other LDAP servers are followed automatically. Available choices: ignore, follow, and throw.
Enable Fast Binding	Set this to "true" to enable fast binding if AD is used for authentication under high load. Default: true
Ignore workstation logon restriction error	Setting this to "true" will allow AD authentication to succeed despite receiving LDAP error code 531 for certain user having their logon restricted to certain workstations (only when user Id/password provided are correct). Default: false
User Attributes Mapping Tab	
This tab contains certain attributes that have been filled with default values for attributes mapping which is good enough to use, as well as other advanced attributes configuration.	
Search Filter	
*Search Context	Specifies the Base DN to use for user search. e.g. "CN=Users,DC=app-sg-isi,DC=i-sprint,DC=com". Click the Fetch Base DNs button to help fill the Search Context attribute value.
Basic Attributes	
*Attribute for UUID	User's UUID attribute, e.g. "objectGUID".

Attribute Name	Description
*Attribute for DN	User's Distinguished Name (DN) attribute, e.g. "distinguishedName".
*Attribute for id	User's Id attribute, e.g. "sAMAccountName". Note: Set value to "distinguishedName" if user Id is to be searched as user DN in the desired LDAP server. Otherwise, set to "email" if user Id is to be searched as email address in the desired LDAP server.
*Attribute for name	User's name attribute, e.g. "name".
Other attributes mapping	Refers to other user attribute's mapping. Format: <AM's attribute name>=<LDAP's attribute name>.
User	
*Search filter for user	Specifies the search filter for user, e.g. "(&(objectClass=user)!(objectClass=Computer)))".
Object class for user	Refers to the object class for user. Default: User
Attribute for email	User's email attribute, e.g. "mail".
Attribute for mobile	User's mobile attribute, e.g. "mobile".
Attribute for account status	User's account status attribute. Default: userAccountControl
Attribute for created time	User's created time attribute, e.g. "whenCreated".
Attribute for modified time	User's modified time attribute, e.g. "whenChanged".
Group	
*Search filter for group	Specifies the filter employed when doing a search for groups, e.g. "(objectClass=Group)".
Object class for group	Refers to the object class for a group. Default: Group
*Attribute for member	Attribute name of a unique member, e.g. "member".
Attribute for member of	Attribute name for group membership, e.g. memberOf".

Attribute Name	Description
Maximum level of nested group to resolve	Refers to the maximum level of nested groups to resolve. "0" means the nested group will not be resolved. Value should be from 0 to 10. Default: 0
Advanced Attributes Configuration	
Returning attributes	(Unused)
Binary attributes	e.g. "objectGUID objectSid". Note: Specify the binary attributes if UUID attributes are in binary form.
Date time attribute format	Refers to LDAP's data time attribute format.
Use shadow UUID	Use shadow UUID instead of AM5 UUID to lookup the LDAP entry. This is used when LDAP data has been migrated without retaining its UUID. The shadow UUID, which is the new LDAP UUID, has to be set as the user's AccessMatrix attribute. This can be done via a database script or via a one-time LDAP migration plugin. Default: false
Others	
Use relative DN to search member	Setting this to "true" will use relative DN to search member. Default: false
Segment/OU Tab	
This tab contains the attributes for mapping the segment to the organization unit (OU) of the LDAP-based server. There are certain attributes that have been defined with default values, which are good enough to use.	
Segment/OU store	By default, this is disabled, which means the user store will not be shown on the left pane with its segment hierarchy if the segment hierarchy license is enabled for the system. To enable it, set the value to "Enabled". Default: Disabled
Search filter for OU	Specifies the filter employed when doing a search for OU, e.g. "(objectClass=organizationalUnit)".
Object class for OU	Refers to OU's object class, e.g. "organizationalUnit". Default: organizationalUnit
Attribute for OU's id	Attribute name of OU Id, e.g. "ou".

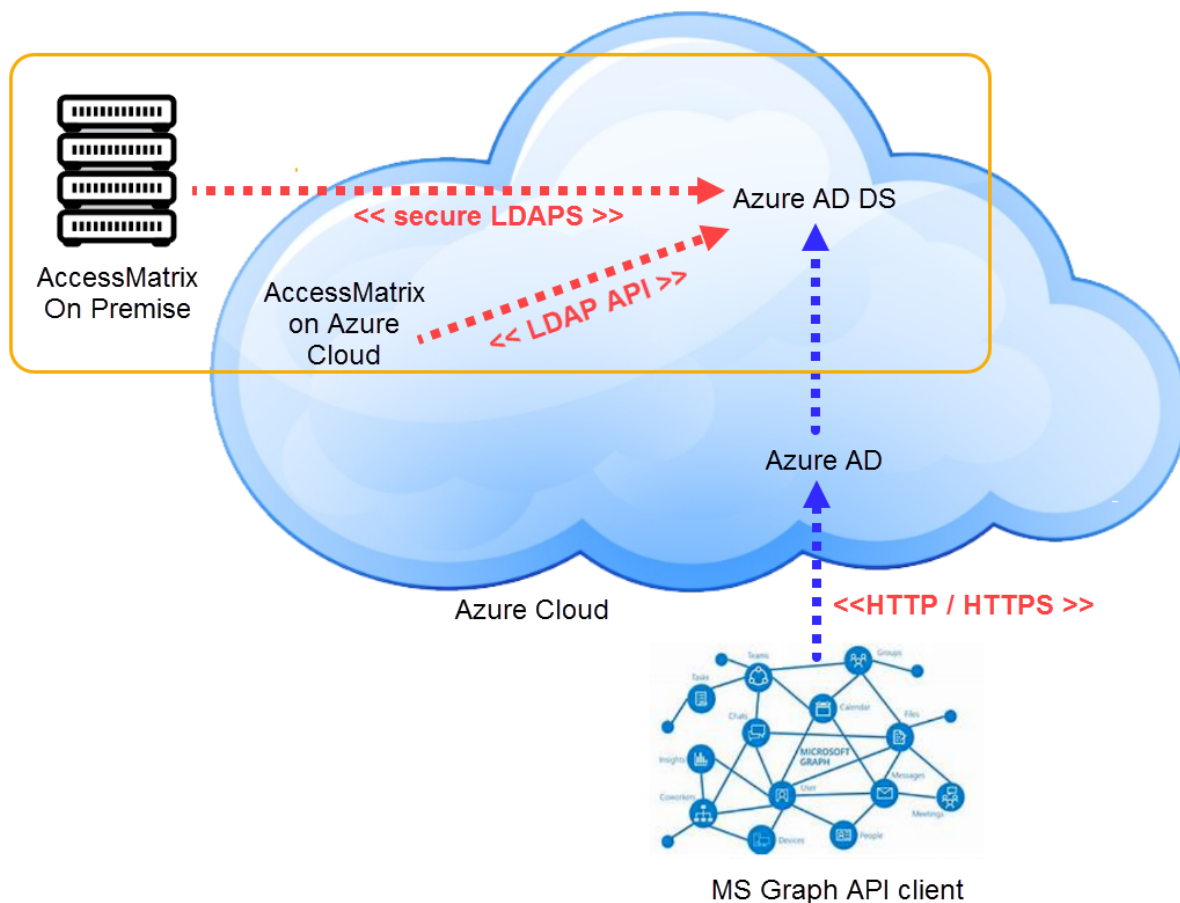
Attribute Name	Description
Extra containers to query	Container's RDNs to query when querying for users or groups in a specific segment. For example, adding "CN=Users" to this list will make the user store to query for users and groups in CN=Users, OU=... in addition to OU=... .
Cache Tab	
Cache Configuration	
Maximum size	Maximum elements that the cache can contain. "0" means do not use the cache. Value should not be less than 0. Default: 5000
Time to idle (seconds)	The maximum amount of time (in seconds) that an element can stay in the cache since last access. "0" means it will not expire due to idle time. Value should not be less than 0. Default: 1800
Time to live (seconds)	The maximum amount of time (in seconds) that an element can stay in the cache from creation time until the element expires. "0" means the element can stay forever. Value should not be less than 0. Default: 3600
Eviction policy	Used to decide which element is to be evicted from the cache when the cache is full. Default: Least Recently Used (LRU)

LDAP Connectivity to Microsoft Entra ID DS (Microsoft Entra ID Domain Services)

This section explains information about Microsoft Entra ID DS that relates to AccessMatrix. This is an excerpt from [Microsoft FAQ on Microsoft Entra ID](#).



Note: Microsoft Entra ID was formerly known as "Azure AD".



Q: Can I set up a secure LDAP connection with Microsoft Entra ID?

A: No. Microsoft Entra ID does not support the Lightweight Directory Access Protocol (LDAP) protocol or Secure LDAP directly. However, it's possible to enable Microsoft Entra ID Domain Services (Microsoft Entra ID DS) instance on your Microsoft Entra ID tenant with properly configured network security groups through Azure Networking to achieve LDAP connectivity.

Based on Microsoft documentation excerpt above, Microsoft Entra ID does not support LDAP. However, Microsoft provides Microsoft Entra ID DS to enable the external application to connect to Microsoft Entra ID by using LDAP or secure LDAP protocol.

Therefore, from the AccessMatrix point of view, Microsoft Entra ID DS is a bridge to connect to Microsoft Entra ID by using LDAP API, similar to how AccessMatrix connects to the local Active Directory. However, there are some notes/drawbacks:

- Both Microsoft Entra ID and Microsoft Entra ID DS maintain their own user store that will be synchronized at a period that is not clearly defined by Microsoft. It can be half an hour or a few

days depends on the number of users.

Excerpt from [Microsoft documentation about Microsoft Entra ID DS synchronization](#): *The initial synchronization may take a few hours to a couple of days, depending on the number of objects in the Microsoft Entra ID directory.*

- User synchronization works one way, from Microsoft Entra ID to Microsoft Entra ID DS.
Excerpt from [Microsoft documentation about Microsoft Entra ID DS synchronization](#): *The synchronization process is one way / unidirectional by design. There's no reverse synchronization of changes from Microsoft Entra ID DS back to Microsoft Entra ID.*
 - Thus, new users or any AD objects being added in Microsoft Entra ID DS will be not synchronized back to Microsoft Entra ID.
- There is no domain admin or administrator user level in Microsoft Entra ID DS. The highest user privilege is "AD DC Administrator", which still has limited privileges, i.e.
 - It cannot add, modify and delete users/groups from Microsoft Entra ID.
 - However, it can add, modify and delete users/group in Microsoft Entra ID DS (that will not be synchronized back to Microsoft Entra ID).
- Following Microsoft Entra ID, there is no OU (Organization Unit) concept to organize users. Therefore, it is not possible to organize users from Microsoft Entra ID into a segment in AccessMatrix.
 - However, "AD DC Administrator" can add local OU in Microsoft Entra ID DS (that will not be synchronized back to Microsoft Entra ID).

Excerpt from [Microsoft documentation about Microsoft Entra ID DS](#)

synchronization: *An Microsoft Entra ID DS managed domain is largely read-only except for custom OUs that you can create.*

- Even if users are synchronized one way (from Microsoft Entra ID to Microsoft Entra ID DS), the passwords are not, which means users must synchronize manually by changing their password in Microsoft Entra ID. Refer to [MS documentation](#). Password synchronization also works one way, from Microsoft Entra ID to Microsoft Entra ID DS. Thus, users cannot change their password to Microsoft Entra ID DS directly.
 - Therefore, some users' operations provided by AccessMatrix are not available, i.e. reset password and change password.

These are prerequisites to use Microsoft Entra ID DS with AccessMatrix, by referring to Microsoft documentation or consult Azure experts/consultants:

- Microsoft Entra ID DS has been set up properly. To verify this, view the user list from any LDAP client, e.g. Microsoft Active Directory Administration Tool from a Windows VM on Azure.
- If AccessMatrix is to be installed on-premise, then Secure LDAP connectivity to Microsoft Entra ID DS is required, and must be set up properly. To verify this, view the user list from any LDAP client e.g. **LDP.exe** from Microsoft Remote Server Administration Tools from a local Windows server.
- Users are required to change their password first on Microsoft Entra ID before using any AccessMatrix service.

Excerpt from [Microsoft documentation](#): *For cloud-only user accounts, users must change their passwords before they can use Microsoft Entra ID DS.*

Create Microsoft Entra ID User Store

Microsoft Entra ID (Active Directory) User Store is a cloud-based directory service up on Azure that Microsoft developed for [Microsoft Entra ID domain services](#). The implementation of the Microsoft Entra ID user store uses the Microsoft Graph API.

The Microsoft Azure Graph API uses the Open Data protocol. Currently, only the query options "\$filter", "\$select", "\$orderby", and "\$top" are implemented. The query option \$filter can be constructed using our AccessMatrix standard Criterion (Restriction class) and the \$top can be specified via the parameter "persistence.query.limit" during the API call.

Query Option	Description	Value	Precedence
\$filter	Filters the response based on a set of criteria.	string	The value passed in API take precedence over Query Option Configuration in Admin Console.
\$top	Specifies and limits the page size of the response or result set.	integer	The Query Option Configuration and QueryLimit in Admin Console take precedence over the value passed in API.
\$select	Returns a set of properties to be included in the response.	string	The value passed in API take precedence over Query Option Configuration in Admin Console.
\$orderby	Sorts the order of the items returned from Microsoft Graph.	string	The value passed in API take precedence over Query Option Configuration in Admin Console.



Notes:

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- The query option \$top will be adjusted to the query limit configuration in user store if the \$top value is higher than query limit because Microsoft Graph API will fail the request if \$top value is higher than the query limit allowed in the license.
 - Attribute DN (Distinguished Name) can be mapped into AM BaseObject but find the object by DN is not supported due to the limitation in Microsoft Graph API can only query directoryObject by id or ids but not for other attributes.
 - The group member query is not traversal means we only implement direct member/memberOf (non-hierarchy).

Configure Microsoft Entra ID User Store

i Note: This section assumes that the administrator has already set up the required Microsoft Entra ID accounts or licenses. The administrator should also have carried out the necessary authentication and authorization steps before implementing the Microsoft Entra ID user store in the AccessMatrix server.

To configure a service and get a token from the Microsoft identity platform endpoint (which your service can use to call Microsoft Graph under its own identity), follow the instructions below:

Register Application:

1. In the Microsoft Microsoft Entra ID web portal, navigate to **Microsoft Entra ID > Manage > App registrations > New registration**. The **Register an application** page is displayed.
2. Perform the following steps:
 - a. Under **Name**, enter a suitable name for your app.
 - b. Under **Supported Account Types**, select the **Accounts in any organizational directory and personal Microsoft accounts** option.
 - c. Click **Register**. The app's **Overview** page is displayed.

Obtain Application and Directory IDs, OIDC Metadata Document URI, and Client Secret:

1. Navigate to your app's **Overview** page.
 2. Locate the following fields and record down their values:
 - a. **Application (client) ID**
 - b. **Directory (tenant) ID**
-

-
3. (For Microsoft login via OIDC only) Next, click **Endpoints** and record the URI displayed under **OpenID Connect metadata document**.



Note: Skip this step if using a non-OIDC authentication method, e.g. Microsoft Microsoft Entra ID login.

4. Next, navigate to **Manage > Certificates & secrets > Client secrets > New client secret**.
5. Perform the following steps:
 - a. Enter an appropriate key description under **DESCRIPTION**.
 - b. Select "In 1 year" under **EXPIRES**.
 - c. Click **Add**. The key value is displayed under **VALUE**.
 - d. Record the key value and store it securely.



Important Note: You must record the key value during this step; it will never be displayed again once you leave the page.

Configure Permissions for Microsoft Graph for your Application:

1. Navigate to **Manage > API Permissions > Add a permission**. The **Request API permissions** window is displayed.
2. Select **Microsoft APIs > Microsoft Graph**.
- 3.

For Microsoft Entra ID Login	a. Under Application permissions, search for the "Directory.Read.All" permission and select it. b. Under Delegated permissions, search for the following permissions and select them: ▪ "Directory.AccessAsUser.All" ▪ "offline_access" ▪ "openid" ▪ "profile" ▪ "User.Read"
For Microsoft Login via OIDC	Under Application permissions , search for the "User.ReadBasic.All" permission and select it.

4. Click **Add permissions**.
 5. Click **Grant admin consent**.
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Note: If the correct permissions were not granted and a consent error is encountered while attempting to log in, simply updating the permissions and grants may not work. In such cases, you should attempt to create a new application in Microsoft Entra ID and reconfigure it as required before trying to log in again.

Create and Configure Microsoft Entra ID User Store in Admin Console

Once the Microsoft Entra ID configurations are complete, you must create a corresponding Microsoft Entra ID user store within the AccessMatrix Admin Console. To do so, perform the following steps:

1. In the Admin Console, navigate to **Administration Dashboard > Configuration > Authentication > User Store**.
2. Click **Create** and select the "Microsoft Entra ID User Store" implementation from the dropdown list.
3. Enter an appropriate name for ***User Store Id** and ***User Store Name**.
4. Under the **Configuration** tab, provide the values you previously recorded from the Microsoft Entra ID App registrations web portal:
 - a. **OAuth2 Authentication URL** - Enter the appropriate OAuth2 authentication URL:

For Microsoft Entra ID Login	Leave the value empty to use the default URL.
For Microsoft Login via OIDC	Enter the URI from the OpenID Connect metadata document mentioned in the previous section. i Note: Refer to UAM OIDC Relying Party Implementation Guide - Configure Lookup/Login Module for Use with Microsoft Entra ID User Store for more information on using the Microsoft Entra ID user store with the UAM OIDC Relying Party lookup/login module for OIDC authentication.

- b. **Directory (tenant) ID** - Enter the **Directory (tenant) ID** from your app's **Overview** page.
- c. **Application (client) ID** - Enter the **Application (client) ID** from your app's **Overview** page.
- d. **Client Secret** - Enter the client secret that you generated for your app.

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Note: This is the key value displayed under the **VALUE** field, which was recorded in step 5 of the earlier section. It is not to be confused with the value displayed under the **Secret ID** field.

5. Under the **User Attributes Mapping** tab, these are the default values you should enter for the following attributes:

- a. **Attribute for UUID** - Enter "id".
 - b. **Attribute for id** - Enter "userPrincipalName".
 - c. **Attribute for name** - Enter "givenName".
6. Configure the remaining attributes as needed and save the changes. Refer to the table in the [Create User Store Page Attribute Description](#) section for the detailed descriptions of each attribute.

Create User Store Page Attribute Description

Attribute Name	Description
*User Store Id	Unique Id of Microsoft Entra ID user store.
*User Store Name	Name of Microsoft Entra ID user store.
Description	Description of Microsoft Entra ID user store.
Version	Microsoft Entra ID user store entry version.
Implementation	Type of Microsoft Entra ID user store implementation. The authentication to Microsoft Graph uses an OAuth 2 Client secret. You must specify the Tenant ID, Application ID, and Client Secret in Admin Console.
Attribute Name	Description
Configuration Tab	
Connection	
OAuth2 Authentication URL	Microsoft's Identity Platform endpoint URL that is used to request an access token for authentication purpose. Note: Do not change this configuration unless you know what to change to. Leave it empty to use the default value. Default: https://login.microsoftonline.com/(tenant)/oauth2/v2.0/token
Microsoft Graph API URL	The URL of Microsoft Graph REST API that enables the user to access Microsoft Cloud service resources. Note: Do not change this configuration unless you know what to change to. Leave it empty to use the default value. Default: https://graph.microsoft.com/
OAuth 2 Authentication	
*Directory (tenant) ID	Refers to the directory tenant that you request permission from. This can be in GUID or friendly name format.

Attribute Name	Description
	Note: GUID (or UUID) is an acronym for "Globally Unique Identifier" (or "Universally Unique Identifier") which is a 128-bit integer number used to identify resources.
*Application (client) ID	Refers to the Application ID that the Azure app registration portal assigned when you registered your app.
*Client Secret	Refers to the application secret that you generated for your application in the application registration portal.
User Attributes Mapping Tab	
Microsoft Graph API	
API version	Refers to Microsoft Graph supported API versions, for example: "v1.0" or "beta". Default: v1.0
Basic Attributes	
*Attribute for UUID	User's UUID attribute.
*Attribute for id	User's id attribute.
*Attribute for name	User's name attribute.
Attribute for DN	User's Distinguished Name (DN) attribute.
Other attributes mapping	Refers to other attributes mapping, where each entry has this format: "<AM's attribute name>=<Azure Directory Object's attribute name>".
User	
Query Options for User	Refers to the query options for user. For example, use "\$select" Query option to return user id, displayName and givenName of the users and sort the result by displayName: "\$select=id,displayName,givenName&\$orderby=displayName asc".
Object class for user	Refers to the object class for user. Default: User
Attribute for email	User's email attribute, e.g. "mail"
Attribute for mobile	User's mobile attribute, e.g. "mobile"
Attribute for created time	User's created time attribute, e.g. "whenCreated"

Attribute Name	Description
Group	
Query Options for Group	Refers to the query options for group. For example, use the "\$filter" Query option to search the first three groups for the displayName attributes start with "HR": "\$filter=startswith(displayName,'HR')&\$top=3". Note: If you query on the user resource, "\$orderby" cannot be combined with the filter expressions.
Object class for Group	Refers to the object class for group. Default: Group
Attribute for group id	Group Id's attribute.
Advanced Attributes Configuration	
Date time attribute format	In Microsoft Entra ID, date time attribute values are represented by strings. Refer to the default example for how to parse or print the date time string. Default: yyyy-MM-dd'T'HH:mm:ss'Z'
Query Limit	Maximum query limit based on the number of Microsoft Entra ID licenses. Default: 999
Cache Tab	
Cache Configuration	
Maximum size	Maximum elements that the cache can contain. "0" means do not use the cache. Value should not be less than 0. Default: 5000
Time to idle (seconds)	The maximum amount of time (in seconds) that an element can stay in the cache since last access. "0" means it will not expire due to idle time. Value should not be less than 0. Default: 1800
Time to live (seconds)	The maximum amount of time (in seconds) that an element can stay in the cache from creation time until the element expires. "0" means the element can stay forever. Value should not be less than 0. Default: 3600
Eviction policy	Used to decide which element is to be evicted from the cache when the cache is full. Default: Least Recently Used (LRU)

(For Microsoft Entra ID Login with CLP Only) Edit CLP Config File

The following steps are only relevant if you are using the Microsoft Entra ID user store with the Microsoft Entra ID login module and CLP. It is assumed that the Microsoft Entra ID login module and Microsoft Entra ID user store have been configured. It is also assumed that an authentication realm with both the Microsoft Entra ID login module and user store has been prepared.

In order to use Microsoft Entra ID login with the Microsoft Entra ID user store and CLP, the CLP config file must be updated:

1. Navigate to `<CLP_HOME>/WEB-INF/classes` and open **am.commonlogin.config.xml**.
2. Look for the `AuthnRealms` element.
3. Under this element, create a new `AuthnRealm` code block containing the Authentication Realm configurations as shown below:

XML

```
<AuthnRealm>
  <id>AzureADLogin</id>
  <idLabel></idLabel>

  <firstLoginModuleNeedsChallenge>false</firstLoginModuleNeedsChallenge>

  <loginModuleIdForChangePassword></loginModuleIdForChangePassword>
  <stateForChangePassword></stateForChangePassword>
  <resetPassword>
    <userStoreId>AzureADUserStore</userStoreId>
    <loginModuleId>AzureADLoginModule</loginModuleId>
  </resetPassword>
  <loginModule>
    <handler>AzureADLogin</handler>
    <loginState>login.start</loginState>
  </loginModule>
</AuthnRealm>
```

Ensure the following parameters are provided:

- `id` - Set this to the Id of the configured authentication realm (e.g. "AzureADLogin")
 - `firstLoginModuleNeedsChallenge` - Set this to "false".
 - `userStoreId` - Set this to the Id of the configured Microsoft Entra ID user store (e.g. "AzureADUserStore").
-

- loginModuleId - Set this to the Id of the configured Microsoft Entra ID login module (e.g. "AzureADLoginModule").
- handler - Set this to "AzureADLogin".
- loginState - Set this to "login.start".

4. Save the changes and restart the CLP server.

i **Note:** Refer to [AccessMatrix Common Login Page Deployment Guide - 1FA](#) for more details on the relevant CLP configurations.

Create JDBC User Store

JDBC (Java Database Connectivity) user store can be an internal user store or an external database store that helps to manage user repository. In general, the table columns must be convertible to Java String (varchar). After table schema information is configured, a **Generate SQL Statement** button will show the SQL statement templates that will be used to query. If the generated SQL statement does not satisfy the requirement, an SQL statement can be configured to override the generated one.

Create User Store Page Attribute Description

Attribute Name	Description
*User Store Id	Unique Id of JDBC user store.
*User Store Name	Name of JDBC user store.
Description	Description of JDBC user store.
Version	JDBC user store entry version.
Implementation	Type of JDBC user store implementation.
Configuration Tab	
Connection	
*DataSource	Refers to the URL used to establish a database connection, e.g. "java://comp/env/jdbc/MasterUserDB".
Updateable	Setting this to "true" will allow AM to create/edit/delete data in JDBC user store. Otherwise, AM can only view JDBC user store's data. Default: false

Attribute Name	Description
Isolation Level Override	This setting is used to override the default database isolation level. However, it is recommended to change the isolation level at the datasource layer if possible. Possible choices: • "User default isolation level" • "TRANSACTION_READ_UNCOMMITTED" • "TRANSACTION_READ_COMMITTED" • "TRANSACTION_REPEATABLE_READ" • "TRANSACTION_SERIALIZABLE"
Enable Caching of External Groups	Specifies if the user's external group membership will be cached. Select "true" to cache the user's external group membership for better performance at the cost of data accuracy. Once cached, the data will only be fetched from the external database when there are changes made on the local server. Select "false" to disable caching for the user's external group membership. This ensures that their external group membership is always up-to-date as data is constantly fetched from the external database. Note: If caching is disabled, frequently retrieving the same user may cause some performance issues.
User Tab	
User Schema	
Table name	The user's table to query the user's information.
User attributes mapping	Maps the user attributes from JDBC user's table to AccessMatrix attributes. Required attributes: UUID, Id, and Name. Map other attributes such as Email, Mobile, Segment/Parent UUID, and other custom attributes, if required.
Column for soft deletion flag	The column to store the soft deletion flag, e.g. "mark_flag". Note: Only if soft deletion is required.
Soft deleted column value	The column value if the user is soft deleted, e.g. "DEL".
Account Status Mapping (required only if Account Status is mapped in attributes mapping)	
Account status mapping for Activated	This value will be mapped to "Activated". Any other value is considered "Disabled". If the user store is updateable, this value will be saved to the database when account status is set to "Activated".
Account status mapping for Disabled	Required only if user store is marked as updateable. This value will be saved to the database when account status is set to "Disabled".
SQL Query Override for User (optional, to override automatically generated SQL)	

Attribute Name	Description
Find user	The SQL query to acquire basic user information. The query returns UUID (unique and immutable), Id, Name and 'User' as an object class. Example: "SELECT p.prn_id,p.prn_login_id,p.prn_name,'User' AS object_class FROM principals p WHERE %1\$s".
Get complete user (with attributes)	The SQL query to acquire basic user information and additional attributes as required. Note: Leave this field empty if there's no additional attribute to acquire. The query returns UUID (unique and immutable), Id, Name and 'User' as object class, and additional attribute (e.g. Email). Example: "SELECT p.prn_id,p.prn_login_id,p.prn_name,a.attr_value as Email,'User' AS object_class FROM principals p LEFT OUTER JOIN prnattrs a ON a.prn_id=p.prn_id AND a.attr_name='%Email' WHERE p.prn_id=?"
Find user (with attribute as criteria)	The SQL query to acquire basic user information with attribute as criteria, e.g. search by Email. Note: Leave this field empty if searching by attributes are not required. The query returns UUID (unique and immutable), Id, Name and 'User' as object class. Example: "SELECT prn_id,prn_login_id,prn_name,'User' AS object_class FROM (SELECT p2.prn_id, p2.prn_login_id, p2.prn_name, a.attr_value as Email FROM principals p2 LEFT OUTER JOIN prnattrs a ON p2.prn_id = a.prn_id and a.attr_name='%Email') WHERE %1\$s"
Prepared Statement for Inserting a New User (optional, to override automatically generated SQL)	
SQL	(Optional) The SQL statement to insert a new user.
SQL Parameters	The SQL parameters to insert into new user.
Group Tab	
Group Schema	
Table name	The group's table to query user's group information.
Group attributes mapping	Maps the group attributes from JDBC group's table to AccessMatrix attributes. Required attributes: UUID , Id , and Name . Map other attributes such as Email , Mobile , Segment/Parent UUID , and other custom attributes, if required.
Column for soft deletion flag	The column to store the soft deletion flag, e.g. "mark_flag". Note: Only if soft deletion is required.

Attribute Name	Description
Soft deleted column value	The column value if the user is soft deleted, e.g. "DEL".
Membership Schema	
Table name	The JDBC table containing membership schema.
User UUID Column	The user UUID value to use when loading the user's groups.
Group UUID Column	The group UUID value to use when loading the group's members.
SQL Query Override for Group (optional, to override automatically generated SQL)	
Find group	The SQL query to acquire basic group information. The query returns UUID (unique and immutable), Id, Name and 'Group' as object class. Example: <code>SELECT g.grn_id,g.grp_name,g.grp_name,g.grp_dn,'Group' AS object_class FROM groups g WHERE %1\$s</code>
Get complete group	The SQL query to acquire basic group information and additional attributes as required. Note: Leave this field empty if there's no additional attribute to acquire. The query returns UUID (unique and immutable), Id, Name, 'Group' as object class, and additional attributes (e.g. Email). Example: <code>SELECT g.grn_id,g.grp_name,g.grp_name,g.grp_dn,g.grp_type,'Group' AS object_class FROM groups g WHERE g.grp_id=?</code>
Find group (with an attribute as criteria)	The SQL query to acquire basic group information with the attribute as criteria, e.g. search by Email. Note: Leave this field empty if searching by attributes is not required. The query returns UUID (unique and immutable), Id, Name and 'Group' as an object class. Example: <code>SELECT g.grn_id,g.grp_name,g.grp_name,g.grp_dn,'Group' AS object_class FROM groups g WHERE %1\$s</code>
SQL Query Override for Group (optional, to override automatically generated SQL)	
Get user's groups	The SQL query to acquire basic information of the groups to which a user belongs. The query returns UUID (unique and immutable), Id, Name and 'Group' as object class. Example: <code>SELECT g.grn_id,g.grp_name,g.grp_name,g.grp_dn,'Group' AS object_class FROM groups g WHERE g.grp_id IN SELECT m.grp_id FROM memberships WHERE m.prn_id=?</code>
Get group's members (users)	The SQL query to acquire basic information of the users which belongs to a group. The query returns UUID (unique and immutable), Id, Name and 'User' as an object class.

Attribute Name	Description
	Example: SELECT p.prn_id,p.prn_login_id,p.prn_name,'User' AS object_class FROM principals p WHERE p.prn_id IN (SELECT m.prn_id FROM memberships WHERE m.grp_id=?)
Prepared Statement for Inserting a New Group (optional, to override automatically generated SQL)	
SQL	The SQL statement to insert a new group.
SQL Parameters	The SQL parameters to insert into new group.
Prepared Statement for Inserting a New Membership (optional, to override automatically generated SQL)	
SQL	The SQL statement to insert a new membership.
Segment Tab	
Segment/OU store	The segment/OU of the AD user store will be displayed if this attribute is set to "Enabled". Default: Disabled
Segment Schema	
Table name	The segment's table to query segment's information.
Segment attributes mapping	Refers to the segment attributes mapping. Require attributes: UUID , Segment/Parent UUID , Id , and Name .
Segment Tree Cache	
Segment tree cache refresh interval (in seconds)	Segment refresh interval. Value should be from 300 to 7200. Default: 1800
Segment may be created and used within short interval	This is to indicate that a segment may be created externally and used within a short interval. For example, create a new segment, followed by the user creation under the segment and the user password reset. This flag tells the user store to refer to the database if a segment is not found in the tree cache and adds the new segment to the tree cache. Adding the new segment to the tree cache is to make sure that hierarchical authorization is consistent. However, if a lot of new multi-level segments are added, the best way is still to refresh the whole tree cache. Enable this ONLY if required as it may affect performance.
SQL Query Override for Segment (optional, to override automatically generated SQL)	

Attribute Name	Description
Find segment	The SQL query to acquire basic segment information. The query returns UUID (unique and immutable), Id, Name, Segment's parent UUID, Description (optional) and 'Segment' as an object class. Example: "SELECT seg_id,seg_uid,seg_name,parent_seg_id,'Segment' AS object_class FROM segments WHERE %1\$s"
Query for segment tree caching	The SQL statement to query UUID and parent UUID for all segments to build segment tree cache. The query returns UUID (unique and immutable) and parent UUID. Example: SELECT seg_id, parent_seg_id FROM segments

Create LDAP User Store

Lightweight Directory Access Protocol (LDAP) user store is used by the user store manager to define the users and his/her roles by defining the attribute that contains the distinguished names (DN) of user objects that are in a group. Objects such as the users, groups, and organizations are stored in the LDAP user directory/store.

Create User Store Page Attribute Description

Attribute Name	Description
*User Store Id	Unique Id of LDAP user store.
*User Store Name	Name of LDAP user store.
Description	Description of LDAP user store.
Implementation	Type of LDAP user store implementation.
Connection Tab	
Connection	
*Urls	Refers to the LDAP server name/IP Address and port number to connect to, e.g. "LDAP://192.168.1.1:389". Note: To make use of AD Global Catalog, configure it as the primary LDAP server. The main AD Domain Service server subsequently can be configured as secondary backup server.
Use SSL	A flag to enable/disable Secure Sockets Layer (SSL) connection. Default: false

Attribute Name	Description
Authentication mechanism	Refers to the authentication mechanism/level used for authentication to the LDAP server. Default: Simple (Login Id + Password)
Login Id	Refers to the user Id of the LDAP credential with the host name, e.g. "WIN2003\demouser".
Password	Refers to the LDAP user's password.
Connection (Advanced)	
Use connection pool	Setting this to "true" will use the values configured in Global Setting > LDAP Connection Pool . Default: true
Connection timeout (seconds)	Specifies the maximum time (in seconds) to wait for connection results. "0" means there is no timeout. Value should be from 0 to 3600.
Read Timeout	Specifies the maximum time (in seconds) to wait for results on a read operation. "0" means there is no timeout. Value should be from 0 to 3600.
Use LDAP paging feature	Setting this to "true" will enable paging when searching (RFC 2696). LDAP paging allows retrieval of search result more than the query limit. The report may not work properly when this feature is disabled. Default: true Note: For older version of LDAP server, set this attribute to "false" (turn off) as this feature is not available.
LDAP Referral	Option specifying whether referrals to other LDAP servers are followed automatically. Available choices: "ignore" "follow" "throw"
User Attributes Mapping Tab	
This tab contains certain attributes that have been filled with default values for attributes mapping which is good enough to use, as well as other advanced attributes configuration.	
Search Filter	

Attribute Name	Description
*Search Context	Specifies the Base DN to use for user search. e.g. "CN=Users,DC=app-sg-isi,DC=i-sprint,DC=com". Click the Fetch Base DNs button to help fill the Search Context attribute value.
Basic Attributes	
*Attribute for UUID	User's UUID attribute, e.g. "objectGUID".
*Attribute for DN	User's Distinguished Name (DN) attribute, e.g. "distinguishedName".
*Attribute for id	User's Id attribute, e.g. "sAMAccountName". Note: Set value to "distinguishedName" if user Id is to be searched as user DN in the desired LDAP server. Otherwise, set to "email" if user Id is to be searched as email address in the desired LDAP server.
*Attribute for name	User's name attribute, e.g. "name".
Other attributes mapping	Refers to other user attribute's mapping. Format: <AM's attribute name>=<LDAP's attribute name>.
User	
*Search filter for user	Specifies the search filter for user, e.g. "(&(objectClass=user)!(objectClass=Computer)))".
Object class for user	Refers to the object class for user. Default: User
Attribute for email	User's email attribute, e.g. "mail".
Attribute for mobile	User's mobile attribute, e.g. "mobile".
Group	
*Search filter for group	Specifies the filter employed when doing a search for groups, e.g. "(objectClass=Group)".
*Object class for group	Refers to the object class for the group. Default: Group
Attribute for group id	Attribute name of group id.
*Attribute for member	Attribute name of unique member, e.g. "member".
Attribute for member of	Attribute name for group membership, e.g. "memberOf".

Attribute Name	Description
Group of URLs (Dynamic Group)	
Object class for group of URLs	Refers to the object class for the group of URLs.
Attribute for member URL	Attribute name of member URL.
Advanced Attributes Configuration	
Returning attributes	(Unused)
Binary attributes	e.g. "objectGUID objectSid". Note: Specify the binary attributes if UUID attributes are in binary form.
Use shadow UUID	Use shadow UUID instead of AccessMatrix UUID to lookup the LDAP entry. This is used when LDAP data has been migrated without retaining its UUID. The shadow UUID, which is the new LDAP UUID, has to be set as a user's AM5 attribute. This can be done via database script or via a one-time LDAP migration plugin. Default: false
Others	
Use relative DN to search member	Setting this to "true" will use relative DN to search member. Default: false
Segment/OU Tab	
This tab contains the attributes for mapping the segment to the organization unit (OU) of the LDAP-based server. There are certain attributes that have been defined with default values, which are good enough to use.	
Segment/OU store	By default, this is disabled, which means the user store will not be shown on the left pane with its segment hierarchy if a segment hierarchy license is enabled for the system. To enable it, set the value to "Enabled". Default: Disabled
Search filter for OU	Specifies the filter employed when doing a search for OU, e.g. "(objectClass=organizationalUnit)".
Object class for OU	Refers to OU's object class, e.g. "organizationalUnit". Default: organizationalUnit
Attribute for OU's id	Attribute name of OU Id, e.g. "ou".

Attribute Name	Description
Cache Tab	
Cache Configuration	
Maximum size	Maximum elements that the cache can contain. "0" means do not use cache. Value should not be less than 0. Default: 5000
Time to idle (seconds)	The maximum amount of time (in seconds) that an element can stay in the cache since last access. "0" means it will not expire due to idle time. Value should not be less than 0. Default: 1800
Time to live (seconds)	The maximum amount of time (in seconds) that an element can stay in the cache from creation time until the element expires. "0" means the element can stay forever. Value should not be less than 0. Default: 3600
Eviction policy	Used to decide which element is to be evicted from the cache when the cache is full. Default: Least Recently Used (LRU)
Group of URLs cache update interval (seconds)	The interval (in seconds) a background thread updates the group of URLs (dynamic group) resolution. This cache is unbounded; thus, the cache maximum size setting does not affect this cache. Value should not be less than 300. Default: 1800

Create Novell eDirectory Server User Store

Create User Store Page Attribute Description

Attribute Name	Description
*User Store Id	Unique Id of Novell eDirectory Server user store.
*User Store Name	Name of Novell eDirectory Server user store.
Description	Description of Novell eDirectory Server user store.
Implementation	Type of Novell eDirectory Server user store implementation.
Connection Tab	
Connection	

Attribute Name	Description
*Urls	Refers to the Novell eDirectory server name/IP Address and port number to connect to, e.g. "LDAP://192.168.1.1:389". Note: To make use of AD Global Catalog, configure it as the primary LDAP server. The main AD Domain Service server subsequently can be configured as secondary backup server.
Use SSL	A flag to enable/disable Secure Sockets Layer (SSL) connection. Default: false
Authentication mechanism	Refers to the authentication mechanism/level used for authentication to the LDAP server. Default: Simple (Login Id + Password)
Login Id	Refers to the user Id of the LDAP credential with the host name, e.g. "WIN2003\demouser".
Password	Refers to the LDAP user's password.
Connection (Advanced)	
Use connection pool	Setting this to "true" will use the values configured in Global Setting > LDAP Connection Pool . Default: true
Connection timeout (seconds)	Specifies the maximum time (in seconds) to wait for connection results. "0" means there is no timeout. Value should be from 0 to 3600.
Read Timeout	Specifies the maximum time (in seconds) to wait for results on a read operation. "0" means there is no timeout. Value should be from 0 to 3600.
Use LDAP paging feature	Setting this to "true" will enable paging when searching (RFC 2696). LDAP paging allows retrieval of search result more than the query limit. The report may not work properly when this feature is disabled. Default: true Note: For older version of LDAP server, set this attribute to "false" (turn off) as this feature is not available.
LDAP Referral	Option specifying whether referrals to other LDAP servers are followed automatically. Available choices: ignore, follow, and throw
User Attributes Mapping Tab	
This tab contains certain attributes that have been filled with default values for attributes mapping which is good enough to use, as well as other advanced attributes configuration.	

Attribute Name	Description
Search Filter	
*Search Context	Specifies the Base DN to use for user search. e.g. "CN=Users,DC=app-sg-isi,DC=i-sprint,DC=com". Click the Fetch Base DNs button to help fill the Search Context attribute value.
Basic Attributes	
*Attribute for UUID	User's UUID attribute. Default: GUID
*Attribute for DN	User's Distinguished Name (DN) attribute. Default: DN
*Attribute for id	User's Id attribute, e.g. "sAMAccountName". Note: Set value to "distinguishedName" if user Id is to be searched as user DN in the desired LDAP server. Otherwise, set to "email" if user Id is to be searched as the email address in the desired LDAP server. Default: cn
*Attribute for name	User's name attribute, e.g. "name". Default: sn
Other attributes mapping	Refers to other user attribute's mapping. Format: <AM's attribute name>=<LDAP's attribute name>.
User	
*Search filter for user	Specifies the search filter for user, e.g. '(&(objectClass=user)(!(objectClass=Computer)))'.
Object class for user	Refers to the object class for user. Default: User
Attribute for email	User's email attribute, e.g. "mail".
Attribute for mobile	User's mobile attribute, e.g. "mobile".
Group	
*Search filter for group	Specifies the filter employed when doing a search for groups, e.g. "(objectClass=Group)".
*Object class for group	Refers to the object class for the group. Default: Group
Attribute for group id	Attribute name of group id.

Attribute Name	Description
*Attribute for member	Attribute name of unique member, e.g. "member".
Attribute for member of	Attribute name for group membership, e.g. "memberOf".
Group of URLs (Dynamic Group)	
Object class for group of URLs	Refers to the object class for group of URLs.
Attribute for member URL	Attribute name of member URL.
Advanced Attributes Configuration	
Returning attributes	(Unused)
Binary attributes	e.g. "objectGUID objectSid". Note: Specify the binary attributes if UUID attribute is in binary form.
Others	
Use relative DN to search member	Setting this to "true" will use relative DN to search member. Default: false
Segment/OU Tab	
This tab contains the attributes for mapping the segment to the organization unit (OU) of the LDAP-based server. There are certain attributes that have been defined with default values, which are good enough to use.	
Segment/OU store	By default, this is disabled, which means the user store will not be shown on the left pane with its segment hierarchy if segment hierarchy license is enabled for the system. To enable it, set the value to "Enabled". Default: Disabled
Search filter for OU	Specifies the filter employed when doing a search for OU, e.g. "(objectClass=organizationalUnit)".
Object class for OU	Refers to OU's object class, e.g. "organizationalUnit". Default: organizationalUnit
Attribute for OU's id	Attribute name of OU Id, e.g. "ou".
Cache Tab	
Cache Configuration	

Attribute Name	Description
Maximum size	Maximum elements that the cache can contain. "0" means do not use cache. Value should not be less than 0. Default: 5000
Time to idle (seconds)	The maximum amount of time (in seconds) that an element can stay in the cache since last access. "0" means it will not expire due to idle time. Value should not be less than 0. Default: 1800
Time to live (seconds)	The maximum amount of time (in seconds) that an element can stay in the cache from creation time until the element expires. "0" means the element can stay forever. Value should not be less than 0. Default: 3600
Eviction policy	Used to decide which element is to be evicted from the cache when the cache is full. Default: Least Recently Used (LRU)
Group of URLs cache update interval (seconds)	The interval (in seconds) a background thread updates the group of URLs (dynamic group) resolution. This cache is unbounded; thus, the cache maximum size setting does not affect this cache. Value should not be less than 300. Default: 1800

Create OpenLDAP User Store

OpenLDAP user store is an open-source Lightweight Directory Access Protocol (LDAP) store that provides storage of information of users, groups, and organizations.

Create User Store Page Attribute Description

Attribute Name	Description
*User Store Id	Unique Id of OpenLDAP user store.
*User Store Name	Name of OpenLDAP user store.
Description	Description of OpenLDAP user store.
Implementation	Type of OpenLDAP user store implementation.
Connection Tab	
Connection	

Attribute Name	Description
*Urls	Refers to the OpenLDAP name/IP Address and port number to connect to, e.g. "LDAP://192.168.1.1:389". Note: To make use of AD Global Catalog, configure it as the primary LDAP server. The main AD Domain Service server subsequently can be configured as secondary backup server.
Use SSL	A flag to enable/disable Secure Sockets Layer (SSL) connection. Default: false
Authentication mechanism	Refers to the authentication mechanism/level used for authentication to the LDAP server. Default: Simple (Login Id + Password)
Login Id	Refers to the user Id of the LDAP credential with the host name, e.g. "WIN2003\demouser".
Password	Refers to the LDAP user's password.
Connection (Advanced)	
Use connection pool	Setting this to "true" will use the values configured in Global Setting > LDAP Connection Pool . Default: true
Connection timeout (seconds)	Specifies the maximum time (in seconds) to wait for connection results. "0" means there is no timeout. Value should be from 0 to 3600.
Read Timeout	Specifies the maximum time (in seconds) to wait for results on a read operation. "0" means there is no timeout. Value should be from 0 to 3600.
Use LDAP paging feature	Setting this to "true" will enable paging when searching (RFC 2696). LDAP paging allows retrieval of search result more than the query limit. The report may not work properly when this feature is disabled. Default: true Note: For older version of LDAP server, set this attribute to false (turn off) as this feature is not available.
LDAP Referral	Option specifying whether referrals to other LDAP servers are followed automatically. Available choices: ignore, follow, and throw
User Attributes Mapping Tab	
This tab contains certain attributes that have been filled with default values for attributes mapping which is good enough to use, as well as other advanced attributes configuration.	

Attribute Name	Description
Search Filter	
*Search Context	Specifies the Base DN to use for user search. e.g. "CN=Users,DC=app-sg-isi,DC=i-sprint,DC=com". Click the Fetch Base DNs button to help fill the Search Context attribute value.
Basic Attributes	
*Attribute for UUID	User's UUID attribute. Default: entryUUID
*Attribute for DN	User's Distinguished Name (DN) attribute. Default: DN
*Attribute for id	User's Id attribute, e.g. "sAMAccountName". Note: Set value to "distinguishedName" if user Id is to be searched as user DN in the desired LDAP server. Otherwise, set to "email" if user Id is to be searched as the email address in the desired LDAP server. Default: cn
*Attribute for name	User's name attribute, e.g. "name". Default: sn
Other attributes mapping	Refers to other user attribute's mapping. Format: <AM's attribute name>=<LDAP's attribute name>.
User	
*Search filter for user	Specifies the search filter for user, e.g. "(&(objectClass=user)!(&(objectClass=Computer)))".
Object class for user	Refers to the object class for user. Default: User
Attribute for email	User's email attribute, e.g. "mail".
Attribute for mobile	User's mobile attribute, e.g. "mobile".
Group	
*Search filter for group	Specifies the filter employed when doing a search for groups, e.g. "(objectClass=GroupOfUniqueNames)".
*Object class for group	Refers to the object class for group. Default: groupOfUniqueNames
Attribute for group id	Attribute name of group id.

Attribute Name	Description
*Attribute for member	Attribute name of the unique member, e.g. "uniqueMember".
Attribute for member of	Attribute name for group membership, e.g. "memberOf".
Group of URLs (Dynamic Group)	
Object class for group of URLs	Refers to the object class for the group of URLs.
Attribute for member URL	Attribute name of member URL.
Advanced Attributes Configuration	
Returning attributes	For example: "entryUUID entryDN cn sn uniqueMember objectClass"
Binary attributes	e.g. "objectGUID objectSid". Note: Specify the binary attributes if UUID attributes are in binary form.
Use shadow UUID	Use shadow UUID instead of AM5 UUID to lookup the LDAP entry. This is used when LDAP data has been migrated without retaining its UUID. The shadow UUID, which is the new LDAP UUID, has to be set as a user's AM5 attribute. This can be done via a database script or via a one-time LDAP migration plugin. Default: false
Others	
Use relative DN to search member	Setting this to "true" will use relative DN to search member. Default: false
Segment/OU Tab	
This tab contains the attributes for mapping the segment to the organization unit (OU) of the LDAP-based server. There are certain attributes that have been defined with default values, which are good enough to use.	
Segment/OU store	By default, this is disabled, which means the user store will not be shown on the left pane with its segment hierarchy if segment hierarchy license is enabled for the system. To enable it, set the value to "Enabled". Default: Disabled
Search filter for OU	Specifies the filter employed when doing a search for OU, e.g. "(objectClass=organizationalUnit)".

Attribute Name	Description
Object class for OU	Refers to OU's object class, e.g. "organizationalUnit". Default: organizationalUnit
Attribute for OU's id	Attribute name of OU Id, e.g. "ou".
Cache Tab	
Cache Configuration	
Maximum size	Maximum elements that the cache can contain. "0" means do not use cache. Value should not be less than 0. Default: 5000
Time to idle (seconds)	The maximum amount of time (in seconds) that an element can stay in the cache since last access. "0" means it will not expire due to idle time. Value should not be less than 0. Default: 1800
Time to live (seconds)	The maximum amount of time (in seconds) that an element can stay in the cache from creation time until the element expires. "0" means the element can stay forever. Value should not be less than 0. Default: 3600
Eviction policy	Used to decide which element is to be evicted from the cache when the cache is full. Default: Least Recently Used (LRU)

Create Sun Directory Server User Store

Create User Store Page Attribute Description

Attribute Name	Description
*User Store Id	Unique Id of Sun Directory Server user store.
*User Store Name	Name of Sun Directory Server user store.
Description	Description of Sun Directory Server user store.
Implementation	Type of Sun Directory Server user store implementation.
Connection Tab	
Attribute Name	Description

Attribute Name	Description
Connection	
*Urls	Refers to the Sun Directory Server name/IP Address and port number to connect to, e.g. "LDAP://192.168.1.1:389". Note: To make use of AD Global Catalog, configure it as the primary LDAP server. The main AD Domain Service server subsequently can be configured as secondary backup server.
Use SSL	A flag to enable/disable Secure Sockets Layer (SSL) connection. Default: false
Authentication mechanism	Refers to the authentication mechanism/level used for authentication to the LDAP server. Default: Simple (Login Id + Password)
Login Id	Refers to the user Id of the LDAP credential with the host name, e.g. "WIN2003\demouser".
Password	Refers to the LDAP user's password.
Connection (Advanced)	
Use connection pool	Setting this to "true" will use the values configured in Global Setting > LDAP Connection Pool . Default: true
Connection timeout (seconds)	Specifies the maximum time (in seconds) to wait for connection results. "0" means there is no timeout. Value should be from 0 to 3600.
Read Timeout	Specifies the maximum time (in seconds) to wait for results on a read operation. "0" means there is no timeout. Value should be from 0 to 3600.
Use LDAP paging feature	Setting this to "true" will enable paging when searching (RFC 2696). LDAP paging allows retrieval of search result more than the query limit. The report may not work properly when this feature is disabled. Default: true Note: For older versions of LDAP server, set this attribute to "false" (turn off) as this feature is not available.
LDAP Referral	Option specifying whether referrals to other LDAP servers are followed automatically. Available choices: "ignore" "follow" "throw"

Attribute Name	Description
User Attributes Mapping Tab	
This tab contains certain attributes that have been filled with default values for attributes mapping which is good enough to use, as well as other advanced attributes configuration.	
Search Filter	
*Search Context	Specifies the Base DN to use for user search. e.g. "dc=yourdomain, dc=com". Click the Fetch Base DNs button to help fill the Search Context attribute value.
Basic Attributes	
*Attribute for UUID	User's UUID attribute. Default: nsUniqueId
*Attribute for DN	User's Distinguished Name (DN) attribute. Default: entrydn
*Attribute for id	User's Id attribute, e.g. "sAMAccountName". Note: Set value to "distinguishedName" if user Id is to be searched as user DN in the desired LDAP server. Otherwise, set to email if user Id is to be searched as the email address in the desired LDAP server. Default: uid
*Attribute for name	User's name attribute, e.g. "name". Default: cn
Other attributes mapping	Refers to other user attribute's mapping. Format: <AM's attribute name>=<LDAP's attribute name>.
User	
*Search filter for user	Specifies the search filter for user, e.g. "(&(objectClass=user)!(objectClass=Computer)))".
Object class for user	Refers to the object class for user. Default: User
Attribute for email	User's email attribute, e.g. "mail".
Attribute for mobile	User's mobile attribute, e.g. "telephoneNumber".
Group	
*Search filter for group	Specifies the filter employed when doing a search for groups, e.g. "(objectClass=groupOfUniqueNames)".

Attribute Name	Description
*Object class for group	Refers to the object class for group. Default: groupOfUniqueNames
Attribute for group id	Attribute name of group id. Default: cn
*Attribute for member	Attribute name of unique member, e.g. "uniqueMember".
Attribute for member of	Attribute name for group membership, e.g. "memberOf".
Group of URLs (Dynamic Group)	
Object class for group of URLs	Refers to the object class for group of URLs.
Attribute for member URL	Attribute name of member URL.
Advanced Attributes Configuration	
Returning attributes	e.g. "nsUniqueID entrydn uid cn uniqueMember mail telephoneNumber ou objectClass".
Binary attributes	e.g. "objectGUID objectSid". Note: Specify the binary attributes if UUID attributes is in binary form.
Others	
Use relative DN to search member	Setting this to "true" will use relative DN to search member. Default: false
Segment/OU Tab	
This tab contains the attributes for mapping the segment to the organization unit (OU) of the LDAP-based server. There are certain attributes that have been defined with default values, which are good enough to use.	
Segment/OU store	By default, this is disabled, which means the user store will not be shown on the left pane with its segment hierarchy if the segment hierarchy license is enabled for the system. To enable it, set the value to "Enabled". Default: Disabled
Search filter for OU	Specifies the filter employed when doing a search for OU, e.g. "(objectClass=organizationalUnit)".

Attribute Name	Description
Object class for OU	Refers to OU's object class, e.g. "organizationalUnit". Default: organizationalUnit
Attribute for OU's id	Attribute name of OU Id, e.g. "ou".
Cache Tab	
Cache Configuration	
Maximum size	Maximum elements that the cache can contain. "0" means do not use cache. Value should not be less than 0. Default: 5000
Time to idle (seconds)	The maximum amount of time (in seconds) that an element can stay in the cache since last access. "0" means it will not expire due to idle time. Value should not be less than 0. Default: 1800
Time to live (seconds)	The maximum amount of time (in seconds) that an element can stay in the cache from creation time until the element expires. "0" means the element can stay forever. Value should not be less than 0. Default: 3600
Eviction policy	Used to decide which element is to be evicted from the cache when the cache is full. Default: Least Recently Used (LRU)
Group of URLs cache update interval (seconds)	The interval (in seconds) a background thread updates the group of URLs (dynamic group) resolution. This cache is unbounded; thus, the cache maximum size setting does not affect this cache. Value should not be less than 300. Default: 1800

Create TAM User Store

Tivoli Access Manager (TAM) from IBM enables secure communication across the network.

The following tables describe the attribute names/objects you can include in the TAM user store.

Create User Store Page Attribute Description

Attribute Name	Description
*User Store Id	Unique Id of TAM user store.

Attribute Name	Description
*User Store Name	Name of TAM user store.
Description	Description of TAM user store.
Implementation	Type of TAM user store implementation.
Connection Tab	
Connection	
*Urls	Refers to the TAM name/IP Address and port number to connect to, e.g. "LDAP://192.168.1.1:389". Note: To make use of AD Global Catalog, configure it as the primary LDAP server. The main AD Domain Service server subsequently can be configured as secondary backup server.
Use SSL	A flag to enable/disable Secure Sockets Layer (SSL) connection. Default: false
Authentication mechanism	Refers to the authentication mechanism/level used for authentication to the LDAP server. Default: Simple (Login Id + Password)
Login Id	Refers to the user Id of the LDAP credential with the host name, e.g. "WIN2003\demouser".
Password	Refers to the LDAP user's password.
Connection (Advanced)	
Use connection pool	Setting this to "true" will use the values configured in Global Setting > LDAP Connection Pool . Default: true
Connection timeout (seconds)	Specifies the maximum time (in seconds) to wait for connection results. "0" means there is no timeout. Value should be from 0 to 3600.
Read Timeout	Specifies the maximum time (in seconds) to wait for results on a read operation. "0" means there is no timeout. Value should be from 0 to 3600.
Use LDAP paging feature	Setting this to "true" will enable paging when searching (RFC 2696). LDAP paging allows retrieval of search result more than the query limit. The report may not work properly when this feature is disabled. Default: true Note: For older version of LDAP server, set this attribute to "false" (turn off) as this feature is not available.

Attribute Name	Description
LDAP Referral	Option specifying whether referrals to other LDAP servers are followed automatically. Available choices: "ignore" "follow" "throw"
User Attributes Mapping Tab	
This tab contains certain attributes that have been filled with default values for attributes mapping which is good enough to use, as well as other advanced attributes configuration.	
Search Filter	
*Search Context	Specifies the Base DN to use for user search. e.g. "CN=Users,DC=app-sg-isi,DC=i-sprint,DC=com". Click the Fetch Base DNs button to help fill the Search Context attribute value. Default: SECAUTHORITY
Basic Attributes	
*Attribute for UUID	User's UUID attribute. Default: ibm-entryUuid
*Attribute for DN	User's Distinguished Name (DN) attribute. Default: distinguishedName
*Attribute for id	User's Id attribute, e.g. "sAMAccountName". Note: Set value to "distinguishedName" if user Id is to be searched as user DN in the desired LDAP server. Otherwise, set to "email" if user Id is to be searched as email address in the desired LDAP server. Default: cn
*Attribute for name	User's name attribute, e.g. "name". Default: sn
Other attributes mapping	Refers to other user attribute's mapping. Format: <AM's attribute name>=<LDAP's attribute name>.
User	
*Search filter for user	Specifies the search filter for user, e.g. "(&(objectClass=user)!(&(objectClass=inetOrgPerson)))".
Object class for user	Refers to the object class for user. Default: inetOrgPerson
Attribute for email	User's email attribute, e.g. "mail".

Attribute Name	Description
Attribute for mobile	User's mobile attribute, e.g. "mobile".
Group	
*Search filter for group	Specifies the filter employed when doing a search for groups, e.g. "(objectClass=groupOfNames)".
*Object class for group	Refers to the object class for group. Default: groupOfNames
Attribute for group id	Attribute name of group id.
*Attribute for member	Attribute name of unique member, e.g. "member".
Attribute for member of	Attribute name for group membership, e.g. "memberOf".
Group of URLs (Dynamic Group)	
Object class for group of URLs	Refers to the object class for group of URLs.
Attribute for member URL	Attribute name of member URL.
Advanced Attributes Configuration	
Returning attributes	e.g. "ibm-entryUuid distinguishedName cn sn member objectClass".
Binary attributes	e.g. "objectGUID objectSid". Note: Specify the binary attributes if UUID attributes is in binary form.
Others	
Use relative DN to search member	Setting this to "true" will use relative DN to search member. Default: false
Segment/OU Tab	
This tab contains the attributes for mapping the segment to the organization unit (OU) of the LDAP-based server. There are certain attributes that have been defined with default values, which are good enough to use.	

Attribute Name	Description
Segment/OU store	By default, this is disabled, which means the user store will not be shown on the left pane with its segment hierarchy if the segment hierarchy license is enabled for the system. To enable it, set the value to "Enabled". Default: Disabled
Search filter for OU	Specifies the filter employed when doing a search for OU, e.g. "(objectClass=organizationalUnit)".
Object class for OU	Refers to OU's object class, e.g. "organizationalUnit". Default: organizationalUnit
Attribute for OU's id	Attribute name of OU Id, e.g. "ou".
Cache Tab	
Cache Configuration	
Maximum size	Maximum elements that the cache can contain. "0" means do not use cache. Value should not be less than 0. Default: 5000
Time to idle (seconds)	The maximum amount of time (in seconds) that an element can stay in the cache since last access. "0" means it will not expire due to idle time. Value should not be less than 0. Default: 1800
Time to live (seconds)	The maximum amount of time (in seconds) that an element can stay in the cache from creation time until the element expires. "0" means the element can stay forever. Value should not be less than 0. Default: 3600
Eviction policy	Used to decide which element is to be evicted from the cache when the cache is full. Default: Least Recently Used (LRU)

4.2.4. Manage Messaging & Response

Users are notified of events occurred such as change of password or request approval according to the Event Notification Policy configured by the administrator. Message Template and Delivery Channel are required to configure the Event Notification Policy. This section describes the management and configurations required.

Configure Message Template

The message template is a standard template that consists of predefined messages in it. Users can add their own messages to this template when sending. The message template is one of the two main components to be used for event notification policy. Upon successful setup of the AccessMatrix system, there are several predefined message templates that the administrator may utilize by applying relevant changes to the message templates, if necessary.

Types of Message Templates supported

Message Template Type	Description
Coordinate-based Template	This coordinate-based message template allows the administrator to specify the message body's contents based on the desired coordinated layout. It is mostly used for the PIN Mailer printing associated with SafeNet PHEFT E2EE PIN Mailer/PDF PIN Mailer.
Default Template Engine	This default template engine-based message template allows the administrator to specify the sender's and recipients' information, as well as message body's contents. It is mostly used for SMS and email.
HTML Content Template Engine	This message template allows the administrator to specify the sender's and recipients' information, as well as message body's contents. It also supports inline image/HTML content in the email's body.
Token PIN Mailer Coordinate-based Template	This token PIN Mailer coordinate-based message template allows the administrator to specify the message body's contents based on the desired coordinated layout, as well as the desired printer driver configuration. Note: This is used for UAS.



Note: The administrator may make use of the predefined entries with necessary changes or create new entries that cater to user requirements.

To navigate to the **Message Template** page, select **Administration Dashboard > Configuration > Messaging & Response > Message Template** on the left pane.

The following tasks can be performed on the Message Templates:

- **Create Message Template**

Click **Create** to display the **Choose Message Template Implementation** page. Select an implementation and click **Next** to open the **Create Message Template** page. Enter the details and click **Save**.

- **Search for Message Template**

Can be searched by message template Id or message template name. All available message templates are displayed by default.

- **View Message Template**

Click a message template Id under the **Search Message Template Result** list to view the details.

- **Edit Message Template**

Click **Edit** on the **View Message Template** page to modify the details and click **Save**.

- **Delete Message Template**

Click **Delete** on the **View Message Template** page and click **OK** in the confirmation pop-up box.

Message Template Page Attribute Description

All templates share the following attributes:

Attribute Name	Description
*Message Template Id	Unique Id of message template.
*Message Template Name	Name of message template.
Description	Description of message template.
Implementation	Type of message template implementation.

The detailed attributes differ according to the type of message template as discussed below.

Coordinate-based Template

Attribute Name	Description
Configuration Tab	
*Delivery Summary	A summary of the message.

Attribute Name	Description
*Body	Contains the message body's content. - To add a text for the Body attribute, specify the desired row and column values as well as the text in the input fields provided and then click on + icon. To remove it, click the x icon. Repeat the steps to add more texts. - After finished specifying the information for the message body, the layout of the message body can be reviewed by clicking on Preview. Click Close to close the window. Note: If a PIN Mailer template in PDF is uploaded, then clicking the Preview button will display the PIN Mailer PDF with coordinates. - PHEFT E2EE PIN Mailer accepts a maximum of 10 entries. However, there is no entry limit for the E2EE PDF PIN Mailer. To specify the PIN in the template for PDF Mailer, use "<<E2EEPIN>>".
Pin Mailer Template In PDF	Refers to the PIN Mailer template.
Encode HTML Tags	If set to "true", HTML encoding will be performed on the value of each macro. Enable this only when the intended message content is HTML. If you expect the macro to contain an HTML tag and it is meant to be rendered as such, set this to "false" so that the HTML tag will not be encoded. Ensure that you have validated all the values of the macros if they are entered by a user or untrusted party. Default: false
	Note: The administrator can add, edit or delete any custom attributes. Refer to Adding Custom Attribute section. Refer to Coordinate-based Template Macros for template macros.

Default Template Engine

Attribute Name	Description
Configuration Tab	
Sender and Recipients	
From	Sender of the message.
*To	Primary recipient(s) of the message.
Cc	List of recipient(s) under CC list.
Bcc	List of recipient(s) under BCC list.
Message Body	

Attribute Name	Description
Subject	Refers to the short summary of the message.
Body	Contains the message body's content.
Allow File Attachment	Enabling this feature will allow file to be attached in the message template. Default: false
Encode HTML Tags	If set to "true", HTML encoding will be performed on the value of each macro. Enable this only when the intended message content is HTML. If you expect the macro to contain an HTML tag and it is meant to be rendered as such, set this to "false" so that the HTML tag will not be encoded. Ensure that you have validated all the values of the macros if they are entered by a user or untrusted party. Default: false
i	Note: The administrator can add, edit or delete any custom attributes. Refer to Adding Custom Attribute section. Refer to Default Template Engine Macros for template macros.

HTML Content Template Engine

Attribute Name	Description
Configuration Tab	
Sender and Recipients	
From	Sender of the message.
*To	Primary recipient(s) of the message.
Cc	List of recipient(s) under CC list.
Bcc	List of recipient(s) under BCC list.
Message Body	
Subject	Refers to the short summary of the message.
Body	Contains the message body's content.
Allow File Attachment	Enabling this feature will allow file to be attached in the message template. Default: false

Attribute Name	Description
In-line Image Content-ID	This is to specify the Content-ID of the in-line image. This Content-ID must match with the "cid" specified in the HTML text of the Body attribute under the tag. E.g. specify the attribute as "inlinelImage1" for . Default: inlinelImage1
Encode HTML Tags	If set to "true", HTML encoding will be performed on the value of each macro. Enable this only when the intended message content is HTML. If you expect the macro to contain an HTML tag and it is meant to be rendered as such, set this to "false" so that the HTML tag will not be encoded. Ensure that you have validated all the values of the macros if they are entered by a user or untrusted party. Default: false
	Note: The administrator can add, edit or delete any custom attributes. Refer to Adding Custom Attribute section. Refer to Default Template Engine Macros for template macros.

Token PIN Mailer Coordinate-based Template

Attribute Name	Description
Configuration Tab	
*Delivery Summary	A summary of the message.
*Body	Contains the message body's content. - To add a text for the Body attribute, specify the desired row and column values as well as the text in the input fields provided and then click on + icon. To remove, click the x icon. Repeat the steps to add more texts. - After finished specifying the information for the message body, the layout of the message body can be reviewed by clicking on Preview. Click Close to close the window.
Printer Driver Type	Options: COM driver and Printer driver Default: COM driver
Printer Driver Configuration	
Note: These settings are necessary only if the Printer Driver Type is "Printer Driver".	
Font	Font to use for printing. Default: Arial Black

Attribute Name	Description
Style	Refers to the font style. Default: ITALIC
Size	Refers to the font size. Default: 20
Encode HTML Tags	If set to "true", HTML encoding will be performed on the value of each macro. Enable this only when the intended message content is HTML. If you expect the macro to contain an HTML tag and it is meant to be rendered as such, set this to "false" so that the HTML tag will not be encoded. Ensure that you have validated all the values of the macros if they are entered by a user or untrusted party. Default: false
i Note: The administrator can add, edit or delete any custom attributes. Refer to Adding Custom Attribute section. Refer to Token PIN Mailer Coordinate-based Template Macros for template macros.	

Configure Delivery Channel

The delivery channel delivers the OTP messages to the target recipient. It is the other main component to be used for event notification policy. Upon successful setup of the AccessMatrix system, there are several predefined delivery channels that the administrator may utilize by applying relevant changes to the delivery channels, if necessary.

Supported Delivery Channels

Delivery Channel	Description
Email Deliverer	This delivery channel makes use of SMTP server via SMTP protocol as a means to deliver the message in the form of an email. A pre-defined delivery channel called "DefaultEmailDeliveryChannel" will be created upon installation.
File Deliverer	This delivery channel saves the message as a file in the specified destination folder.
HTTP Deliverer	This delivery channel makes use of HTTP server via HTTP protocol as a means to deliver the message in the form of short message service (SMS). Two pre-defined delivery channels called "DefaultHTTPODeliveryChannel" and "DPMobileHTTPODeliveryChannel" will be created upon installation.
RawHttpDeliverer	This delivery channel supports SMS gateways that use XML-based HTTP POST to deliver the message in the form of short message service (SMS).

Delivery Channel	Description
SMPP Deliverer	This delivery channel makes use of SMPP server via TCP/IP protocol as a means to deliver the message in the form of short message service (SMS). A pre-defined delivery channel called "DefaultSMPPDeliveryChannel" will be created upon installation.
SNMP Deliverer	This delivery channel makes use of SNMP trap protocol to notify the management station of significant events by way of an unsolicited SNMP message. A pre-defined delivery channel called "DefaultSNMPDeliveryChannel" will be created upon installation.
 Note: The administrator may make use of the predefined entry with necessary changes or create new entries that cater to the user requirements.	

To locate the **Delivery Channel** page, navigate to **Administration Dashboard > Configuration > Messaging & Response > Delivery Channel** on the left pane.

The following tasks can be performed on the delivery channels:

- **Create Delivery Channel**
Click **Create** to display the **Create Delivery Channel** page, enter the details and click **Save**.
- **Search for Delivery Channel**
Can be searched by using the delivery channel Id or delivery channel Name. All available delivery channels are displayed by default.
- **View Delivery Channel**
Click a delivery channel Id under the **Search Delivery Channel Result** list to view the details.
- **Edit Delivery Channel**
Click **Edit** on the **View Delivery Channel** page to modify the details and click **Save**.
- **Delete Delivery Channel**
Click **Delete** on the **View Delivery Channel** page and click **OK** in the confirmation pop-up box.

Delivery Channel Page Attribute Description

All implementation types share the following attributes:

Attribute Name	Description
*Delivery Channel Id	Unique Id of delivery channel.

Attribute Name	Description
*Delivery Channel Name	Name of delivery channel.
Description	Description of delivery channel.
Implementation	Type of delivery channel implementation.

The detailed attributes differ according to the type of delivery channel as discussed below.

Email Deliverer

Attribute	Description
Configuration Tab	
SMTP Server Information	
*Host	The server name or IP Address of the SMTP server.
Port	Port of SMTP server. Default: 25
Secure Channel	Set to "true" to use SMTP SSL/TLS to connect to the server. Default: false
Enable STARTTLS	Set to "true" to enable STARTTLS for SMTP over SSL/TLS. Disable it if SMTPS is enabled. Default: true
Connection Timeout	Socket connection timeout value in seconds. "0" value sets an infinite timeout. Default: 0
Read Timeout	Socket read timeout value in seconds. "0" value sets an infinite timeout. Default: 0
Write Timeout	Socket write timeout value in seconds. "0" value sets an infinite timeout. Default: 0
Authentication Tab	
SMTP Server Authentication	

Attribute	Description
Does the server require authentication?	Setting to "true" will enable SMTP server authentication and will require username and password for authentication. Default: false
User Name	Refers to the SMTP username.
User Password	Refers to the SMTP password.
OAuth Authentication	
Use OAuth Authentication	To enable OAuth authentication, both this setting and "Does the server require authentication?" in the SMTP Server Authentication section must be enabled. Default: false
OAuth Token Endpoint	The token endpoint of the OAuth client.
OAuth Token Scope(s)	The token scope(s) of the OAuth client.
Client Id	The client id of the OAuth client.
Client Secret	The client secret of the OAuth client.
Others	
Print debug information	Setting to "true" will print the debug information to standard out. Default: false
Email Body Content Type	Specifies the email's body content type. Either "Text" or "HTML" content. Default: Text
Proxy Server Tab	
HTTP Forward Proxy	
Enabled	Setting to "true" will enable the HTTP Proxy. Default: false
Port	Refers to the HTTP Proxy port value. Note: Value should not be less than 0.
User Name	Username to be used for proxy authentication.
User Password	User's password to be used for proxy authentication.

Attribute	Description
NTLM Authentication	The name of the Windows domain that the user belongs to, used during NTLM proxy authentication.
Domain Name	Domain name of the proxy.
HTTP Reverse Proxy	
Replace OpenID Provider Token Endpoint Host Name	The host name to replace the token endpoint's original host in OpenID Provider (OIDC) communication.



Note: To see a more detailed set up for OAuth Authentication for SMTP Server in regards to Microsoft Online and Office 365, please check under OAuth Authentication for SMTP Server for Microsoft Online and Office 365.

- **OAuth Authentication under Client Credential flow for SMTP Server for Microsoft Online and Office 365**
 - Reference: <https://learn.microsoft.com/en-us/exchange/client-developer/legacy-protocols/how-to-authenticate-an-imap-pop-smtp-application-by-using-oauth>
 - After registering service principals in exchange, set Does the server require authentication, and OAuth Authentication to true.
 - Configure User Name, OAuth Token Endpoint, OAuth Token Scope(s), Client ID and Client Secret information, referencing the created App in Microsoft Entra under App Registrations.
 - For the token endpoint, it must be configured with <https://login.microsoftonline.com/<<your tenant ID>>/oauth2/v2.0/token> to support client credential flow.

File Deliverer

Attribute	Description
Configuration Tab	
Destination Folder	The folder that the file will be saved to. Notes: • Please ensure that the appropriate permissions have been granted for the AccessMatrix server to save the file to the destination folder. • Ensure that there is sufficient space within the destination folder. • Housekeeping of the saved files is not handled by the AccessMatrix server.

Attribute	Description
File Name Attribute	Specifies the file name attribute of the files to be handled by the File Deliverer. Possible values: "fileName" - The default file name attribute. Unless otherwise specified, use this attribute. "pdfFileName" - This attribute is used for PDF PIN Mailer PDF files. Refer to <i>UAS PDF Pin Mailer Implementation Guide - Configure PDF Pin Mailer Delivery Channel</i> for details on the PDF Pin Mailer.

HTTP Deliverer

Attribute	Description
Configuration Tab	
HTTP Information	
*URL	Refers to the URL of HTTP server.
Method	Refers to the HTTP request method. Possible values: "POST" "GET" Default: POST
Response code for successful scenario	Specifies the response code to indicate successful scenario. Default: 200
HTTP Authentication	
Use basic authentication	Setting to "true" will enable HTTP basic access authentication. Default: false
User Name	Username to be used for authentication.
User Password	User's password to be used for authentication.
HTTP Parameters	
Parameter name for recipient	Defines the HTTP parameter name for the recipient of this delivery channel. Note: The values of this parameter are taken from the attribute values of the To, Cc and, Bcc recipients of the Message Template.
Parameter name for sender	Defines the HTTP parameter name for the sender of this delivery channel. Note: The value of this parameter is based on the attribute value of From attribute of the Message Template.

Attribute	Description
Parameter name for message body	Defines the HTTP parameter name for the message body of this delivery channel. Note: The value of this parameter is based on the attribute value of body attribute of the Message Template.
HTTP parameter to attribute mapping	Allows the setting of multiple HTTP parameters which must be defined from Custom Attributes. The custom attributes must not be part of the predefined attributes used by the system. Each entry must in this format: <HTTP parameter name>=<AM's attribute name>, (e.g. "userId=attrUserId", where userId is the HTTP parameter name and attrUserId is the attribute name).
Other HTTP parameters mapping	Refers to other parameters mapping, where each entry is a parameter name. The value of these parameters is passed by the client application. Example client code using Java: <pre>Map<String, String> nParams = new HashMap<String, String>(); nParams.put("phoneNumber", "123456"); nParams.put("homeAddress", "ChaiChee"); generateOTP(sessionToken, "", "DefaultSmsOtpPolicy", new String[]{""}, nParams);</pre>
HTTP Proxy	
Enabled	Setting to "true" will enable the HTTP Proxy. Default: false
URL	Refers to the HTTP Proxy URL.
Port	Refers to the HTTP Proxy port value. Note: Value should not be less than 0.
User Name	Username to be used for proxy authentication.
User Password	User's password to be used for proxy authentication.
Others	
Verify Response Body	Setting to "true" will perform verification on the HTTP Response Body through regular expression matching. Define the regular expression in Response Body Regular Expression attribute. Default: false
Response Body Regular Expression	This is used to perform regular expression matching for HTTP Response Body verification based on the Java regular expression defined, e.g. "^[A-Z0-9].*", "\b01010\b", etc. Note: Only required if "Verify Response Body" is set to "true".
Audit response body	Refers to the audit response body. Maximum response length is 1KB (automatically truncated), including HTML tags, newline and space characters. Default: false

Attribute	Description
Response headers to audit	Names of the response headers to be audited.
Content type character set	Refers to the character encoding to use. Default: UTF-8
TLS Version	Specify the TLS version(s) to be used by the HttpDeliverer for HTTPS connection, e.g. TLSv1, TLSv1.1, TLSv1.2. Default: TLSv1, TLSv1.1, TLSv1.2
Connection Timeout	The maximum amount of time waiting to connect to URL. Note: Specified in milliseconds. Default: 0
Response Timeout	The maximum amount of time waiting to get a response from URL. Note: Specified in milliseconds. Default: 0



Note:

- After entering all the mandatory information, the administrator may click on the **Simulate With Message Template** button to simulate the message template. Select the desired message template for the simulation, and then click on **Simulate** to view the simulation result.
To allow using HTTP proxy without any authentication (proxy or basic authentication), set **Use basic authentication** to "false" and set the HTTP Proxy enabled with URL and port values but leave the **User Name** and **User Password** fields to empty values.
- The administrator can define other custom attributes. This is described in [Adding Custom Attribute](#).

RawHttpDeliverer

Define the SMS server and response message information in this section.

Attribute Name	Description
Configuration Tab	
SMS Server Information	
*SMS Gateway Host	Refers to the host name or IP Address of the SMS Gateway.

Attribute Name	Description
Method	Refers to the HTTP request method. Possible values: "POST" "GET" Default: POST
Request Content Type MIME type	Refers to the MIME type. Options: "application/json" "application/x-www-form-urlencoded" "application/xml" "application/soap+xml" "text/html" "text/xml" "text/plain" Default: text/xml
Request Content Type Charset	Refers to the character encoding to use. Default: UTF-8
SMS Response Message Information	
HTTP Response Verification Way	Specifies the method for detecting failures when sending a message. "HTTP Response Code" means it is based on the HTTP Response Code. "HTTP Response Header" means it is based on the existence of certain response header. "HTTP Response Body" means it is based on the occurrence of certain string in the response body. Options: "HTTP Response Code" "HTTP Response Header" "HTTP Response Body" Default: HTTP Response Body
HTTP Response Status Header Name	The HTTP header name which indicates the delivery status. Only required if "HTTP Response Header" is selected as HTTP Response Verification Way.
HTTP Response Status Header Value	The value of the HTTP Status Header. Only required if "HTTP Response Header" is selected as HTTP Response Verification Way.
Success Response Code	The response HTTP status code indicating successful sending. The attribute is used only if "HTTP Response Code" is selected as HTTP Response Verification Way.

Attribute Name	Description
Success Response Message	The text contained in the response body indicating successful sending. Only required if "HTTP Response Body" is selected as HTTP Response Verification Way. Example: "<hostReturnCode>0000</hostReturnCode>"
Request Headers	
Header to Attribute Mapping	Allows the setting of multiple HTTP headers which must be defined in Custom Attributes. The custom attributes must not be part of the predefined attributes used by the system. Each entry must be in this format: <HTTP parameter name>=<AM's attribute name> (e.g. "Authorization=authzHeaderAttr", where "Authorization" is the HTTP header name and "authzHeaderAttr" is the attribute name).
Others	
TLS Version	Specifies the TLS version(s) used by HttpDeliverer for HTTPS connection. e.g. "TLSv1, TLSv1.1, TLSv1.2" Default: TLSv1, TLSv1.1, TLSv1.2
Connection Timeout	The maximum amount of time waiting to connect to SMS Gateway Host. Note: Specified in milliseconds. Default: 0
Response Timeout	The maximum amount of time waiting to get a response from SMS Gateway Host. Note: Specified in milliseconds. Default: 0
Advanced	
Maximum Connections Total	The total number of connections allowed for a single port or URL. Minimum of 2 connections.
Certificate Verification	Enables/disables domain verification in CA certificates. Default: Enabled
	Note: The administrator can define other custom attributes. This is described in Adding Custom Attribute .

SMPP Deliverer

Attribute Name	Description
Configuration Tab	

Attribute Name	Description
SMS Center	
*Host	Refers to the SMPP server host name or IP Address.
*Port	Refers to the SMPP server port value.
*System id	Refers to the SMPP system Id.
*Password	Refers to the SMPP password.
Message Parameters	
Validity Period	Expiration time (in milliseconds) after which the message should be discarded if not delivered to destination. Default: 180000
Delivery Status	Request delivery status. Possible values: "Not Requested" "Requested" "Requested on failure" Default: Requested
Connection Parameters	
Session Timeout	The time duration when connection to the SMS center will remain open (in milliseconds) waiting for delivery status. Default: 300000
 Note: The administrator can define other custom attributes. This is described in Adding Custom Attribute .	

SNMP Deliverer

Attribute Name	Description
Configuration Tab	
SMS Center	
Host	Refers to the host or IP Address of the SNMP server. Default: localhost

Attribute Name	Description
Port	Refers to the SNMP server port value. Value should be from 1 to 65535. Default: 162
OID	Refers to the identifier of a network object. Default: 1.2.3.4.5
SNMP Version	Refers to the SNMP versions. Possible values: "2" - version2c "3" - version3 Default: 2
Security Level	This is required if SNMP Version is set to "3". Possible security levels: "NOAUTH_NOPRIV" "AUTH_NOPRIV" "AUTH_PRIV "
Context Name	This is required if Security Level is set to "AUTH_NOPRIV" or "AUTH_PRIV".
Security Name	This is required if Security Level is set to "AUTH_NOPRIV" or "AUTH_PRIV".
Authentication Passphrase	This is required if Security Level is set to "AUTH_NOPRIV" or "AUTH_PRIV".
Authentication Passphrase Encryption Protocol	Encryption protocol for authentication passphrase. Possible values: "MD5" "SHA-1"
Privacy Passphrase	This is required if Security Level is set to "AUTH_PRIV".
Privacy Passphrase Encryption Protocol	Encryption protocol for privacy passphrase. Possible values: "MD5" "SHA-1"



Note: The administrator can define other custom attributes. This is described in [Adding Custom Attribute](#).

Configure Event Notification Policy

Event notification policy administration completes the entire process of message administration after administering both message templates and delivery channels. Each event notification policy contains the message template and delivery channel together with the desired event types defined to form a complete policy in the system.

i **Note:** System pre-defined Events Notification Policies are available for use for administrator's convenience, however, some policies required to be activated as they are disabled by default upon successful installation of AccessMatrix.

To locate the **Event Notification Policy** page, select **Administration Dashboard > Configuration > Messaging & Response > Event Notification Policy** on the left pane.

The following tasks can be performed on an event notification policy:

- **Create Event Notification Policy**

Click **Create** to display the **Create Event Notification Policy** page, enter the details and click **Save**.

- **Search for Event Notification Policy**

Can be searched by using an event notification policy Id or an event notification policy name. All available event notification policies are displayed by default.

- **View Event Notification Policy**

Click an event notification policy Id under the **Search Event Notification Policy Result** list to view the details.

- **Edit Event Notification Policy**

Click **Edit** on the **View Event Notification Policy** page to modify the details and click **Save**.

- **Delete Event Notification Policy**

Click **Delete** on the **View Event Notification Policy** page and click **OK** in the confirmation pop-up box.

View Event Notification Policy Page Attribute Description

Attribute Name	Description
*Event Notification Policy Id	Unique Id of event notification policy.
*Event Notification Policy Name	Name of event notification policy.
Version	Event notification policy entry version.
Description	Description of event notification policy.
Status	Defines if the status is "Activated" or "Disabled". "Activated" means it will listen based on the event types and send the notification automatically. "Disabled" means it will not listen based on the event types. Instead the only way to invoke this event notification is by specifying this event notification specifically in other module's configuration.
Attributes Tab	
Event Filter	
Event Types	Type of triggering event. Refer to Appendix B. Event Category and Types for more information.
First Template and Delivery Channel	
Message Template	Associated message template: DefaultTokenInventoryTemplate
Delivery Channel	Associated delivery channel: DefaultEmailDeliveryChannel
Second Template and Delivery Channel	
Message Template	Associated message template.
Delivery Channel	Associated delivery channel.
Third Template and Delivery Channel	

Attribute Name	Description
Message Template	Associated message template.
Delivery Channel	Associated delivery channel.
Note: Three different message templates and delivery channels can be configured to serve different purposes.	

Configure User Response

The administrator can configure the AccessMatrix system to return certain user-specific information upon successful completion of the use cases that make use of the user responses. Some frequently used cases are illustrated below:

- For successful user authentication, a JSON object containing user-related attributes can be returned by using the user response. Object names can be specified with **Name (Claim Name)**.
- Depending on the SSO flow, user information can be returned by using the user response. For example, this can be returned in the form of OIDC token claims, or assertion attribute names in SAML.

Note: For more illustrations and specific configurations on using the user response in SSO use cases, refer to the [UAM Administration Guide - User Response Configuration](#).

To locate the **User Response** page, navigate to **Administration Dashboard > Configuration > Messaging & Response > User Response**. The following tasks can be performed on the **User Response** page:

- Create User Response**
Click **Create** and select "User Information Response Collector" from the **Implementation** drop-down list, and then click **Next** to display the **Create User Response page**. Enter the details and save.
- Search User Response**
Can be searched by entering a user response Id or a user response name in the empty fields,

and click **Search**. All available user responses are displayed by default on the **Search User Response Result** page.

- **View User Response**

Click a user response Id under the **Search User Response Result** page to view the details.

- **Edit User Response**

Click **Edit** on the **View User Response** page to modify the details and save.

- **Delete User Response**

Click **Delete** on the **View User Response** page and select **OK** on the **Confirm** dialog box.

 **Notes:**

- The search can be based on a wildcard (*) or exact phrase match and it is case-sensitive.
e.g. If "DEMO*" is entered for a search field, the entry with an Id of "DEMO123" will be retrieved, however, not "123DEMO". Enter "*DEMO" to retrieve "123DEMO", or leave the search field blank to search for all available results.
- The search criterion is also based on 'AND' condition.
e.g. If the administrator enters values for both the **Id** and **Name** fields, then only the results matching both values will be retrieved.

View User Response Page Attribute Description

 **Note:**

' * ' denotes mandatory attribute.

Attribute Name	Description
*User Response Id	Unique Id of user response.
*User Response Name	Name of user response.
Description	Description of user response.
Version	User response entry version.
Implementation	Type of user response implementation.
User Tab	
Basic User Fields and User Info	

Attribute Name	Description
Mapping of basic user fields and user info to the return name of user response.	
Name (Claim Name)	The return name of the user response, or the claim of the scope in OIDC.
Attribute Name	The basic user field or user info attribute to be retrieved. This attribute can be retrieved from a user, session, WEB SSO's Access Agent and User, or OIDC token and user. For a list of supported attributes, refer to Basic User Field and User Info Attributes .
Delimiter	Sequence of characters used to separate individual values in the returned data. The delimiter can be any character, e.g. "," or ":-:". For example, if "," is specified as the delimiter, the data will be returned as "VALUE1,VALUE2,VALUE3...".
Pre-Defined User Attributes / User Custom Attributes	
Mapping of system pre-defined user attributes to the return name of user response.	
Name (Claim Name)	The return name of the user response, or the claim of the scope in OIDC.
Attribute Name	The pre-defined user attribute to be retrieved. This attribute can be retrieved from user, session, WEB SSO's Access Agent and User, or OIDC token and user. For a list of supported attributes, refer to Pre-Defined User Attributes / User Custom Attributes .
Delimiter	Sequence of characters used to separate individual values in the returned data. The delimiter can be any character, e.g. "," or ":-:". For example, if "," is specified as the delimiter, the data will be returned as "VALUE1,VALUE2,VALUE3...".
User Custom Attributes (Custom Plug-In Class)	
Retrieve response values through a custom implementation class. The custom implementation class contains a CustomResponseCollector interface that collects the specified data. The custom class will be loaded during AccessMatrix runtime, where the response value can be subsequently returned by using the getResponseString method. It can be deployed within the <ACMATRIX_ROOT>/tomcat/webapps/am5/WEB-INF/classes/ or <ACMATRIX_ROOT>/tomcat/webapps/am5/WEB-INF/lib/ file paths.	
Name (Claim Name)	The return name of the user response, or the claim of the scope in OIDC.

Attribute Name	Description
Class Name	Custom implementation class name.
Delimiter	Sequence of characters used to separate individual values in the returned data. The delimiter can be any character, e.g. "," or ":-:". For example, if "," is specified as the delimiter, the data will be returned as "VALUE1,VALUE2,VALUE3...".
Literal	
Mapping of hard-coded values to the return name of user response.	
Name (Claim Name)	The return name of the user response, or the claim of the scope in OIDC.
Hard Coded Value	Attribute that allows any specified value to be returned in the user response.
Delimiter	Sequence of characters used to separate individual values in the returned data. The delimiter can be any character, e.g. "," or ":-:". For example, if "," is specified as the delimiter, the data will be returned as "VALUE1,VALUE2,VALUE3...".

Attribute mappings can be configured as necessary:

Note: Once the changes have been confirmed, click on the **Save** button.

- To add a new attribute mapping, complete the necessary fields in the corresponding row and click the **Add** button.
- To remove a mapped attribute, click on the **X** button in the corresponding row.

For each mapped attribute to be returned in the User Response, either of the following must be adhered to when sending a request:

- The name of the mapped attribute must be used as the variable/parameter name in the request header.
- The claim name of the mapped attribute must be specified accordingly.

Basic User Fields and User Info Attributes

Under the **Basic User Fields and User Info Attributes** section, select an attribute using the drop-down list corresponding to the **Attribute Name**.

Refer to the following for a list of attribute names:

Attribute Name	Attribute Value Source	Description
ActivatedTokenInstance	User	The active token instance for a user.
ApplicationRoles	User	The application roles owned or inherited by a user.
AuthenticatedRealmId	User Session	The user's authenticated realm Id based on the user's session.
GroupMembership	User	The group's membership of a user.
LastAuthenticatedLoginModule	User Session	The last authenticated login module of a user.
LastFailedLoginDate	User	The user's last failed login date based on the user's most recent login events or history. Note: This information may not be accurate if retrieved right after the user's authentication since the user's login history is saved asynchronously when the user is authenticated.
LastSuccessfulLoginDate	User	The user's last successful login date based on the user's most recent login events or history. Note: This information may not be accurate if retrieved right after the user's authentication since the user's login history is saved asynchronously when the user is authenticated.
LoginAccounts	User	The login accounts of the users
SecondaryAccountID	User & Federated SSO Application or UAM Application	The secondary account identity of a user.
SecondaryAccounts	User & Federated SSO Application or UAM Application	The secondary account information of a user, filtered by external app UUID. Selecting this enables the return of the following information for all of the user's secondary accounts associated with a specific user UUID and external app UUID:

Attribute Name	Attribute Value Source	Description
		• Profile name - The name of the profile, e.g. "Profile 1". • External application Id - The Id of the client application, e.g. "Client1". • Application Object Class - The application object class, e.g. "OAuthClient", "SAMLServiceProvider" or "UserApplication". ○ For OIDC RP application, the possible values are: "Lookup Module", "Login Module" • Id - The Id of the profile, e.g. "Profile1".
SecondaryAccountsByAppObjClassAndUser	User & Federated SSO Application or UAM Application	The secondary account information of a user, filtered by app object class. Selecting this enables the return of the following information for all of the secondary accounts associated with a specific user UUID and app object class: • Profile name - The name of the profile, e.g. "Profile 1". • External application Id - The Id of the client application, e.g. "Client1". • Application Object Class - The application object class, e.g. "OAuthClient", "SAMLServiceProvider" or "UserApplication". ○ For OIDC RP application, the possible values are the "Lookup Module", "Login Module". • Id - The Id of the profile, e.g. "Profile1".
SessionIdleTimeoutInSeconds	Global Setting	The user's idle session timeout in seconds. (Global Setting)

Attribute Name	Attribute Value Source	Description
SessionLifeTimeoutInSeconds	Global Setting	The user's life or active session timeout in seconds. (Global Setting)
TokenModel	User	The model of a token.
TokenSerialNumber	User	The serial number of a token.
TokenStatus	User	The status of a token.
TokenVendor	User	The vendor of a token.
UserDN	User	The Distinguished Name of a user.
UserDescription	User	The description of a user.
UserID	User	The identity of a user.
UserName	User	The name of a user.
UserSegmentId	User	The segment Identity of a user.
UserSegmentName	User	The segment name of a user.
UserSegmentUUID	User	The segment UUID (Universally Unique Identifier) of a user.
UserSessionExpiryTime	User Session	The user's session expiry time.
UserUUID	User	The user's UUID (Universally Unique Identifier).

i **Note:** Refer to [UAM Administration Guide](#) for the UAM-related user response.

Pre-Defined User Attributes / User Custom Attributes

Under the **Pre-Defined User Attributes / User Custom Attributes** section, select an attribute using the drop-down list corresponding to the **Attribute Name**.

Refer to the following table for a list of supported attributes:

i **Note:** The following list is not exhaustive. It includes only some identical attributes with different retrieval patterns.

Attribute Name	Type	Description
Custom	String	If selected, specify the name of the user's custom attribute. This attribute will be returned in the user response. Note: For more information on how to configure custom attributes for users, refer to AccessMatrix Common Administration Guide - User Management - View User Page Attribute Description.
email	String	End-User's preferred email address. Its value must conform to the RFC 5322 [RFC5322] addr-spec syntax.
externalStatus	String	End-User's external status.
mobile	String	End-User's preferred telephone number. E.164 [E.164] is RECOMMENDED as the format of this Claim, for example, +1 (425) 555-1212 or +56 (2) 687 2400. If the phone number contains an extension, it is RECOMMENDED that the extension be represented using the RFC 3966 [RFC3966] extension syntax, for example, +1 (604) 555-1234;ext=5678.

4.2.5. System Configuration

Certain System and Server settings can be configured to accommodate the organization's requirements.

Configure Global Setting

This feature allows the administrator to configure certain system settings related to primary key management, user login session, etc.

These settings are applicable globally throughout the entire system. The administrator is not required to modify the settings unless necessary. The administrator is only allowed to view and edit the global setting entry.

To locate the **Global Setting** page, navigate to **Administration Dashboard > Configuration > Global Setting**. Global Settings can be edited by clicking **Edit** on the **View Global Setting** page.



Note: Administrator will not be able to edit the global settings if there is a pending/rejected admin request pertaining to that Global Setting entry.

Global Setting Page Attribute Description

Attribute Name	Description
Primary Key Management Tab	
Key Length (Base64)	
Prefix key with server id integer	Key length for Prefix key. If set to "true", prefix key with hex-encoded server id integer (0-3F). This can be important to avoid uniqueness conflict in database replication. Default: False
User, Credential and 2FA Token objects	Key length for User, Credential and 2FA Token objects. - Set to "10" or "12" if you have more than 1 million users or tokens. - Use "16" if more than 5 million. - Use "22" only if you require a globally unique key. Default: 10
Group & Segment objects	Key length for Group & Segment objects. Set to above "8" only if you have more than 50,000 groups or segments. Default: 8

Attribute Name	Description
USO & UAM Application & Token Group objects	Key length for USO & UAM Application & Token Group objects. Default: 4
Authentication Configuration Objects	Key length for Authentication Configuration objects. Make sure you create your database with case sensitivity on. Chance of collision of random mix-case alphanumeric is about 2/1k for 3 characters, 5/10k for 4 characters, 6/100k for 5 characters. Default: 2
Default for other objects	Key length for all other objects. Default: 5
User ID Format	
Min User/Group Id Length	Minimum number of characters allowed for a single user/group Id. Value should be from 1 to 32. Default: 2
Max User/Group Id Length	Maximum number of characters allowed for a single user/group Id. Value should be from 8 to 64. Default: 20
User/Group Id Charset Restriction	Type of characters allowed for a user/group Id. The range of allowed characters should be specified as a regular expression. It is strongly recommended that you only allow the use of ASCII characters. Default: [a-zA-Z0-9]+
Segment Id Charset Restriction	Type of characters allowed for a segment Id. The range of allowed characters should be specified as a regular expression. It is strongly recommended that you only allow the use of ASCII characters. Default: [a-zA-Z0-9._]+
Module Implementation Tab	
Login Module Implementation	
This tab contains the list of user-defined login module implementations. If necessary, administrator may click on Reload Implementation to reload all the login module implementations. A "Registered handlers have been reloaded at the server." information message will be displayed upon successful reload.	

Attribute Name	Description
Session Tab	
Session	
Session cache size	Specifies the size of the session cache. Value should be from 5000 to 200000. Default: 20000
Invalidated Session cache size	Specifies the invalidated session cache size. Default: 20000
Session Idle Time (in seconds)	Specifies the maximum session idle time. This is the validity of the user logged-in session when the user is idle or has no activity. In terms of API, this is the validity of the sessionToken. sessionToken has last access time and idle timeout is calculated based on this last access time. A refresh session API can be called to generate a new sessionToken to extend the session (It has an updated last access time). Value should not be less than 120. Default: 600
Session Lifetime (in seconds)	Specifies the maximum session lifetime. This is the maximum validity of the user logged-in session regardless of idleness/activity. In terms of API, this is the maximum validity of sessionToken regardless of last access time. The validity checking is calculated based on the logged-in time, instead of last access time. Value should be from 120 to 864000. Default: 28800
E2EE Session Lifetime (in seconds)	Specifies the maximum E2EE session lifetime. This refers to the validity of E2EE session. As part of the E2EE authentication process, E2EE session is generated before the user enters an E2EE credential. If E2EE session has expired or invalidated, the E2EE authentication will fail although the user has entered a correct E2EE credential. E2EE session is meant to be one-time use only. After E2EE authentication is done, the E2EE session is invalidated. This E2EE session has nothing to do with the user login session (sessionToken). Note: Require AM server restart for changes to take effect. Value should be from 300 to 3600. Default: 600
Others	

Attribute Name	Description
Display legal banner in login window	Specifies whether to display the legal banner in the login window. If "true", <AM5 Web Application>\WEB-INF\classes\legalbanner.html will be loaded. Default: true
Concurrent Session Policy	
• Every authenticated AccessMatrix user will have a session token. If a user authenticates more than once before the existing session expires, then this user will have concurrent sessions. AccessMatrix handles concurrent sessions in various ways by using concurrent session policy. • AccessMatrix session can be set as a client state token (i.e. no persistence) or it can be persisted to the database. If the policy is set to use non-persistent sessions, then one user can have unlimited sessions. However, if the policy is set to use persistent sessions, then a policy to set the maximum number of concurrent logins per user should be defined. A user may only have one or up to 10 concurrent logins.	
Session Concurrency	Specifies whether to allow concurrent logins. Allow concurrent logins do not persist session in the database; thus, it allows much faster performance compared to other options. Limiting concurrent logins per user will persist session in the database for session tracking. Options: • "Allow concurrent logins" - A user may log in concurrently with many active sessions without restriction. Option when reaching max. number of concurrent logins (per user) • "Reject new login session" - A user may log in concurrently with a specified maximum of active sessions (refer to Max. number of concurrent login's value), after which the new session will be rejected and the user will not be allowed to log in. • "Terminate user's oldest session to allow the new one" - A user may log in concurrently with a specified maximum of active sessions, after which a new login will terminate the oldest active session so that the total number of concurrent sessions do not exceed the specified maximum value (refer to Max. number of concurrent login's value). • "Ask user's confirmation to terminate the oldest session to allow the new one" - A user may log in concurrently with a specified maximum of active sessions (refer to Max. number of concurrent login's value), after which a new login will prompt

Attribute Name	Description
	the user to terminate the oldest session to allow the new one. ○ If the user accepts, the server will terminate the oldest active session so that the total number of concurrent sessions do not exceed the specified maximum. ○ If the user rejects, the login will terminate without success immediately and the existing concurrent sessions will not be affected.
LDAP Connection Pool Tab	
ldap.connect.pool.initsize	The number of connections per connection identity to create for initial connection creation for the identity. Initial size. Default: 1
ldap.connect.pool.maxsize	The maximum number of connections per connection identity that can be maintained concurrently. Maximum size. Default: 10
ldap.connect.pool.prefsiz	The preferred number of connections per connection identity that should be maintained concurrently. Preferred size. Default: 5
ldap.connect.pool.timeout	The number of milliseconds that an idle connection may remain in the pool without being closed and removed from the pool. Idle connection timeout (in milliseconds). Default: 1800000.
ldap.connect.pool.authentication	A list of space-separated authentication types of connections that may be pooled. Valid types are "none", "simple", and "DIGEST-MD5". Default: None and simple.
ldap.connect.pool.protocol	A list of space-separated protocol types of connections that may be pooled. Valid types are "plain" and "ssl". Default: Plain.
ldap.connect.pool.debug	A string that indicates the level of debug output to produce.
Token Management Tab	

Attribute Name	Description
This tab contains attributes related to token management. This is specific to UAS product and will only be displayed provided the user has subscribed to the UAS features for the AccessMatrix license key. The associated attributes will be discussed in details in the UAS Administration Guide .	
External URLs Tab	
Menu Title	Name of Menu Title for the External URL Tab. Default: External Links.
Features parameter for opening browser window	Browser window size specifications. Default: Height=480, width=640.
Logout Succeeded URL	Logout URL.
Add URL	External URL to connect to.

Configure Server

Server administration is used to set up new AccessMatrix system, which covers managing AccessMatrix server information and server licensing. The administrator can find out AM features granted based on the server license purchased.


Notes:

- To set up more than one AccessMatrix system (i.e. having more than one AccessMatrix server) which are accessing a centralized registry, the administrator is required to create new server entry for every AccessMatrix server instance. This can be done via one of the AccessMatrix systems that have been granted access to the centralized registry.
- Administrator can configure the current server instance or a new server instance.

To locate the **Server** page, select **Administration Dashboard > System > Server & Cache > Server** on the left pane. The following tasks can be performed on the servers:

- **Create Server**

Click **Create** to display the **Create Server** page, enter the details and click **Save**.

- **Search for Server**

Can be searched by using a server Id or a server name. All available servers are displayed by default.

- **View Server**

Click a server Id under the **Search Server Result** list to view the details.

- **Edit Server**

Click **Edit** on the **View Server** page to modify the details and click **Save**.

- **Delete Server**

Click **Delete** on the **View Server** page and click **OK** in the confirmation pop-up box.

View Server Page Attribute Description

Attribute Name	Description
*Server Id	Unique Id of server.
*Server Name	Name of server.
Description	Description of the server.
Version	Server entry version.
*IP address	IP address of the server.
Id Integer	System generated server Id in integer.
MAC Address	MAC address of the Server.
Violation Message	Displays the error message for any license error/violation encountered.
Primary Radius Configuration Tab	
Enable radius server	Setting this to "true" will enable AM5 to serve as a RADIUS server. Default: true
Primary Radius server port	The AccessMatrix RADIUS server port. Value should be from 1 to 65535. Default: 1812
Replication Tab	
Cache replication port	Cache replication port is used for replication and notification of changes between AM5 servers. Value should be from 1024 to 65535. Default: 40000

Attribute Name	Description	
Remote object port	The port number on which the remote objects bound in the registry receive calls. Value should be from 1024 to 65535. Default: 40001	
Licensed Features Tab		
License Module	Feature	Description
CORE FEATURE	Customer Id	Refers to the recipient of the issued license.
	Remark	Specifies a description of the license.
	Host	Refers to the IP Address of the host machine where AccessMatrix server is running and where the license will be applied.
	MAC Address	Refers to the MAC Address of the host machine where AccessMatrix server is running and where the license will be applied.
	Check IP/MAC Addr	Setting to "Yes" will check either the Host or MAC Address value specified in the license. Note: This is set to "No" only by special requests from customers who run AccessMatrix server in a cloud environment.
	Max Users	This value refers to allowable user counts in a system and the checking is based on the conditions as follows: - This user count refers to the login users of the system, not the total users created or the total users existing in the various user stores, as it is not feasible or possible to determine total users in the external user stores. Max Users checking is based on a figure derived from the estimation of distinct active login users during the last 3 months. - If this derived figure exceeds Max Users specified in the license, the License violation prompt is triggered where: - AccessMatrix Admin Console will only present a restricted menu until a new valid license is applied; - Normal user functions like user login and all API functions will continue to work properly as usual.

Attribute Name	Description	
		Note: If the value is set to "0", then there is no limit with the number of users.
	Start Date	Refers to the validity start date of the license.
	End Date	Refers to the validity end date of the license.
	Dual Control	Setting to "Yes" will enable dual control feature.
	Segment Hierarchy (Internal Default Store)	Setting to "Yes" will display segment hierarchy for the internal default store.
	Segment Hierarchy (External Segment Stores)	Setting to "Yes" will enable will display segment hierarchy for the external segment store.
UAS FEATURE	Token Vendors	Refers to the token vendors that will be supported. Note: Only those selected token vendors in license generator will be displayed.
	VASCO Digipass for Web	Setting to "Yes" will enable support for VASCO Digipass for Web.
	VASCO Virtual Digipass	Setting to "Yes" will enable support for VASCO Virtual Digipass.
	VASCO EMV-CAP	Setting to "Yes" will enable support for VASCO EMV-CAP.
	VASCO Matrix Card	Setting to "Yes" will enable support for VASCO Matrix Card.
	VASCO Double DPX	Setting to "Yes" will enable support for VASCO Double DPX.
	VASCO OTP Verification in HSM	Setting to "Yes" will enable support for VASCO OTP Verification in HSM.
	Radius PAM	Sets the maximum number of Radius PAM that can be created.
	Max Radius Clients (0 means disabled)	Sets the maximum number of radius clients. Note: If the value is "0", then Radius feature will be disabled.
	Max Token Batches	Sets the maximum number of token batches. Note: If the value is "0", then no token batch can be created.

Attribute Name	Description	
	Max Token Groups	Sets the maximum number of token groups. Note: If the value is "0", then no token group can be created.
	E2EEA Base	Setting to "Yes" will enable the E2EE Base feature.
	E2EEA Pin Mailer	Setting to "Yes" will enable the E2EEA Pin Mailer feature.
	E2EEA Remote Printing	Setting to "Yes" will enable the E2EEA Remote Printing feature.
	E2EE Data Protection	Setting to "Yes" will enable the E2EE Data Protection feature.
	Biometric Login	Refers to the biometric vendor(s) that will be supported.
	Biometric Portal	Setting to "Yes" will enable the Biometric Portal feature.
	Enable Token Usage Billing	Setting to "Yes" will enable the Token Usage Billing feature.
	Contextual Authentication	Setting to "Yes" will enable the Contextual Authentication feature.
	Max users per login module handler	It displays the maximum number of users that can be assigned to specific login modules. The Remarks column displays the count of users at a specific day/time per login module. For example, if the users' count assigned to System Password Basic reached 85% of 100000 counts, a warning will be displayed when you log in or check in this licensed tab. If the assignment exceeds 100%, there will be a restriction imposed against assigning new users to the affected login module.
	Max assigned tokens per vendor/model	It displays the maximum number of assigned tokens based on specific token vendor and model combination. The Remarks column displays the count of assigned tokens at a specific day/time per token vendor and model. There will be a warning message if assigned tokens exceed 85%(non-configurable) of the license. If the assigned tokens exceeds 100%, there will be a restriction imposed against assigning the affected token model.
USO FEATURE	Server	Setting to "Yes" will enable the USO server feature.
	Max USO Applications	Sets the maximum number of USO Applications that can be created.

Attribute Name	Description	
		Note: If the value is "0", then there is no limit of USO applications that can be created.
UCM FEATURE	PSM	Set to "Yes" will enable Privileged Session and Activities Manager (PSM) feature. Default: No
	Max connectors	Sets the maximum number of connectors that can be created per connector type.
	Max Hosts	Sets the maximum number of unique hostname/IP Address that is used during the creation of connectors.
	Enable these advanced connectors:	Allows support of the advance connectors. Note: Only those selected advance connectors in license generator will be displayed.
	Reporting Hierarchy Based Approval	Setting to "Yes" will enable Reporting Hierarchy Based Approval feature.
	Password Consumer	Setting to "Yes" will enable Password Consumer feature.
	Max UCM Gateway Clients	Sets the maximum UCM Gateway clients.
UAM FEATURE	Max User Applications	Sets the maximum user applications. Note: If the value is "0", then there is no limit of user applications that can be created.
	Application Type	Specifies the application type supported.
	OAuth Client	Sets the maximum number of OAuth Client. Note: If the value is "0", then there is no limit of OAuth clients that can be created.
	SAML: Service Provider	Sets the maximum number of SAML Server Provider. Note: If the value is "0", then there is no limit of service providers that can be created.
	Social Network Login	Specifies the social networks supported.
UIM FEATURE	Max target resources	Sets the maximum UIM resources that can be created regardless of the implementation type.

Attribute Name	Description	
		Note: If the value is "0", then no target resource can be created.
LICENSE VIOLATIONS		Display the license violation message, if there is any.



Notes:

- The **Primary Radius Configuration** tab is used to enable AM5 to be a RADIUS server to serve RADIUS authentication. RADIUS is normally used by VPN or network appliances.
- The **Replication** tab is used to configure replication and object ports for AM5 cache replication. It requires a restart of all server instances for changes to take effect.
- The **Licensed Features** tab is for License upload and viewing. An evaluation license of 1 month is issued during AM5 installation but with limited features. If a valid license is issued, refer to [Load AccessMatrix Server License](#). It requires the user to log out and log in again in order for a newly uploaded license to take effect.
- The **Licensed Features** values are entered and selected in the license generator. Values cannot be modified.

Configure Common Login Page (CLP)

Common Login Page (CLP) is a client authentication component that serves as a common login page for AccessMatrix Universal Sign-On (USO) application and others. The administrator can configure the current server instance or a new server instance.

To navigate to **Common Login Page** page, select **Administration Dashboard > Configuration > Common Login Page**. **Common Login Page** can be edited by clicking **Edit** on the **View Configuration** page on the right pane.

View Configuration Page Attribute Description

Attribute Name	Description
*Configuration Id	Unique Id of Common Login Page.
*Configuration Name	Name of Common Login Page configuration.
Description	Description of Common Login Page.

Attribute Name	Description
Version	Common Login Page entry version.
Implementation	Type of Common Login Page implementation.
Attributes Tab	
Client Type and Product Target URL	Refers to the application/client type and the target SSO URL. Note: The following are already pre-defined: • USO Server, /usoserver/start.uso • USO SSF, /usossf/start.ssf • UCM URL values, /ucmweb.index.jsp • TAP (Token Activation Portal), /uastap/clp.jsp
Token Validity Period (in seconds)	This indicates how much time the user has to complete the login process. The difference between the time the login request is issued and the time the login request is received must be within this configured time period. This is only valid for login request via AccessMatrix components, such as USO Server, USO SSF, and UAS TAP. Value should not be less than 300. Default: 360
Cache	
Session Nonce Cache Size	The number of session nonce to be kept in the cache. The cache is used to verify the login session. In general, there is no need to modify this unless it is expected that there are many users try to log in at the same time to AccessMatrix components, such as USO Server, USO SSF, and UAS TAP. Value should not be less than 1000. Default: 5000

4.2.6. Key Management

AccessMatrix Key Management is used to manage the encryption keys used in the AccessMatrix system, including but not limited to:

- Attribute Encryption Key (AEK)
- Session Encryption Key (SEK)
- Token Encryption Key (TEK)
- Pin Encryption Key (PEK)
- URL Encryption Key (UEK)

AccessMatrix allows some keys (Encryption Keys) to be created using various cryptographic system (Key Providers), such as the Hardware Security Module (HSM). It allows, for example, that Token Encryption Key (TEK) is using HSM-stored key where the cryptographic operation is done within the HSM.

AccessMatrix Key Management manages the key providers and encryption keys in a cryptosystem. Key Provider is a key manager that defines different implementations of cryptography key provisioning.

Generating a cryptographic key in the AccessMatrix system, a key provider needs to be created first to indicate where and how the key would be created. The key providers can be implemented in hardware such as the Hardware Security Module (HSM), software, or both. Each implementation includes the configuration of how and where the cryptography key(s) are created, stored and used. Also, it includes the configuration for an HSM (vendor-specific), or if the cryptography standard is explicitly used by specific country (e.g. SM cryptography for China) with or without an HSM.

Encryption Key is used to represent the key itself, which may be software-based or HSM-based, depending on the Key Provider. AccessMatrix Key Management also supports a wrapping key to provide key hierarchy. For example, an Attribute Encryption Key (AEK) is a software-based key that allows for fast cryptographic operations within the AccessMatrix process. However, the AEK can still be wrapped by a wrapping Encryption Key created using an HSM-based Key Provider (this wrapping key resides in HSM) to protect the AEK while in storage securely.

Configure Key Provider

To locate the **Key Provider** page, navigate to **Administration Dashboard > Security Policy > Key Management > Key Manager > Key Provider** on the left pane. The following tasks can be performed on the key providers:

- **Create Key Provider**

Click **Create** to display the **Choose Key Provider Implementation** page. Select an implementation and click **Next** to open the **Create Key Provider page**. Enter the details and click **Save**.

- **Search for Key Provider**

Can be searched by using the key provider Id or the key provider name. All available key providers are displayed by default.

- **View Key Provider**

Click a key provider Id under the **Search Key Provider Result** list to view the details.

- **Edit Key Provider**

Click **Edit** on the **View Key Provider** page to modify the details and click **Save**.

Types of Key Providers

Key Provider	Description
System Key Provider	This is the default key provider that does not rely on external software or hardware such as HSM. System Key Provider contains the configuration for active Attribute Encryption Key (AEK) and controls the Session Encryption Key (SEK) rollover process. Upon successful setup of the AccessMatrix system, a System Key Provider called "SystemKeyProvider" will be created by default. There can be only one entry of the System Key Provider throughout the entire AccessMatrix System.
AWS KMS Key Provider	This key provider uses AWS Key Management Service (KMS) to create and store symmetric keys and perform cryptographic operations. It uses the Customer Master Keys (CMKs) that are created and stored inside AWS KMS to protect your keys. You can create one or multiple entries of AWS KMS Key Provider throughout the entire AccessMatrix System. Note: This option will be displayed for selection after you deploy AWS KMS Key Provider Plugin in the AccessMatrix system.
Azure Key Vault Key Provider	This key provider uses Azure Key Vault Cloud HSM to create and store asymmetric keys (wrapping keys) and perform cryptographic operations. It stores the Azure Key Vault URL, directory tenant Id, client application Id, and client secret information that are created in Azure Cloud for communication purpose. You can create one or multiple entries of the Azure Key Vault Key Provider throughout the entire AccessMatrix System. Note: This option will be displayed for selection after you deploy the Azure Key Vault Key Provider Plugin in the AccessMatrix system. Refer to the AccessMatrix Azure Key Vault HSM Implementation Guide to learn how to deploy Azure Key Vault Key Provider Plugin.

Key Provider	Description
Gemalto Luna Key Provider	This key provider uses both JCE Luna Provider and Luna API. Luna Provider uses Keystore to access the keys inside the partition. The keystore is tied to a slot. Multiple keystores can be created for the same slot. It creates LunaSAUtil as a wrapper for calling Luna HSM. It uses Luna asymmetric key for AM key management (as wrapping key) to wrap AEK or SEK.
Google Key Provider	This key provider uses Google Cloud HSM to create and store symmetric keys and perform cryptographic operations. It stores the project Id, location Id, and key ring name information that are created in Google Cloud for communication purpose. You can create one or multiple entries of Google Key Provider throughout the entire AccessMatrix System. Note: This option will be displayed for selection after you deploy the Google Key Provider Plugin in the AccessMatrix system. Refer to the AccessMatrix Google Cloud HSM Implementation Guide to learn how to deploy Google Key Provider Plugin.
Hashicorp Vault Transit Key Provider	This key provider uses Hashicorp Vault server to create and store symmetric keys and perform cryptographic operation. The administrator needs to pre-create the key in the vault server and enable the vault transit key operation. See Vault Transit Secrets for more information. It does not have external dependencies; only AccessMatrix libraries are used. Note: This option will be displayed for selection after you deploy the Hachicorp Vault Transit Key Provider Plugin in the AccessMatrix system.
PKCS11 Key Provider	This key provider supports storage of keys within a PKCS11 device such as a Hardware Security Module (HSM) and smart card (or software HSM that supports PKCS11). It relies on the JVM's PKCS11 JCE provider to communicate with the PKCS11 device. The administrator is allowed to create only one entry of the PKCS11 Key Provider throughout the entire AccessMatrix System.
RSA BSAFE Key Provider	This key provider offers versions of popular cryptographic algorithms optimized for both small code size and high performance. The administrator is allowed to create one or more entry of RSA BSAFE Key Provider throughout the entire AccessMatrix System.
SafeNet Jcprov Key Provider and SafeNet PHEFT Key Provider	This key provider supports storage of keys within the SafeNet PHEFT HSM. AccessMatrix does not have a fixed key hierarchy. Thus, AEK can be wrapped by the HSM key directly. The administrator is allowed to create only one entry of SafeNet PHEFT Key Provider but one or more entry of SafeNet JCProv Key Provider throughout the entire AccessMatrix System. Note: SafeNet PHEFT Key Provider is deprecated and will not be supported in future.
SM Key Provider	This key provider supports SM cryptography, i.e. using SM2 and SM4 algorithms.

Key Provider	Description
	The administrator is allowed to create one or more entry of SM Key Provider throughout the entire AccessMatrix System.
TASS HSM JCE Key Provider	This key provider supports SM4/ECB/NoPadding algorithm. The administrator is allowed to create only one entry of TASS HSM JCE Key Provider throughout the entire AccessMatrix System.
Thales payShield Key Provider	This key provider uses Thales payShield 9000\10000 Hardware Security Module (HSM) to create and store symmetric keys and perform cryptographic operations. It is catered for TLS (cryptographic protocol) connection that provides end-to-end communications security over networks and is widely used for internet communications and online transactions. It uses Thales payShield 9000\10000 HSM in response to call commands from AccessMatrix server to create the keys in the HSM, which it receives via a data link to protect your keys. You can create one or multiple entries of Thales payShield Key Provider throughout the entire AccessMatrix System.



Notes:

- The administrator is only allowed to create, view and edit but not allowed to delete the key provider entry once created.
- The administrator will not be able to edit the key provider settings if there is a pending/rejected admin request of that key provider entry.

View Key Provider Page Attribute Descriptions

All implementation types share the following attributes:

Attribute Name	Description
*Key Provider Id	Unique Id of key provider.
*Key Provider Name	Name of key provider.
Description	Description of key provider.
Version	Key provider entry version.
Implementation	Type of key provider implementation.

The detailed attributes differ according to the type of key provider, as discussed below.

System Key Provider

Attribute Name	Description
Configuration Tab	
Active Keys	
Active Key for Attribute Encryption	Encryption key in use for Attribute Encryption. Default: AEK1
Active Key for Token and Biometric Template Encryption	Encryption key in use for Token and Biometric Template Encryption. Default: TEK1
Active Key for Session Encryption	Encryption key in use for Session Encryption. Default: SEK00000001
Active Key for Client State Encryption	Encryption key in use for Client State (productized applications) Encryption. Default: CSK0000001
Session Encryption Key Rollover - Require restart to take effect	
Enabled	To enable or disable Encryption Key Rollover. Default: False
Rollover interval (in minutes)	The interval for Encryption Key Rollover to take place. Default: 120
Number of previous keys to be kept	Number of previous keys to be kept for decryption purpose. Default: 10
Key Size	Size of a key in bytes. Default: 128
Session Encryption Key Wrapper - Require restart to take effect	
Encoding format	Encoding format used to wrap key. Note: Select "RAW" if no Key Wrapper is used. Default: Wrapped by domain master key.
Encoder wrapper key id	Encryption Key Id, if Encoding format is "Wrapped by other Key".

AWS KMS Key Provider

Attribute Name	Description
Configuration Tab	
Access Key ID	It is the access key identity (ID). The access keys are long-term credentials for an Identity & Access Management (IAM) user or the AWS account root user. Access Keys consist of two parts: an access key ID and a secret access key, as like a user name and password, you must use both together to authenticate your requests.
Secret Access Key	Secret Access Key is used together with the Access Key ID to authenticate requests. The secret access key is available only at the time of creation.
AWS Region	A collection of AWS resources in a geographical area, i.e. "ap-southeast-1". Each region is isolated and independent of other regions. The keys created in one region do not exist in another region.

Azure Key Vault Key Provider

Attribute Name	Description
Configuration Tab	
*Azure Key Vault URL	The URL of Azure Key Vault. E.g. "https://sample-vault.azure.net/".
*Directory (tenant) ID	The directory tenant that you want to request permission from. This can be in GUID or friendly name format.
*Application (client)ID	The application ID that the Azure app registration portal assigned when you registered your app.
*Client Secret	The application secret that you generated for your app in the app registration portal.
HTTP Proxy	
Enabled	Setting this to "true" will enable the HTTP Proxy. Default: false
IP address	Refers to the IP address of the proxy server.
Port	Refers to the port of the proxy server.
User Name	Refers to the username used to authenticate to the proxy server.
User Password	Refers to the user password used to authenticate to the proxy server.

Gemalto Luna Key Provider

Attribute Name	Description
Configuration Tab	
Note: Configure the user PIN of the slot in the amsystem.properties file. Modification requires a server restart to take effect.	
*Slot label	Slot/token label refers to the partition label.
Wrapping key label	HSM key used to wrap generated host-stored keys (including host-stored asymmetric keys). Wrapping key must be AES and must allow wrap/unwrap operations.

Google Key Provider

Attribute Name	Description
Configuration Tab	
*Location Id	Refer to Google documentation for Google Cloud KMS (Key Management Service) service location, e.g. "asia-southeast1", "asia-east1", "asia-east2", "asia-northeast1", "global" and others. Google Cloud KMS resources can be created at a geographical location or multiple locations. There is a special location for Google cloud KMS resources called "global" whereby your cloud KMS resources are located at multiple locations.
*Project Id	The identity of your Google cloud project. It is the project identity that you provide when you perform the settings, permissions, and other metadata to describe your application as a GCP (Google Cloud Platform) resource.
*Key Ring Name	The name of a key ring. A key ring belongs to a Google cloud project is used to group several cryptography keys for organizational purposes. Grouping by related permissions of the keys allows you to manage the permissions at the key ring level to prevent acting on each key individually.

Hashicorp Vault Transit Key Provider

Attribute Name	Description
Configuration Tab	
*API Url	The Vault HTTP API gives you full access to Vault via HTTP. Uses API Url to access Vault. Specifies the API address (full URL) of the vault server. All API routes are prefixed with /v1/. Example: "https://10.1.42.201:8201/v1"

Attribute Name	Description
*Vault Token	Specifies the token for authentication to the vault server.

PKCS11 Key Provider

Attribute Name	Description
Configuration Tab	
Note: Modification requires a server restart to take effect.	
*PKCS11 configuration file	Specifies the PKCS11 configuration file.
Security provider implementation	Specifies the PKCS11 security provider. Default: Auto-detect
HSM User PIN	User PIN for the PKCS11 slot. If not specified, it will check the am.crypto.pkcs11.userpin property in the amsystem.properties file.

RSA BSAFE Key Provider

Attribute Name	Description
Configuration Tab	
Note: Leave the following configurations empty to use JAVA device. Otherwise, PKCS11 device will be used.	
Library name	Refers to the RSA BSAFE PKCS11 library name.
Token label	Refers to the token/partition name. PIN is configured in the amsystem.properties file (am.crypto.jcprov.<KeyProviderId>.userpin).
Note: Session pool configuration (for PKCS11 device only, server restart is required).	
Maximum session pool size	Sets the maximum session pool size. Value should be from 1 to 50. Default: 10
Session pool recycle interval in minutes	Sets the session pool's recycle interval."0" means no recycling. Value should be from 0 to 1440. Default: 60

Attribute Name	Description
Maximum number of retries	Sets the maximum number of retries on specific errors. "0" means no retry. Value should be from 0 to 10. Default: 0
Retry only if exception message satisfies this regular expression	Specifies the regular expression for retry condition.
Waiting time between retries in seconds	Sets the waiting time between retries. Value should be from 1 to 30. Default: 1

SafeNet Jcprov Key Provider

Attribute Name	Description
Configuration Tab	
Note: Configure the user PIN of the slot in the amsystem.properties file. A server restart is required.	
*Slot index	The slot index value. Value should not be less than 0.
Slot label	The slot or token labels. If the label is not unique, it can be prefixed with HSM serial number, e.g. "408801: Slot01". If defined, it will overwrite the Slot index configuration. This setting allows multiple token labels to be specified to enable failover feature where the first token label is pointing to the primary HSM, second token label is pointing to secondary HSM, and so on.
Server-based HSM Slot label	This is to define the slot labels based on serverID. It is to allow specifying different Primary HSM/HSM priority based on the AccessMatrix server instance. If the slot labels are specified for a serverID, then that server will use this configuration to overwrite Slot index and Slot label configurations.
Note: Host-stored/unmanaged keys configuration (used by CryptoApp module.)	
Wrapping key label	HSM key used to wrap generated host-stored keys.

SafeNet PHEFT Key Provider

i	Note: SafeNet PHEFT Key Provider is deprecated and will not be supported in future.
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Attribute Name	Description
Configuration Tab	
Login Module Id	Uses the connection of this module. Module Id such as SafeNetPHEFTE2EE.
Connection	It is applicable and mandatory only if connecting to HSM via Direct Socket Protocol, specify as "host:port" format, e.g. "192.168.1.44:80".
Connection Timeout	Socket connection timeout value in seconds. "0" means an infinite timeout. Default: 30

SM Key Provider

Attributes Tab
There is no available attributes, however, custom attribute(s) may be added by following the instructions described in Adding Custom Attribute .

TASS HSM JCE Key Provider

Attribute Name	Description
Configuration Tab	
*Config File	Set the path to the configuration file.

Thales payShield Key Provider

Attribute Name	Description
Configuration Tab	
Connection	
*Thales payShield HSM Host, Port and Number of Connections	Defines the connection settings to the HSM. Specify the value in "host:port:number of connections" format, e.g. "192.168.1.44:80:2". For the number of connections, consult the administrator for Thales payShield HSM. Unless set otherwise by the HSM administrator, the default number of connections should be set to 2.
*Message Header	Value can be any length from 1 to 255 characters and it is configured at HSM installation.

Attribute Name	Description
Enable TLS	Enables TLS for mutual authentication between AccessMatrix and HSM. Once enabled, the HSM server certificate must be added into the default JRE truststore or javax.net.ssl.truststore. Default: false
Keystore	Path to the keystore file containing the HSM client certificate.
Keystore Passphrase	The Passphrase of the keystore.
Private Key Passphrase	Client private key passphrase in the keystore.
Connection timeout (seconds)	Socket connection timeout value in seconds. "0" means an infinite timeout. Value should not be less than 0 Default: 15
Heartbeat Interval (minutes)	Interval of heartbeat connection (in minutes) to HSM to keep the connection alive. The value must be lower than the idle timeout value configured in HSM. If the socket connection is unhealthy (temporarily disconnected), this value will also be used as the time interval for reconnecting the socket. Value should not be less than 1 Default: 5
Maximum Connection Retry	Specify the maximum number of retry attempts for HSM connection. "0" means no retry. Value should not be less than 0 Default: 3


Note: Administrator is allowed to add, edit or delete any custom attribute. Refer to [Adding Custom Attribute](#) section for details.

Configure Encryption Key

To navigate to Encryption Key page, select **Administration Dashboard > Security Policy > Key Management > Encryption Key** on the left pane. The following tasks can be performed on the encryption keys:

- **Create Encryption Key**

Click **Create** button to display the **Choose Encryption Key Template** page, select the key provider and the key-encryption template, click **Next**, enter the details and click **Save**.

- **Search for Encryption Key**

Can be searched by using the encryption key Id or the encryption key name. All available encryption keys are displayed by default.

- **View Encryption Key**

Click an encryption key Id under the **Search Encryption Key Result** list to view the details.



Notes:

- You may click **Test KeyInfo** to verify the key information. The "Key is successfully tested" message is displayed if testing is successful. The **Test KeyInfo** button is using random plaintext for encryption key testing.
- You may click **View KCV** to view the associated key check value (KCV). For Asymmetric key, click **View Public Key** to see the associated public key.



Note: The administrator is not required to create any Attribute Encryption Key (AEK), Session Encryption Key (SEK), PIN Encryption Key (PEK) and Token Encryption Key (TEK) apart from the existing keys created during setup.

Types of Encryption Keys

Encryption Key	Description
Domain Master Key (DMK)	This is the default wrap key for AEK and SEK. It is a derived key stored in the AccessMatrix database. A derived key is a key, which may be calculated (derived) by a well-defined algorithm, usually referred to as a key derivation function, from an input consisting of the public as well as secret data (e.g. a master key or primary key).
Attribute Encryption Key (AEK)	This is the key that uses Advanced Encryption Standard (AES) to encrypt the user-defined encrypted object attributes, e.g. principal attributes, group attributes and others in the AccessMatrix security registry. New AEK can be created via the AccessMatrix Admin Console and wrapped by DMK by default. Meantime, new AEK can also be created and wrapped by an HSM key. AEK is randomly generated based on the algorithm provided by the JCE. An active AEK can be selected in the System Key Provider. The encrypted AEK is stored in AccessMatrix security registry. Under the System Key Provider configuration, the administrator can select the AEK to be used. When an active AEK is used to encrypt data, the AEK key index will be stored together with the encrypted data. If the active AEK changes, the old encrypted data can still be decrypted using the old AEK. Upon successful setup of the AccessMatrix system, an AEK called "AEK1" associated with System Key Provider will be created by default. The administrator is not allowed to edit or delete such entry.
Session Encryption Key (SEK)	This is the key that uses AES to encrypt sensitive information within the session token used by AccessMatrix server for session management. DMK encrypts the SEK. SEK is randomly generated based on the algorithm provided by the JCE. The encrypted SEK is stored in AccessMatrix security registry. SEK is automatically managed by the AccessMatrix server where the key will

Encryption Key	Description
	be rolled-over based on the settings defined under System Key Provider configuration as described below: • The server instance on which the rollover will take place • The frequency (in minutes) the rollover will take place • Total number of SEKs to keep in the database A rollover means that each SEK has a time window, once it expires, the next encryption key will be used. Depending on the validity period of the SEK, there may be multiple valid SEKs. The administrator must ensure that relevant SEK is available for decryption concerning the session expiry policy. A background thread in the AccessMatrix server will automatically manage the SEKs seamlessly. Upon successful setup of the AccessMatrix system, several SEKs called "SEK0000001" to "SEK0000010" associated with System Key Provider will be created by default. The administrator is not allowed to edit or delete such entries.
PIN Encryption Key (PEK)	This is the key that uses AES to encrypt the PIN in the AccessMatrix security registry. DMK wraps PEK by default. PEK is randomly generated based on the algorithm provided by the JCE. The encrypted PEK is stored in AccessMatrix security registry. Upon successful setup of the AccessMatrix system, a PEK called "PEK1" associated with System Key Provider will be created by default. The administrator is not allowed to edit or delete such entry.
Token Encryption Key (TEK)	This is the key that uses AES to encrypt the token in the AccessMatrix security registry. DMK wraps TEK by default. TEK is randomly generated based on the algorithm provided by the JCE. The encrypted TEK is stored in AccessMatrix security registry. Upon successful setup of the AccessMatrix system, a TEK called "TEK1" associated with System Key Provider will be created by default. The administrator is not allowed to edit or delete such entry.
Client State Encryption Key (CSK)	This key is managed similarly to SEK where there is CSKMaintainer class that initializes and maintains the key. The key configuration follows SEK (e.g. key size). However, the rollover feature is not implemented yet, and access restriction is also not added yet. The AccessMatrix server does not use the key at all, but AccessMatrix acts as the key manager and storage only. Upon successful setup of the AccessMatrix system, a CSK called "CSK0000001" associated with System Key Provider will be created by default. The administrator is not allowed to edit or delete this entry. Just like SEK, AdminConsole can only be used to see the active CSK (under SystemKeyProvider) and also to list all CSKs (under Encryption Key).
Audit Integrity Key (AuditSHA256)	This key is used for HMAC operation to calculate audit integrity (mac) value, which is used to verify the audit data whether it has been altered. DMK wraps audit Integrity Key by default. It is randomly generated based on the

Encryption Key	Description
	algorithm provided by the JCE. The Audit Integrity Key is stored in the AccessMatrix security registry.
Wrapper Key	This key uses symmetric encryption to encapsulate key material. It protects keys at rest or transmits over insecure networks. Key wrapping is the process of encrypting one key using another key to store it or transmit it over an untrusted channel securely. Key wrapping may rely on either symmetric or asymmetric cryptography, depending on the context.
i Note: Administrator is only allowed to create and view but is not allowed to edit/delete the encryption key entry once created.	

Types of Encryption Key Template

Encryption Key Template	Description
Symmetric key (secret key)	This template consists of detailed attributes, storage and key wrapping, and access control information for the symmetric key (contains secret information for encrypting and decrypting messages).
Asymmetric key (key pair)	This template consists of detailed attributes, storage and key wrapping, and access control information for an asymmetric key pair (contains a pair of a private and public key for encrypting and decrypting messages).
Upload Java keystore for PDF Pin Mailer Signing	This template is used for uploading an RSA private key and certificate from a Java keystore, for providing a digital signature to the PDF Pin Mailer. Select a generated keystore (JKS or keystore) file from the 3rd party application or authority, and then enter the certificate alias, keystore passphrase and private key passphrase to start the import process.
Data Protect Key	This template consists of detailed attributes, storage and key wrapping, and access control information for the data protection key.
Asymmetric key (wrapping key)	This template consists of detailed attributes, purpose (key wrapping), key provider id, and the key name information for an asymmetric key pair that will be stored in Azure Key Vault.
Asymmetric Elliptic Curve key (key pair)	This template consists of detailed attributes, storage and key wrapping, and access control information for the asymmetric elliptic curve key. This key is used for OIDC purposes.

View Encryption Key Page Attribute Descriptions

All Encryption Key templates share the following attributes:

Attribute Name	Description
*Encryption Key Id	Unique Id of encryption key.
*Encryption Key Name	Name of encryption key.
Description	Description of encryption key.
Version	Encryption key entry version.
Status	Status of the encryption key.
Created Time	Time of encryption key creation.
Access Control Tab	
Agent-based Access Restriction	Access restriction is only applicable to General Purpose key. This is to restrict which application (non-interactive user/agent) may use the key. Option: "USO Agent"

The detailed attributes differ according to the selected type of key provider and encryption key template, as shown below.

System Key Provider

Supported templates: Symmetric key (secret key), Asymmetric key (key pair), Upload Java keystore for PDF Pin Mailer Signing, Asymmetric Elliptic Curve key (key pair)

Attribute Name	Description
Keystore	Allows to upload a java keystore that consists of a private key and certificate(s) for signing the PDF Pin Mailer. Note: Only applicable for the "Upload Java keystore for PDF Pin Mailer" template.
Certificate Alias	The alias that identifies a specific certificate to upload. Note: Only applicable for the "Upload Java keystore for PDF Pin Mailer" template.
Keystore Passphrase	The passphrase of the Java keystore. Note: Only applicable for the "Upload Java keystore for PDF Pin Mailer" template.
Private Key Passphrase	The passphrase of a private key. Note: Only applicable for the "Upload Java keystore for PDF Pin Mailer" template.
Attributes Tab	
Basic Key Attributes	

Attribute Name	Description				
*Purpose	Type of encryption. Possible values based on encryption template: <u>Symmetric key (secret key)</u> Options: "Attribute Encryption" "PIN Encryption" "Token Encryption" "Key Wrapping" "General Purpose" "Key Derivation" <u>Asymmetric key (key pair)</u> Options: "General Purpose" "Key Wrapping" <u>Upload Java keystore for PDF Pin Mailer</u> Option: "Data Integrity" <u>Asymmetric Elliptic Curve key (key pair)</u> Options: "General Purpose" "Key Wrapping" "JWT Encryption of JWT Authentication" "JWT Signing of JWT Authentication" "OIDC Token Encryption" "OIDC Token Signing"				
Key Provider Id	Unique Id of System Key Provider. Note: Applicable for all System Provider templates. Default: SystemKeyProvider				
Elliptic Curve Standard Name	Elliptic curve is used for Elliptic curve key. Below are the supported curves: <table> <tr> <th>Curve Name</th><th>Alias Name</th></tr> <tr> <td>secp256r1</td><td>NIST P-256</td></tr> </table>	Curve Name	Alias Name	secp256r1	NIST P-256
Curve Name	Alias Name				
secp256r1	NIST P-256				

Attribute Name		Description	
	secp384r1	P-384	
	secp521r	NIST P-521	
	Default: secp256r1 Note: This attribute is only applicable for the "Asymmetric Elliptic Curve key (key pair)" encryption template.		
JCE Provider			
JCE Provider	Java Cryptography Extension Provider		
	Possible values based on encryption template: <u>Symmetric (secret key)/Asymmetric key (key pair)</u> Options: "Default JCE Provider" "SUN" "IBM" "Bouncy Castle" <u>Upload Java keystore for PDF Pin Mailer</u> N/A <u>Asymmetric Elliptic Curve key (key pair)</u> N/A		
Key Parameters			
Algorithm	Encryption algorithm		
	Possible values based on encryption template: <u>Symmetric key (secret key)</u> "AES" "DESede"		

Attribute Name	Description
	Default: AES <u>Asymmetric key (key pair)</u> Default: RSA <u>Upload Java keystore for PDF Pin Mailer</u> N/A <u>Asymmetric Elliptic Curve key (key pair)</u> Default: EC
Key size	Size of a key in bytes.
	Possible values based on encryption template: <u>Symmetric key (secret key)</u> Options: "128" "192" "256" Default: 128 <u>Asymmetric key (key pair)</u> Options: "1024" "2048" "4096" Default: 1024 <u>Upload Java keystore for PDF Pin Mailer</u> N/A <u>Asymmetric Elliptic Curve key (key pair)</u> N/A
Mode	Algorithm to apply block cipher.
	Possible values based on encryption template:

Attribute Name	Description
	<u>Symmetric key (secret key)</u> Options: • "CBC" • "CFB" • "OFB" • "PCBC" • "ECB" Default: CBC <u>Asymmetric key (key pair)</u> Options: • "NONE" • "ECB" Default: NONE <u>Upload Java keystore for PDF Pin Mailer</u> N/A <u>Asymmetric Elliptic Curve key (key pair)</u> N/A
Padding	Key padding to be used. Possible values based on encryption template: <u>Symmetric key (secret key)</u> Options: • "No Padding" • "PKCS5Padding" Default: PKCS5Padding <u>Asymmetric key (key pair)</u> Options: • "NoPadding" • "OAEPWithSHA1AndMGF1Padding" • "OAEPWithSHA256AndMGF1Padding" • "PKCS1Padding"

Attribute Name	Description
	Default: OAEPWithSHA1AndMGF1Padding <u>Upload Java keystore for PDF Pin Mailer</u> N/A <u>Asymmetric Elliptic Curve key (key pair)</u> N/A
Initialization vector scheme	Initialization vector to be used.
	Possible values based on encryption template: <u>Symmetric key (secret key)</u> Default: SHA1_IV <u>Asymmetric/Upload Java keystore for PDF Pin Mailer</u> N/A <u>Asymmetric Elliptic Curve key (key pair)</u> N/A
Storage and Key Wrapping Tab	
Encoding format	Encoding format used to wrap key. Select "RAW", if no Key Wrapper is used. Options: • "Wrapped by Domain Master Key" • "Wrapped by other key" • "RAW" Default: Wrapped by Domain Master Key Note: Applicable for all System Key Provider templates.
Encoder wrapper key id	Key Id to wrap JCE Key. Note: Applicable for all System Key Provider templates.
Access Control	
Agent-based Access Restriction	This attribute is used to restrict the use of the encryption key to specific applications only. Specify the non-interactive user/agent representing the application that will be allowed to use the key. Note: Access restriction is only applicable for General Purpose keys.

Different key creation options are available when using the system key provider:

- **System generation**

A key is automatically generated by the system.

This is the default method that is used when a user creates a new system key provider encryption key in the Admin Console.



Note: This default method is not used when the "Upload Java keystore for PDF Pin Mailer Signing" template is selected - see the next method.

- **Import from uploaded Java key store file (for asymmetric keys only)**

A key (already generated outside of AccessMatrix) is imported from the user-uploaded .jks (Java key store) file.

This method is used by selecting the "Upload Java keystore for PDF Pin Mailer Signing" template when creating a new System Key Provider encryption key in the Admin Console. It is used to import keys for PDF file signing purposes.

- **Import from user-entered key components (for symmetric keys only)**

A key (already generated outside of AccessMatrix) is imported by manually entering its hex-encoded key component(s). This method is used by clicking the **Enter key components (instead of generating key)** button in the **Create Encryption Key** page. To enter a key component, type in the hex-encoded key component string in the text field and click **Add**. When all of the components have been added, click **OK**.

This method is supported for symmetric keys using the AES and DESede algorithms.



Note: Each entered key component must be a hexadecimal string of valid length. This varies depending on the algorithm used, as well as the **Key size** selected. Invalid inputs will result in an error message.

AWS KMS Key Provider

Supported Template: Symmetric key (secret key)

Attribute Name	Description
*Customer Master Key (CMK) ID	The IDentity of a Customer Master Key (CMK). • If the connection to AWS KMS is established, AccessMatrix will automatically fetch the CMK IDs, and then display the list of crypto key IDs (CMK IDs) in the drop-down list box for selection. Note: AccessMatrix will fetch mainly the "Enabled" CMK IDs in AWS KMS, and will not fetch the "Disabled" CMK IDs. • If the connection to AMS KMS is unsuccessful, an error message will be displayed.

Attribute Name	Description
	If there is no crypto key (not yet created) found, the message "No Crypto Key Found" will be displayed in the list box.
Attributes Tab	
Basic Key Attributes	
*Purpose	Type of encryption. Options: "Attribute Encryption" "Key Wrapping" "PIN Encryption" "Token Encryption" "Key Derivation"
Key Provider Id	Unique Id of AWS KMS Key Provider.

Azure Key Vault Key Provider

Supported Template: Asymmetric key (wrapping key)

Attribute Name	Description
*Key Name	Key Name in Azure Key Vault. Example: sample. - If the connection to Azure Key Vault is established, AccessMatrix will automatically fetch the key names, and then display the list of crypto key names in the drop-down list box for selection. Note: AccessMatrix will fetch mainly the "Enabled" key names in the Key Vaults, and will not fetch the "Disabled" key names. - If the connection to Azure Key Vault is unsuccessful, an error message will be displayed. - If there is no crypto key (not yet created) found, the message "No Crypto Key Found" will be displayed in the list box.
Attributes Tab	
Basic Key Attributes	
*Purpose	Type of encryption. Option: "Key Wrapping"
Key Provider Id	Unique Id of Azure Key Vault Key Provider.

Google Key Provider

Supported Template: Symmetric key (secret key)

Attribute Name	Description
*Crypto Key Name	This is a named object in Google Cloud Platform representing a cryptographic key used for a specific purpose such as encryption, decryption or signing. - If the connection to Google Cloud is established, AccessMatrix will automatically fetch the crypto key names, and then display the list of crypto key names in the drop-down list box for selection. Note: AccessMatrix will fetch mainly the "Enabled" crypto key names in the Google cloud KMS, and will not fetch the "Disabled" crypto key names. - If the connection to Google Cloud is unsuccessful, an error message will be displayed. - If there is no crypto key (not yet created) found, the message "No Crypto Key Found" will be displayed in the list box.
Attributes Tab	
Basic Key Attributes	
*Purpose	Type of encryption. Options: "Attribute Encryption" "Key Wrapping" "PIN Encryption" "Token Encryption" "Key Derivation"
Key Provider Id	Unique Id of Google Key Provider.

Hashicorp Key Provider

Supported Template: Symmetric key (secret key)

Attribute Name	Description
*Transit Key Name	This is a named transit key in Hashicorp server representing a cryptographic key used for encryption or decryption. If the connection to Hashicorp server is established, AccessMatrix will automatically fetch the transit key names, and then display the list of transit key names in the drop-down list box for selection.

Attribute Name	Description
	If the connection to Hashicorp server is unsuccessful, an error message will be displayed.
Attributes Tab	
Basic Key Attributes	
*Purpose	Type of encryption. Options: "Attribute Encryption" "Key Wrapping" "PIN Encryption" "Token Encryption" "Key Derivation"
Key Provider Id	Unique Id of Hashicorp Vault Transit key Provider.

PKCS11 Key Provider

Supported Template: Symmetric key (secret key), Asymmetric key (key pair)

Attribute Name	Description
Attributes Tab	
Basic Key Attributes	
*Purpose	Type of encryption. Possible values based on encryption template: <u>Symmetric key (secret key)</u> Options: "Attribute Encryption" "PIN Encryption" "Token Encryption" "Key Wrapping" "General Purpose" "Key Derivation" <u>Asymmetric key (key pair)</u> Options:

Attribute Name	Description
	"General Purpose" "Key Wrapping" "Data Integrity"
Key Provider Id	Unique Id of PKCS11 Key Provider. Note: Applicable for all PKCS11 Key Provider templates. Default: SystemKeyProvider
Key Parameters	
Key alias	Key alias/label as listed in the keystore. If not specified, KeyInfo Id is used as the alias. Note: Only applicable for Symmetric key (secret key) template.
Private key alias	Key alias/label as listed in the keystore. If not specified, KeyInfo Id is used as the alias. Note: Only applicable for Asymmetric key (key pair) template.
Certificate alias	Key alias/label as listed in the keystore. If not specified, KeyInfo Id is used as the alias. Note: Only applicable for Asymmetric key (key pair) template.
Algorithm	Encryption algorithm
	Possible values based on encryption template: <u>Symmetric key (secret key)</u> Options: "AES" "DESede" Default: AES <u>Asymmetric key (key pair)</u> Default: RSA
Mode	Algorithm to apply block cipher.
	Possible values based on encryption template: <u>Symmetric key (secret key)</u> Default: CBC

Attribute Name	Description
	<u>Asymmetric key (key pair)</u> Options: "NONE" "ECB" "SSL" Default: ECB
Padding	Key padding to be used. Possible values based on encryption template: <u>Symmetric key (secret key)</u> Options: "NoPadding" "PKCS5Padding" Default: NoPadding <u>Asymmetric key (key pair)</u> Options: "NoPadding" "OAEPWithSHA1AndMGF1Padding" "PKCS1Padding" Default: PKCS1Padding
Time Stamping Authority URL	The URL to the Time Stamping Authority that will issue cryptographically secure timestamps related to certificate signing.
Storage and Key Wrapping Tab	
Encoding format	Encoding format used to wrap key. Select "RAW", if no Key Wrapper is used. Note: Applicable for all PKCS1 templates. Default: RAW
Encoder wrapper key id	Key Id to wrap JCE Key. Note: Applicable for all PKCS1 Provider templates.

RSA BSAFE Key Provider

Supported Template: Symmetric key (secret key), Asymmetric key (key pair)

Attribute Name	Description
Attributes Tab	
Basic Key Attributes	
*Purpose	Type of encryption.
	Possible values based on encryption template: <u>Symmetric key (secret key)</u> Options: • "Attribute Encryption" • "PIN Encryption" • "Token Encryption" • "Key Wrapping" • "General Purpose" • "Key Derivation" <u>Asymmetric key (key pair)</u> Options: • "Key Wrapping" • "General Purpose"
Key Provider Id	Unique Id of RSA BSAFE Key Provider. Note: Applicable for all RSA BSAFE Key Provider templates. Default: SystemKeyProvider
*Transformation	Algorithm, block mode, and padding used for cryptographic operations.
	Possible values based on encryption template: <u>Symmetric key (secret key)</u> Options: • "AES/CBC/PKCS5Padding" • "AES/CFB/PKCS5Padding" • "3DES_EDE/CBC/PKCS5Padding" • "3DES_EDE/CFB/PKCS5Padding"

Attribute Name	Description
	<u>Asymmetric key (key pair)</u> Option: "RSA/PKCS1Block02Pad"
Key size	Size of a key in bytes.
	Possible values based on encryption template: <u>Symmetric key (secret key)</u> Options: • 128" • "168" • "192" • "256" <u>Asymmetric key (key pair)</u> Options: • "1024" • "2048" • "4096"
Initialization vector scheme	Initialization vector to be used.
	Possible values based on encryption template: <u>Symmetric key (secret key)</u> Options: • "SHA1_IV" • "Random number and digest as IV" Default: Random number and digest as IV <u>Asymmetric key (key pair)</u> N/A
Secure random algorithm	Specifies the secure pseudo-random number generator algorithm.
	Possible values based on encryption template:

Attribute Name	Description
	<u>Symmetric key (secret key)</u> Options: "MD5Random" "SHA1Random" "X931Random-6" Default: SHA1Random <u>Asymmetric key (key pair)</u> Options: "X931Random-6" "HWRandom/SHA1"
Key Import	
Encoded key data	AccessMatrix 3-encoded key data. The import process will make use of the key wrapping configuration specified for this KeyInfo. Note: Only applicable for Symmetric key (secret key) template.
Encoded public key data	AccessMatrix 3-encoded key data. The import process will make use of the key wrapping configuration specified for this KeyInfo. Note: Only applicable for Asymmetric key (key pair) template.
Encoded private key data	AccessMatrix 3-encoded key data. The import process will make use of the key wrapping configuration specified for this KeyInfo. Note: Only applicable for Asymmetric key (key pair) template.
Storage and Key Wrapping Tab	
Encoding format	Encoding format used to wrap key. Select "RAW", if no Key Wrapper is used. Note: Applicable for all RSA BSAFE Key Provider templates. Options: "Wrapped by Domain Master Key" "Wrapped by other keys" "RAW" Default: Wrapped by Domain Master Key
Encoder wrapper key id	Key Id to wrap JCE Key. Note: Applicable for all RSA BSAFE Key Provider templates.

SafeNet Jcprov Key Provider

Supported Template: Symmetric key (secret key), Asymmetric key (key pair)

Attribute Name	Description
Attributes Tab	
Basic Key Attributes	
*Purpose	Type of encryption. Possible values based on encryption template: <u>Symmetric key (secret key)</u> Options: "Attribute Encryption" "PIN Encryption" "Token Encryption" "Key Wrapping" "General Purpose" "Key Derivation" <u>Asymmetric key (key pair)</u> Option: "General Purpose"
Key Provider Id	Unique Id of SafeNet Jcprov Key Provider. Note: Applicable for all SafeNet Jcprov Key Provider templates. Default: SystemKeyProvider
Key Parameters	
Key Label	Key label/alias of the key inside the HSM. When creating keys in HSM, assign the same key label/alias to both private and public keys. Note: Applicable for all SafeNet Jcprov Key Provider templates.
Mechanism	Algorithm, block mode, and padding to use for encryption/decryption. Possible values based on encryption template: <u>Symmetric key (secret key)</u> Default: AES/CBC/PKCS5Padding

Attribute Name	Description
	<u>Asymmetric key (key pair)</u> Options: "RSA with PKCS1 v1.5 padding" "RSA with OAEP padding" Default: RSA with OAEP padding
Initialization vector scheme	Initialization vector to be used. Possible values based on encryption template: <u>Symmetric key (secret key)</u> Options: "SHA1_IV" "Random number and digest as IV" Default: Random number and digest as IV <u>Asymmetric key (key pair)</u> N/A

SafeNet PHEFT Key Provider



Note: SafeNet PHEFT Key Provider is deprecated and will not be supported in future.

Supported Template: Data Protect Key (DPK)

Attribute Name	Description
Attributes Tab	
Basic Key Attributes	
*Purpose	Type of encryption. Options: "Attribute Encryption" "PIN Encryption" "Token Encryption"

Attribute Name	Description
	"Key Wrapping" "General Purpose" "Key Derivation"
Key Provider Id	Unique Id of SafeNet PHEFT Key Provider. Default: SystemKeyProvider
Key Parameters	
*DPK key index	Value should be from 1 to 99.
Storage and Key Wrapping Tab	
Encoding format	Encoding format used to wrap key. Default: RAW
Encoder wrapper key id	Key Id to wrap JCE Key.

SM Key Provider

Supported Template: Symmetric key (secret key)

Attribute Name	Description
Attributes Tab	
Basic Key Attributes	
*Purpose	Type of encryption. Options: "Attribute Encryption" "PIN Encryption" "Token Encryption" "Key Wrapping" "General Purpose" "Key Derivation"
Key Provider Id	Unique Id of SM Key Provider. Default: SystemKeyProvider
Key Parameters	

Attribute Name	Description
Key size	Size of a key in bytes. Default: 128
Algorithm, Mode and Padding	Mechanism to use for encryption/decryption. Default: SM4_CBC_PKCS5Padding
Storage and Key Wrapping Tab	
Encoding format	Encoding format used to wrap key. Options: "RAW" "Wrapped by Domain Master Key" "Wrapper by another key" Default: Wrapped by Domain Master Key with SM4
Encoder wrapper key id	Key Id to wrap JCE Key.

Thales payShield Key Provider

Supported Template: Symmetric key (secret key)

Attribute Name	Description
Attributes Tab	
Basic Key Attributes	
*Purpose	Type of encryption. Options: "General Purpose" "Attribute Encryption" "Key Derivation" "Key Wrapping" "PIN Encryption" "Token Encryption"
Key Parameters	
LMK Index	Consult the administrator for Thales payShield HSM to get the value. Default: 00

Attribute Name	Description
Algorithm	Encryption algorithm. Default: AES
Key size	Size of a key in bytes. Default: 256
Mode	Algorithm to apply block cipher. Default: CBC
Padding	Key padding to be used. Default: PKCS5Padding
Initialization vector scheme	Initial vector scheme to be used. Default: SHA-256

4.2.7. Background Task Configurations

Background tasks are carried out by pre-scheduled settings or by trigger actions, which are used for monitoring, sync-ing information or purging purposes.

Scheduled Task

Locate the **Scheduled Tasks** page by navigating to **Administration Dashboard > System > Server & Cache > Scheduled Task** on the left pane.

Scheduled task management displays a summary of the background tasks running in the AccessMatrix System. It shows the next execution time, the last result, the start and end time, as well as the elapsed time.

Housekeeping

Housekeeping module is used to housekeep data or records in the database. The data that is meant to be housekept includes old audit data, deleted records, etc. Apart from housekeeping data, it also can be used to update the database records according to the housekeeping policy defined.

- **Housekeeping Task Administration**

Housekeeping tasks can be configured to run based on the user-defined schedule in terms of execution interval. By default, the system-defined housekeeping task

"DefaultHousekeepingTask" will be created upon successful installation of AccessMatrix. The administrator should configure properly the housekeeping policy prior to configuring the housekeeping task.

 **Note:** Housekeeping Task runs on any one of the active servers.

- **Housekeeping Policy Administration**

Housekeeping policy governs the behaviour of the housekeeping module in terms of the activities to be performed by the housekeeping tasks, e.g. suspending users, deleting suspended users, purging deleted user records, etc. By default, the system-defined housekeeping policy "DefaultHousekeepingPolicy" will be created upon successful installation of AccessMatrix.

To locate the **Housekeeping Configurations** page, navigate to **Administration Dashboard > System > Housekeeping > Housekeeping Task / Housekeeping Policy** on the left pane. Housekeeping policies

and tasks can be edited by clicking **Edit** on the **View Housekeeping Policy** or the **View Housekeeping Task** pages respectively.

View Housekeeping Task Page Attribute Description

Attribute Name	Description
*Housekeeping Task Id	Unique Id of housekeeping task.
*Housekeeping Task Name	Name of housekeeping task.
Description	Description of housekeeping task.
Version	Housekeeping task entry version.
Configuration Tab	
Policy	Refers to the housekeeping policy, i.e. DefaultHousekeepingPolicy.
Simulation	If set to "true", then no actual data will be created, modified or deleted in the database during task execution. Default: false
Task execution interval (hour)	The time interval before the next task execution. Default: 24
Task first time execution	Refers to the task's first time execution.

View Housekeeping Policy Page Attribute Description

Attribute Name	Description
*Housekeeping Policy Id	Unique Id of housekeeping policy.
*Housekeeping Policy Name	Name of housekeeping policy.
Description	Description of housekeeping policy.
Version	Housekeeping policy entry version.
Attributes Tab	
General Purging	

Attribute Name	Description
Purge deleted objects	Determines if the deleted objects will be purged or not. Default: true
Purge deleted objects older than (days)	Only those deleted objects older than the specified value will be purged. Value should be from 7 to 10000. Default: 30
Events and Audits Purging	
Purge method	WARNING: By choosing SQL Delete on the purge method, the database engine will purge the audit tables using the SQL Delete command. It will use database transactions and will require a huge size of the transaction log, therefore requires a vast amount of computing power and might degrade overall system performance. SQL Truncate command is non-transaction based. It will wipe off all the audit tables. It is a fast process, select this command when you want to empty all the audit tables. It does not require much computing power, therefore provides better system performance. Default: SQL Truncate
Purge audit data older than (days)	Only those audit data older than the specified value will be purged. Value should be from 7 to 10000. Default: 30
Users	
Enable inactive users suspension	Setting this to "true" will enable suspension of inactive users. This only applies to users of user stores that support user status change (Default Store, LDAP User Store, and its variants, updateable JDBC User Store with mapped user status). For other user stores except for LDAP user store, the Admin modification is considered as a user activity to prevent a user that is enabled back by an administrator from being suspended again. Default: false
Inactive users suspension policy	If enabled, this setting will suspend the inactive users of the defined user store.
Enable inactive users deletion	If enabled, this setting will enable deletion of inactive users. Notes: - Admin modification is considered a user activity. - Only applies to updateable user store (Default Store and updateable JDBC User Store). Default: false
Inactive users deletion policy	If enabled, this setting will delete the inactive users of the defined user store.

Attribute Name	Description
Enable user reconciliation	If enabled, this setting will delete the user-related information of a user that has been deleted in the external store. Notes: - User reconciliation task helps to housekeep unused user data, e.g. login accounts from the default database after confirming that such user does not exist anymore in an external user store. - During the execution of the user reconciliation task, the token will be unassigned from the deleted user. The token status will be changed to the status set in Token Status After Unassign attribute in the Global Setting > Token Management tab, only if all the users assigned with the token are deleted. Default: false
Recon users who have not logged-in within (days)	Only those users who have not logged in within the specified value will be reconciled. Users that still exists in the external store will not be deleted. This setting is only applicable if the Enable user reconciliation attribute is enabled. Value should be from 7 to 10000. Default: 90
Batch per transaction for user reconciliation	Number of users to reconcile per transaction. Value should be from 10 to 300. Default: 100
Administrative Workflow Request	
Purge completed requests	Delete requests that have been completed ("Approved", "Withdrawn" or "Deleted" status). Default: true
Purge completed requests older than (days)	Based on modified time (the time when the request was last modified). Value should be from 1 to 365. Default: 30
Purge uncompleted requests	Delete uncompleted requests (normally those with "Pending" or "Rejected" status). Objects that are locked by workflow will be forced unlocked. Default: true
Purge uncompleted requests older than (days)	Based on request time (the time when the request was last submitted by the requestor). Value should be from 1 to 365. Default: 60
Bulk Job	
Purge executed bulk jobs	Determines if the executed bulk jobs will be purged or not. Default: true

Attribute Name	Description
Purge executed bulk jobs older than (days)	Only those executed bulk jobs older than the specified value will be purged. Value should be from 1 to 60. Default: 7
Retry	
Number of retries	Number of retries when there is a failure. "0" means no retry. Value should be from 0 to 10. Default: 2
Waiting time between retry (seconds)	The time interval before the next try. Value should be from 1 to 120. Default: 15
Contextual Authentication History	
Number of history to keep per user	Number of history to keep for each user for audit purposes. Default: 10
UAS Tab	
This tab contains attributes related specifically to the UAS product and will only be displayed provided user has subscribed to the UAS features for the AccessMatrix license key. The associated attributes will be discussed in details in the UAS Administration Guide .	
Tokens	
Purge SMS tokens	Determines if the SMS tokens will be purged or not. Default: true
Purge SMS tokens older than (days)	Only those SMS tokens older than the specified value will be purged. The value should be from 1 to 365. Default: 3
Purge non-SMS token if it is in a certain status for longer than the specified period	Specifies how long non-SMS tokens can remain in a certain status before they are purged. Define the status for the token, i.e. "Battery Expired", and its period in months, i.e."24", and then add the status and its period to the list. Note: Period (months) should be between 1 and 1200.
Change non-SMS token status if it is not used for longer than the specified period	Specifies how long non-SMS tokens can remain unused before their current status is changed to a new status. Define the affected token group, the new status to change to, i.e. "Deactivated", and the time period for the tokens to remain unused before their status is changed to the new status.

Attribute Name	Description
	This attribute only affects tokens which meet the following requirements: - Token must be assigned to a user. - The current token status must have been defined as operational or usable. By default, this would be "Activated" and "Preactivation". The list of statuses can be configured using the attribute Token Status for usable functions like verify OTP/signature/response etc. For more information on how to configure this, refer to UAS Administration Guide - Global Setting - Token Management. Note: Period (months) should be between 1 and 1200.
Purge Token Instances in 'PreActivation' status	Setting this to true will purge VASCO Multi-Device token instances that are in "PreActivation" status for a certain defined period. Default: true
Purge Token Instances in 'PreActivation' status older than (days)	Only those token instances with "PreActivation" status older than the specified value will be purged. Value should be from 1 to 365. Default: 3
Purge 'Deactivated' Token Instances	Setting this to true will purge VASCO Multi-Device token instances that have been "Deactivated" for a certain defined period. Default: true
Purge 'Deactivated' Token Instances older than (days)	Only those "Deactivated" token instances older than the specified value will be purged. Value should be from 1 to 365. Default: 30
Purge Unassigned UAF Tokens (Deprecated)	Setting this to "true" will purge unassigned UAF tokens. Default: true
Records committed per transaction	Number of records to purge per transaction. Value should be from 1 to 2500. Default: 1000
PIN Mailer	
Purge ended PIN mailers	Determines if the ended PIN mailers will be purged or not. Default: true
Purge ended PIN mailers older than (days)	Only those ended PIN mailers older than the specified value will be purged. Value should be from 1 to 365. Default: 30

Attribute Name	Description
Records committed per transaction	Number of ended PIN Mailers to purge per transaction. Value should be from 1 to 100. Default: 5
User Device	
Purge deleted User Device	Determines if the deleted user device will be purged or not. Default: true
Purge deleted User Device older than (days)	Only deleted User Devices older than the specified value will be purged. Value should be from 1 to 365. Default: 30
Purge deleted User Device records committed per transaction	Number of deleted User Device records to purge per transaction. Default: 1000
UCM Tab	
This tab contains attributes related specifically to UCM product and will only be displayed provided the user has subscribed to the UCM features for the AccessMatrix license key. The associated attributes will be discussed in details in the <i>UCM Administration Guide</i> .	
Administrative UCM Workflow Request	
Note: See the <i>UCM Administration Guide - Housekeeping</i> section for attribute details.	

User Reconciliation Task

The user reconciliation task helps to housekeep unused user data (e.g. login accounts from the default database) after confirming that the user no longer exists in an external user store. This feature makes use of [Housekeeping Policy](#).

To locate the **Reconciliation Task** page, navigate to **Administration Dashboard > System > User Reconciliation Task**. User reconciliation task can be edited by clicking **Edit** on the **View User Reconciliation Task** page.



Note: User reconciliation task will only be executed if the **Enable user reconciliation** attribute of the Housekeeping Policy is set to "true".

Bulk Job

AccessMatrix provides a mechanism to queue bulk jobs that normally take a long time to finish, e.g. Token Import. A background thread will run periodically (i.e. every 5 minutes) to check whether there is any bulk job to run. The system also provides a Bulk Job configuration that allows the Administrator to define when the bulk job can be processed, e.g. process bulk jobs only during off-peak hours.

Bulk Job configuration has a **Time Restriction** attribute that sets when the bulk job can be executed. If the time restriction is not specified, then the bulk job will run on the next background thread interval. Moreover, a running job will not be stopped if it falls outside of the time restriction.

To locate the **Bulk Job** page, navigate to **Administration Dashboard > System > Bulk Job**. Bulk Job can be edited by clicking **Edit** on the **View Configuration** page.

4.2.8. Log and Monitoring

Events occurred in the system can be logged and monitored for audit purposes. The administrator is allowed to configure the settings accordingly.

Cache

Locate the **Cache Management Page** by navigating to **Administration Dashboard > System > Server & Cache > Cache** on the left pane.

This page displays the list of cache names, its current sizes and max sizes, its time to idle and time to live, a flag indicator to show if asynchronous replication is configured or not, allowing the administrator to monitor whether each cache is useful. Clicking a cache name will display **Calculate In Memory Size** dialog box where a **Calculate** button will be displayed to compute cache in memory size. Moreover, if the **Replication Configured** column is set to "true", then a **Replication Configuration** section will be displayed in the **Calculate In Memory Size** dialog box.

A **Calculate Total In-Memory Size for All Caches** button is provided to calculate the estimated total size of the cache. Take note that this is just an estimate calculation and may not be accurate. The actual cache memory size could be bigger. Below are some use cases for EhCache Replication:

- **Object & Hibernate Cache**

To ensure performance, we make use of cache for our configurations/policies and some other persisted information. If replication is not set up, changes to our configurations/policies will only be reflected to other AccessMatrix server instances after the cache entries have expired. By default, these caches are configured to be replicated.

- **UAS SMS OTP Memory Cache**

To ensure SMS OTP can be generated and verified by different AccessMatrix instance when in-memory SMS is used.

By default, SMS OTP uses MemoryTokenStore, which is not persisted in the database. To enable generateOtp and verifyOtp on different AccessMatrix server instances, locate the following in *WEB-INF/classes/am5-ehcache.xml*:

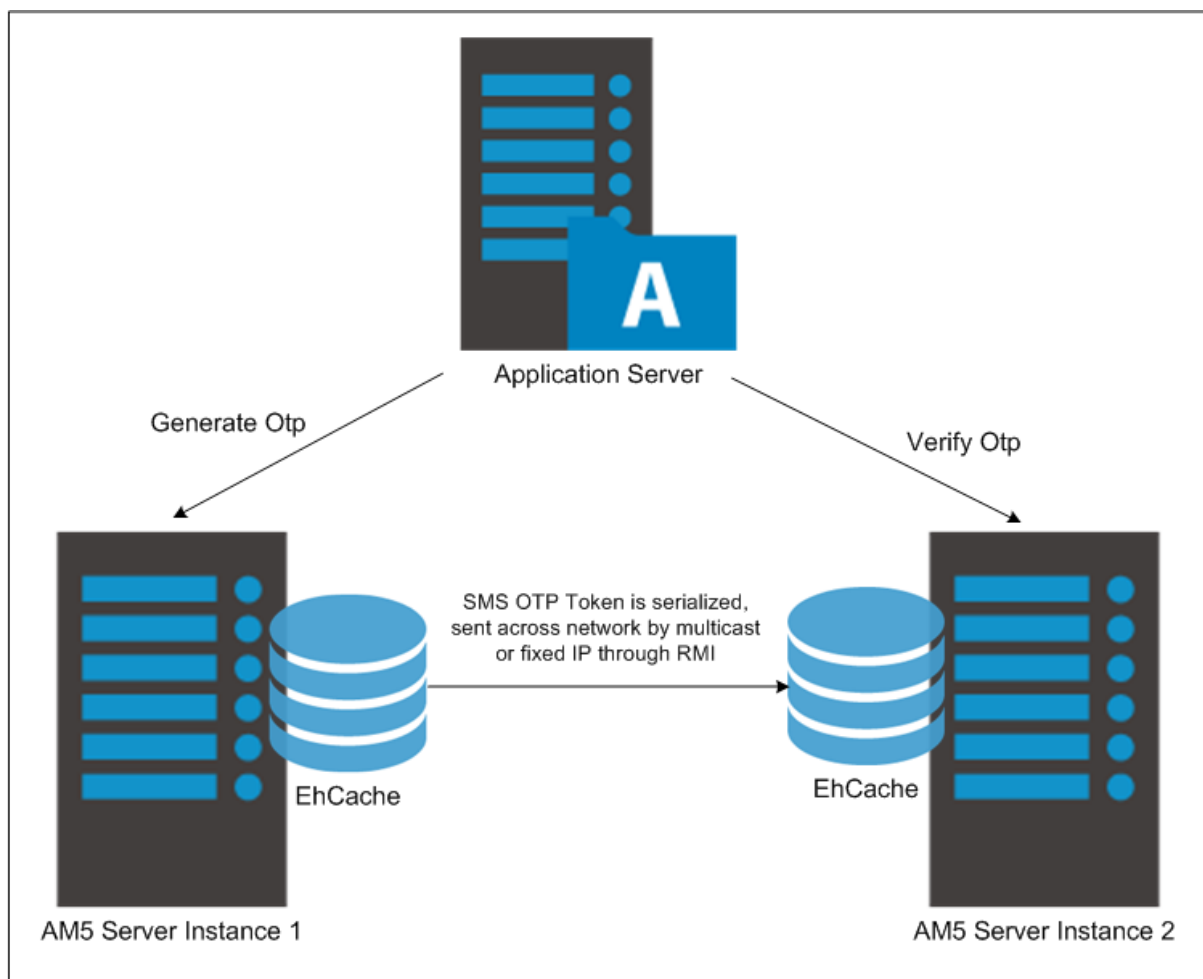
XML

```

<cache name="MemoryTokenStore"
    maxElementsInMemory="30000"
    eternal="false"
    timeToIdleSeconds="0"
    timeToLiveSeconds="900"
    overflowToDisk="false">
    <!-- if generateOtp/verifyOtp could end up with
different instances of AM5 servers, replication must be
enabled and tested -->
    <cacheEventListenerFactory
class="net.sf.ehcache.distribution.RMICacheReplicatorFactory"
    properties="replicateAsynchronously=true,
    replicatePuts=true,
    replicateUpdates=true,
    replicateUpdatesViaCopy=true,
    replicateRemovals=true,
    asynchronousReplicationIntervalMillis=100"/>
</cache>

```

Alternatively, SystemTokenStore can be considered, i.e. SMS OTP token persisted to database.



- **UAS E2EE Session Cache**

A UAS E2EE session is not persisted to the database. It is kept in a cache with the entry expiry

controlled by a global E2EE session expiry setting within the Admin Console. Since E2EE projects normally involve web applications with a web load balancer configured to be web session sticky (i.e. same web request of same web session will always reach the same web application server), it does not normally require clustering.

To ensure the E2EE session object can be found in the rare scenario that pre-authenticate request reaches instance 1 but authenticate request reaches instance 2 (i.e. session stickiness not present), locate and configure the following in *WEB-INF/classes/am5-ehcache.xml*:

XML

```
<cache
name="com.isprint.am.core.auth.E2EESessionRepository.sessionCache"
maxElementsInMemory="30000"
eternal="false"
timeToIdleSeconds="0"
timeToLiveSeconds="360"
overflowToDisk="false">
  <!-- if preauthenticate/authenticate could end up with
different instances of AM5 servers, uncomment the following to
enable replication and test -->
  <cacheEventListenerFactory
class="net.sf.ehcache.distribution.RMICacheReplicatorFactory"
properties="replicateAsynchronously=true,
replicatePuts=true,
replicateUpdates=true,
replicateUpdatesViaCopy=true,
replicateRemovals=true,
asynchronousReplicationIntervalMillis=100"/>
</cache>
```

- **RADIUS Session Cache**

Radius server cache is required if RADIUS challenge response protocol is used. It is defined as:

com.isprint.am.core.auth.EhCacheSessionRepository.sessionCache

No replication is necessary, as normally radius client will stick to same RADIUS server during the entire RADIUS challenge-response session.

Cache Peers

Locate the **Cache Peers Management** page by navigating to **Administration Dashboard > System > Server & Cache > Cache Peers** on the left pane.

This page displays information according to the type of cache replication (RMI/ JMS/ Redis) being used.



Note: For more information on how to configure cache replication, or how to activate multicast, please refer to the [AccessMatrix Common Deployment Guide - Installation and Deployment - Cache Configurations](#).

RMI/JMS Cache Replication

The system displays a list of IP addresses, replication ports, and replicated caches of the Ehcache peers. The information displayed is organized according to the respective cache groups.



Notes:

- If JMS cache replication is configured, the **Total Active Peers** column will display zero.
- If RMI cache replication is configured, the **Total Active Peers** column will display the count of active peers.

Redis Cache Replication

The system indicates if the current AccessMatrix server instance has subscribed to the Redis Pub/Sub channel. The information displayed includes the:

- Connected Redis server's IP address and port
- Connection status with the Redis cluster
- Connected Redis Pub/Sub channel
- Number of subscribers

Server Monitoring

Server monitoring configuration is a useful feature that allows the administrator to configure the AccessMatrix system to send out alert messages for certain system monitoring events. Some of these events include JVM free memory warning or critical threshold, event manager thread pool queue size, etc.

These settings are applicable globally throughout the entire AccessMatrix system. The administrator is not required to modify the settings unless necessary. The administrator is only allowed to view and edit the server monitoring configuration entry.

To locate the **Server Monitoring** page, navigate to **Administration Dashboard > System > Server Monitoring**. Server Monitoring can be edited by clicking **Edit** on the **View Configuration** page.

Server Monitoring Page Attribute Description

Attribute Name	Description
*Configuration Id	System pre-defined server monitoring Id.
*Configuration Name	Name of server monitoring configuration.
Description	Description of server monitoring.
Version	Server monitoring entry version.
Implementation	Server monitoring implementation.
Attributes Tab	
SafeNet PHEFT HSMs	A MonitoringCritical event will be reported if health check on the HSMs listed here failed.
SafeNet PSG/PSE HSMs	A MonitoringCritical event will be reported if health check on the HSMs listed here failed.
nCipher HSMs	Specify nCipher HSM's Electronic Serial Number (ESN). A MonitoringCritical event will be reported if health check on the HSMs listed here failed.
Thales payShield HSMs	A MonitoringCritical event will be reported if a health check on the HSMs listed here failed. Specify as "host: port" format, e.g. "192.168.1.45:80". Only non-TLS ports are accepted.
PKCS#11 HSMs	A MonitoringCritical event will be reported if a health check on the HSMs listed here failed. Specify as PKCS#11 Key Provider.
JVM Free Memory Warning Threshold (MB)	A MonitoringWarn event will be reported if JVM free memory is less than this value. Note: Administrator can configure event notification policy to send out an alert. Default: 100
JVM Free Memory Critical Threshold (MB)	A MonitoringCritical event will be reported if JVM free memory is less than this value. Note: Administrator can configure event notification policy to send out an alert. Default: 50
Event Manager Thread Pool Queue Size	A MonitoringWarn event will be reported if event manager thread pool size is greater or equal than half of this value, or we will notify a MonitoringCritical event if the thread pool size is greater or equal than this value. Note: Administrator can configure event notification policy to send out

Attribute Name	Description
	an alert. Default: 60
Hibernate Session Threshold	A MonitoringWarn event will be reported if the number of session to the database is greater or equal than this value. Note: Administrator can configure event notification policy to send out an alert. Default: 20
Average Authentication Response Time Warn Threshold	A MonitoringWarn event will be reported when average authentication response time (measured over last 1000 samples per server instance) exceeds this value. Specified in milliseconds. Default: 88
Average Authentication Response Time Critical Threshold	A MonitoringCritical event will be reported when average authentication response time (measured over last 1000 samples per server instance) exceeds this value. Specified in milliseconds. Default: 5000
UAM Check Authorization Response Time Warning Threshold	A Stopwatch trace will be logged as WARN in am.log if UAM WSA checkAuthz response time exceeded this value in milliseconds. Default: 88
Tomcat Connector Name	Specify the Tomcat Connector Name for threads monitoring. It requires a restart for the change to take effect. Default: https-jsse-nio-8443
Tomcat Thread Busy Count Warn Threshold	A MonitoringWarn event will be reported if Tomcat's thread busy count exceeded this value. The default value is based on a blocking connector. Tomcat monitoring by default is enabled, the attributes (protocol, port) can be configured in a server.xml file Default: 50
Tomcat Thread Busy Count Critical Threshold	A MonitoringCritical event will be reported if Tomcat's thread busy count exceeded this value. The default value is based on a blocking connector. Default: 100
Trigger critical events only once for HSM(s)	This is a special handling for HSM related indicators, such as SafeNet PHEFT HSMs and SafeNet PSG/PSE HSMs, where its critical events and event notifications (if any) would only be triggered when the events are critical and different from the previous status. Default: false

Logging

AccessMatrix utilizes an Apache Log4j2 module (a Java-based logging utility) as its logging framework. The logging utility allows the system to specify the level of logging as well as the messages to be logged. It also provides a flexible way for the system to configure its log level such as via a property or XML file or on the fly. With this flexibility, the administrator can configure the log level either through the default **log4j2.properties** file used by the AccessMatrix system or through the administration console.

About Log Files



Note: The term `<ACMATRIX_ROOT>` refers to the root or installation directory of AccessMatrix on both Windows and Unix platforms.

The log file is a text file that contains detailed information about the results of the task or event for each individual file. The logging configuration page displays the list of logging options grouped in a different category: Core, USO, UCM, UAS, UAM, UIM, API and Third Party. The default log levels are retrieved from **amlog4j2.properties/amlog4j2.properties.xml**. After you start up the AccessMatrix application, the log files can be found in the `<ACMATRIX_ROOT>\tomcat\logs` folder. Each log filename is prefixed with the hostname where AccessMatrix is running. For example, if the hostname is AccessMatrix-PC, then the log file names will be as follow.

- AccessMatrix-PC_am.commonlogin
 - AccessMatrix-PC_am
 - AccessMatrix-PC_am.uastap
 - AccessMatrix-PC_am.usoagent
 - AccessMatrix-PC_am.usosf
 - AccessMatrix-PC_audit
 - AccessMatrix-PC_event
 - AccessMatrix-PC_housekeeping
 - AccessMatrix-PC_offline-report
 - AccessMatrix-PC_request
 - AccessMatrix-PC_usoserver
 - AccessMatrix-PC_velocity
-

By default, a system-defined logging configuration will be created upon successful installation of AccessMatrix. The administrator is allowed to view and configure the logging options on the fly. To locate the **Logging** page, navigate to **Administration Dashboard > System > Logging**. On the **Logging Configuration** page, click **Edit** to modify the logging configurations. See the list of logging level and its descriptions below.

Level	Description
ALL	All levels
DEBUG	Indicates detailed and informational messages that are useful to debug
ERROR	Indicates the result errors that might allow the application to continue running
FATAL	Indicates very severe error messages that might cause the application to terminate
INFO	Indicates informational messages that tell the progress of the application at a coarse-grained level
OFF	No logging
TRACE	Highly detailed tracing messages more than DEBUG
WARN	Indicates potential problems

The following tasks can be performed during Edit mode:

- **Add Log**

Click the **Add Log** button to display the **Add Log** dialog box, enter the details and click **OK**.

- **Reset to Default Level**

Click the **Reset to Default Level** button to reset the log level of the logging options back to default.



Note: Any modification on log configurations will not be persisted. All logging configurations will return to the default configuration as defined in **amlog4j2.properties/amlog4j2.properties.xml**.

Manage Log Files

The log files will be propagated in the `<ACMATRIX_ROOT>\tomcat\logs\` folder for each product's logging. By default, the AccessMatrix system will roll over these log files every day and compress them to reduce

the files' size. The default compression is using GZ format. You can optionally change the format to ZIP. See below.

You can find current log files, as well as compressed ones (GZ) that contain the previously rolled-over log file for each logging, in the same folder. See below for more information. The compression and rollover of log files are applicable for AccessMatrix products such as AM5, CLP, UASTAP, USOSERVER and USOSSF.

- AccessMatrix-PC_am.commonlogin
- AccessMatrix-PC_am.commonlogin.log.2019-09-02.gz
- AccessMatrix-PC_am
- AccessMatrix-PC_am.log.2019-09-02.gz
- AccessMatrix-PC_am.uastap
- AccessMatrix-PC_am.uastap.log.2019-09-02.gz
- AccessMatrix-PC_am.usoagent
- AccessMatrix-PC_am.usoagent.log.2019-09-02.gz
- AccessMatrix-PC_am.usossf
- AccessMatrix-PC_am.usossf.log.2019-09-02.gz
- AccessMatrix-PC_audit
- AccessMatrix-PC_event
- AccessMatrix-PC_event.log.2019-09-02.gz
- AccessMatrix-PC_housekeeping
- AccessMatrix-PC_offline-report
- AccessMatrix-PC_request
- AccessMatrix-PC_usoserver
- AccessMatrix-PC_usoserver.log.2019-09-02.gz
- AccessMatrix-PC_velocity



Note: The date of the latest log file is included in the compressed file name. This is customizable in the respective product's **log4j2.properties** file.

For Windows platform, you can use an appropriate program to open the GZ compressed file and view the log files for debugging or troubleshooting. Should you need to change the compression method, you can customize the **amlog4j2.properties** file in the folder <ACMATRIX_ROOT>\tomcat\webapps\am5\WEB-

INF\classes by uncommenting the line below to change the compressed file type from a GZ to a ZIP compressed file.

Properties (.properties files)

```
#appender.LOG_ROOT.filePattern =  
logs/${hostname}_am.log.%d{yyyy-MM-dd}.zip
```

Comment the following line and restart the AccessMatrix server for the change to take effect.

Properties (.properties files)

```
appender.LOG_ROOT.filePattern =  
logs/${hostname}_am.log.%d{yyyy-MM-dd}.gz
```

For other products, modify the related properties files for each product at the locations as listed below.

Product	Properties file	Folder
Common Login Page (CLP)	clplog4j2	<ACMATRIX_ROOT>\tomcat\webapps\clp\WEB-INF\classes
UAS Token Activation Portal (UASTAP)	uastap-log4j2	<ACMATRIX_ROOT>\tomcat\webapps\uastap\WEB-INF\classes
USO Server (USOSERVER)	usoagentlog4j2	<ACMATRIX_ROOT>\tomcat\webapps\usoserver\WEB-INF\classes
USO Self Service Function (USOSSF)	usossflog4j2	<ACMATRIX_ROOT>\tomcat\webapps\usossf\WEB-INF\classes

i **Note:** You can also modify the maximum number of log files and more in each product's **log4j2.properties** file.

Audit Policy

Audit policy governs the behavior of the audit module e.g. enable audit event filter, suppress all audit events, etc. The outcome of the audit information can be viewed through the audit report generation.

By default, a system-defined audit policy will be created upon successful installation of AccessMatrix. The administrator is allowed to view and configure the audit policy information. To locate the **Audit Policy** page, navigate to **Administration Dashboard > Security Policy > Admin & Audit > Audit Policy**

on the left pane. Audit policy configurations can be edited by clicking **Edit** on the **View Audit Policy** page.

View Audit Policy Page Attribute Description

Attribute Name	Description
*Audit Policy Id	System pre-defined audit policy Id.
*Audit Policy Name	Name of audit policy.
Description	Description of audit policy.
Version	Audit policy entry version.
Attributes Tab	
Audit Data Integrity Check	
Data Integrity Key	Key used to sign the audit data fields for the purpose of data integrity checking.
Audit events that will not be recorded	
Filter out the following event categories and types without logging	The specified events in this field will be filtered out from database audit logs. • Click the Add button to add a new event category and type. The newly added value(s) would be highlighted. • Click the Add Default Values button to add the default list of events in addition to the current value(s), if there's any. • Click the Remove button to delete the selected event(s). Multiple selection of events is allowed. Default: Token-Modify, UCM-CredVerifierStarted, Token-Add, Server-MonitoringCritical, UCM-CheckoutOverdueTaskStarted, UCM-CheckoutOverdueTaskCompleted, UCM-CheckinOverdueTaskStarted, Server-MonitoringWarn, UCM-CredUpdaterCompleted, Server-MonitoringInfo, UCM-CredUpdaterStarted, Persistence-All, UCM-CredUpdaterPaswdExpCompleted, UCM-CredVerifierCompleted, UCM-CsvImportTaskStarted, UCM-CredUpdaterPaswdExpStarted, OAuth-Authz, UCM-CsvImportTaskCompleted, OAuth-RevokeToken, UCM-CheckinOverdueTaskCompleted
Audit events output redirections	
Redirect all audit events	If set to "true", then all audit events specified in the Audit events that will be redirected attribute will be redirected to the selected value in the File only, File

Attribute Name	Description
	and DB attribute. Default: false
Audit events that will be redirected	List of events that will be redirected.
File only, File and DB	Options: "File only" - Saves the audit events to a file. "File and DB" - Saves the audit events to a file and database. Default: File only
Other options	
Turn on audit to capture the transaction data of token digital signature	If enabled, the transaction data of token digital signature will be captured. Default: false
Audit right checking option	Options: "Normal Security Without Masking" - Allows viewing of the audit record if one is authorized to view either actor or target of the audit record. "Normal Security With Masking" - Allows viewing of the audit record if one is authorized to view either actor or target of the audit record, the Id of the entity that is not authorized to be seen would be masked. "Medium High Security Without Masking" - Shows audit event if and only if the login user is authorized to view actor of the audit record. "High-Security" - Does not show audit event unless the login user is authorized to view both actor and target of the audit record. Default: Normal Security Without Masking
Default Query Limit	Sets the default value of the query limit for Audit Reports. Default: 1000
Maximum Query Limit	Specifies the maximum query limit that can be set and is used to validate against the user-entered query limit in Audit reports. Default: 10000

Attribute Name	Description
Biometric Image File Location	Specify the absolute directory path to store the image files during biometric enrollment/verification. The image file should be stored in shared file system storage where it is writable and accessible by all AM instances.

4.2.9. Restriction Management

AccessMatrix has provided features to control user access by means of time and location restrictions.

Time Restriction

It is a useful feature for controlling user access to the AccessMatrix system by means of time restriction. Such access control is based on the date/time which the user accesses the system and is governed by the restriction rules defined by the administrator. It is optional for the administrator to perform such Time Restriction Administration.

To locate the **Time Restriction** page, navigate to **Administration Dashboard > Security Policy > Restriction > Time Restriction** on the left pane. The following tasks can be performed on the time restrictions:

- **Create Time Restriction**

Click **Create** to display the **Choose Time Restriction Implementation** page. Select an implementation and click **Next** to open the **Create Time Restriction** page. Enter the details and click **Save**.

- **Search for Time Restriction**

Time restriction can be searched by using the time restriction Id or the time restriction name. All available time restrictions are displayed by default.

- **View Time Restriction**

Click a time restriction Id under the **Search Time Restriction Result** list to view the details.

- **Edit Time Restriction**

Click **Edit** on the **View Time Restriction** page to modify the details and click **Save**.

- **Delete Time Restriction**

Click **Delete** on the **View Time Restriction** page and click **OK** in the confirmation pop-up box.

Location Restriction

It is another useful feature for controlling user access to the AccessMatrix system by means of location restriction in terms of IP addresses. Such access control is based on the location from which the user accesses the system and is governed by the restriction rules defined by the administrator. It is optional for the administrator to perform such Location Restriction Administration.

To locate the **Location Restriction** page, navigate to **Administration Dashboard > Security Policy > Restriction > Location Restriction** on the left pane. The following tasks can be performed on the location restrictions:

- **Create Location Restriction**

Click **Create** to display the **Choose Location Restriction Implementation** page. Select an implementation and click **Next** to open the **Create Location Restriction** page. Enter the details and click **Save**.

- **Search Location Restriction**

Location restriction can be searched by using the location restriction Id or the location restriction name. All available location restrictions are displayed by default.

- **View Location Restriction**

Click a location restriction Id under the **Search Location Restriction Result** list to view the details.

- **Edit Location Restriction**

Click **Edit** on the **View Location Restriction** page to modify the details and click **Save**.

- **Delete Location Restriction**

Click **Delete** on the **View Location Restriction** page and click **OK** in the confirmation pop-up box.

4.3. Report Dashboard

This dashboard provides users with real-time operational data, historical audit data in the form of reports and system monitoring information. Report dashboard of Admin Console user interface contains the first-level navigation bar with **Home**, **System & Admin**, **Segment & User**, **Authentication**, **Enterprise SSO & Web SSO**, **Token & SMS**, **UCM**, and **About** options.

Selecting this dashboard will display the user's last successful login date and time based on the user's most recent login events or history.

The reporting function of AccessMatrix allows the administrator to generate real-time and historical reports on user and system activities. AccessMatrix keeps track of all activities occurred via audit, which can be generated through various reports available under the Reporting Dashboard.

All reports can be generated into an HTML, PDF or CSV file. Filtering options are available to generate a clear and precise report with only the required chunk of information. Different filtering options are available for each reporting function providing dynamism and maximum efficiency of each report.

AccessMatrix also has an audit record digital signature feature. All audit records generated are digitally signed for tamper-proof validation, ensuring that the audit reports and their contents have not been modified by an external party.



Notes:

- While generating reports, leaving a field blank or entering a ' * ' wildcard will search for all available results.
- Attributes marked with ' * ' denote mandatory; a value must be entered to be able to generate the report.

4.3.1. System & Admin Reports

System & Admin Reports include system configuration and administration-related information.

Real Time Reports

Real-time reports allow the administrator to perform live monitoring of the system and user events.

Below are the system and admin related real-time reports.

Monitoring Report

This report provides information on the number of concurrent sessions in the AccessMatrix system and other system information.

To locate the **Monitoring Report** page, navigate to **Report Dashboard > System & Admin > Real Time > Monitoring** on the left pane. Click on the desired report from the various monitoring reports available to display the report graph. See below for the report details:

Report	Description
System Information	
JVM Maximum Memory	Maximum JVM allocated.
JVM Free Memory	Available JVM Memory. This value fluctuates due to garbage collection.
JVM Thread Count	Total number of JVM threads.
Event Thread Pool Active Count	Indicates non-blocking events processing concurrently (e.g. event notification, audit).
Event Thread Pool Queue Size	Once the event manager thread pool runs out of free threads, new events will be put in a queue, and the queue size will start to grow as more events come in.
Event Thread Pool Rejected Count in Time Window	When the event thread pool queue is full, additional events coming in will be rejected. This indicates the number of rejected events within a time frame.
Event Thread Pool Rejected Count	Total number of rejected events.

Report	Description
Authentication Response Time (Average)	The average time it takes to perform authentication.
Authentication Response Time (90th percentile)	90th percentile of authentication response time is usually more reliable than the average response time, as the minimal occurrence of the anomaly may affect the average number to reflect inaccurately.
Tomcat Busy Thread Count (https-jsse-nio-8443)	Currently active Tomcat thread count. Indicates concurrency of web service requests. A constant high number indicates persistent load or slow response time at the database.
Tomcat's Current Thread Count (https-jsse-nio-8443)	Total number of Tomcat thread count.
Hibernate	
Hibernate Session Count	Indicates the level of the concurrent database I/O.
Tomcat DBCP	
Active Connections	The number of currently active connections.
Idle Connections	The number of idle connections.
All Connections	The number of total connections.
Delivery Channel	
Delivery Channel	Information regarding the delivery channel.
HSM Information	
Thales payShield - HSM Host and Port	Indicates the host and port of Thales payShield HSM(s), i.e. "192.168.50.101:1500".
Initialize	Indicates if Thales payShield HSM initializes appropriately.

Admin Privilege Report

This report provides information on the user's admin roles assignment (s) or group(s).

To locate the **Admin Privilege Report** page, navigate to **Report Dashboard > System & Admin > Real Time > Admin Privilege** on the left pane. Enter relevant search criteria to generate the report. The generated report reflects user information, account status, and admin role entitlements assigned. Refer to [Admin Privilege Report Sample](#). See below for the attributes description of the generated report:

Attribute Name	Description
Role Id	The Id of the admin role assigned to the user.
Role Name	The name of the admin role assigned to the user.
Entitlement	Indicates whether the admin role was assigned to the user as an individual or as a part of group entitlement.
Delegation Type	Indicates the degree of user's delegation ability, i.e. whether users can delegate an admin role for themselves and to fellow users.
Restricted Segment Id(s)	Restricted Segments where the admin role is applicable.
Restricted Token Group Id(s)	Restricted Tokens where the admin role is applicable.

Token Assignment Count By Token Vendor Report

This report displays if only the UAS license feature **Enable Token Usage Billing** is set to "Yes". It is a UAS Module-specific report. Refer to the [UAS Administration Guide](#) for more details.

System Summary Report

This report presents the current statistical information about the system. To locate the **System Summary Report** page, navigate to **Report Dashboard > System & Admin > Real Time > System Summary** on the left pane. On the right pane, select the type of output report file (HTML or PDF) on the dropdown list, and then click the **Generate** button. Refer to [System Summary Report Sample](#). See below for the attributes and its descriptions of the generated report.

Attribute Name	Description
AccessMatrix version	Current version of the AccessMatrix product.
JVM version	Current version of the JAVA virtual machine.
OS Version	Current version of the Operating System platform used in the server(s).
Computer RAM	Memory of the computer data storage that stores data and machine code currently being used.
JAVA Memory	JAVA heap memory that uses to store objects.

Attribute Name	Description
Number of processor	Number of the processor chips found in the computer to receive input and provide the appropriate output.
Web Container	It is also known as a servlet container, a web server component that interacts with Java servlets.
Number of Servers	Number of servers currently in use.
User count per Login Module	The last checked date and time of user count per Login Module.
Login Module Handler	Count of users based on each Login Module Handler.
Count of assigned Tokens	The last checked date and time of the assigned tokens.
Token vendor (model)	Count of assigned tokens based on the token vendor and token model.
Number of users in Default userstore	Count of users currently in the default user store (Default Store).

Historical Reports

Historical reports allow the administrator to analyze the past system and user events. Below are the system and admin related historical reports.

Events Report

This report provides information regarding the AccessMatrix system's events within the specified range of date and time.

To locate the **Events Report** page, navigate to **Report Dashboard > System & Admin > Historical > Events** on the left pane. Enter relevant search criteria to generate the report. If the **Include actions done by background tasks** attribute is set to "Yes", the report will also include the system triggered events, on top of user-triggered events. Identifying the **IP Address** attribute will only generate events triggered from that specific IP address. Refer to [Events Report Sample](#). See below for the attributes description of the generated report:

Attribute Name	Description
Audit Id	System-generated Id for audit purposes.

Attribute Name	Description
Server Id	Refers to the server Id.
Actor User Id@User Store	Id of the user who performed the particular action and the name of the User Store the user belongs to.
Actor IP Address	IP address of user who performed the particular action.
Actor Segment Id@Segment Store	Name of the segment and the segment store the action performing user belongs to.
Agent Segment Id@Segment Store	Name of the agent segment and the segment store the agent segment belongs to. Agent segment is a non-interactive middle application that connects/accesses the Access Matrix server.
Reported Time	The time the event took place.
Event Category	Category of the event, such as "Session Timeout" or "Login Authentication".
Target Id	Id of the target user that the event was performed on.
Target Class	The Class of the Targeted user.
Target Segment Id@Segment Store	Name of the segment and the segment store the target user belongs to.
Target Field (New Value)	Name of the fields where the event had affected and the fields' new values.
(A - Authenticated, NA = Not Authenticated)	If the event is an authentication event, it indicates what kind of authentication and the status of the authentication result.
Event Result	Indicates whether the event was successful or failed.
Message	Short description of the event.
Result Code	Short description/code of the result.

Modification History Report

This report provides the modification history based on User, Group or Policy within the specified range of date and time.

To locate the **Modification History Report** page, navigate to **Report Dashboard > System & Admin > Historical > Modification History** on the left pane. Enter relevant search criteria to generate the report. For User and Group based reports, it is possible to generate the report by specifying a particular User or

Group Id. However, for a policy-based report, it is currently not possible to filter by a particular policy name. Refer to [Modification History Report Sample](#). See below for the attributes description of the generated report:



Note: The report display is based on date (from earliest to latest) and by Target Id.

Attribute Name	Description
Audit Id	System-generated Id for audit purposes.
Server Id	Refers to the server Id.
Actor User Id@User Store	Id of the user who performs the report generation and the user store that the performing user belongs to.
Actor IP Address	IP address of the machine requesting the report generation.
Reported Time	Time the modification was done.
Event Category	Type of modification performed.
Target Id	Id of the user/account the modification was performed on.
Target Class	The Class of the target user.
Target Field (Old Value) (New Value)	The name of the fields modifications were done. Previous value and changed value of the target fields.
Event Result	Indicates whether the event was successful or failed.

Admin Workflow Request Report

This report provides the admin workflow request history and status within the specified range of date and time.

To locate the **Admin Workflow Request Report** page, navigate **Report Dashboard > System & Admin > Historical > Admin Workflow Request** on the left pane. Enter relevant search criteria to generate the report. Refer to [Admin Workflow Report Sample](#). See below for the attributes description of the generated report:

Attribute Name	Description
Audit Id	System-generated Id for audit purposes.
Server Id	Refers to the server Id.

Attribute Name	Description
Actor User Id@User Store	Id of requesting user and the User Store's name the user belongs to.
Actor IP Address	IP address of the user who performed the request.
Actor Segment Id@Segment Store	Name of the segment and the segment store the action performing user belongs to.
Agent Segment Id@Segment Store	Name of the agent segment and the segment store the agent segment belongs to. Agent segment is a middle application that connects/access the Access Matrix server.
Reported Time	The time the request took place.
Event Type	Type of event such as "Add" or "Process".
Requested Id	System generated request number.
Request Content Type	Type of request such as "Role" or "User".
Content Summary	Summary of the content, such as Role Name or User Name.
Workflow Type	The workflow type for the new request. Only the "Default Dual Control" workflow is currently available.
Request Action	The request action such as "Submit", "Approve", "Reject", etc.
Request Status	The current status of the request such as "Approved", "Pending", "Withdrawn".
Changes Version	Number of entry version.
Field Scope/Field Name/Old Value/New Value	Fields whose values are affected by the request, old and new field values.
Event Result	Indicates whether the event was successful or failed.
Message	Short description of the event.
Result Code	Short description/code of the result.

4.3.2. Segment & User Reports

Segment & User Reports provide all information regarding users and user activities.

Real Time Reports

Real-time reports allow the administrator to perform live monitoring of the system and user events.

Below are the segment and user-related real-time reports.

Inactive User Report

This report provides a list of inactive user(s) based on the user's last login date. For users who have never login before, the calculation of the inactive days would be based on user-created time.

To locate the **Inactive Users Report** page, navigate to **Report Dashboard > Segment & User > Real Time > Inactive User** on the left pane. Enter relevant search criteria to generate the report. See below for the attribute descriptions of the generated report:

Attribute Name	Description
User Id	The user Id of the inactive user.
Segment Id	Id of the segment the inactive user belongs to
Status	User account status: "Activated" or "Disabled".
Last Login	Last login date and time of the user
Last Modified Time	Date and time the user account was last modified
Inactive Days	Number of days user has been inactive

User Login Account Status Report

This report provides the user the status of the user login account.

To locate the **User Login Account Status Report** page, navigate to **Report Dashboard > Segment & User > Real Time > User Login Account Status** on the left pane. Enter relevant search criteria to generate the report. See below for the attribute descriptions of the generated report:

Attribute Name	Description
User Id	Login account user Id.
User Store	Name of user store where the login user belongs to.
Segment Id	Id of the segment the login user belongs to.
Account Status	User account status: "Activated" or "Disabled".
External Status	User account status in the external user store.
Authentication Realm	Type of authentication realm assigned to the login account.
Realm Status	Status of authentication realm: "Activated" or "Disabled".
Last Successful Login	Last successful login date and time.
Last Failed Login	Last failed login date and time.

User Grouped By Segment Report

This report provides information about the list of users under a segment.

To locate the **User Grouped By Segment Report** page, navigate to **Report Dashboard > Segment & User > Real Time > User Grouped By Segment** on the left pane. Enter relevant search criteria to generate the report. See below for the attribute descriptions of the generated report:

Attribute Name	Description
Segment Id	Unique Id of the segment.
Segment Name	Name of segment.
User Account Status	User account status: "Activated" or "Disabled".
User Id	User Id(s) that belongs to the segment.
Realm Id (Status)	Unique Id of the authentication realm and status of the Realm: "Activated" or "Disabled".
Login Module Id (Status)	The login module Id that the user Id is associated with and the status of the login module Id.

Attribute Name	Description
Total By Segment and User Account Status	Total number of users in that segment categorized by user account status.
Total By Segment	Total number of users in that segment.

User Summary By Segment Report

This report provides information about the summary of the users assigned to a segment.

To locate the **User Summary By Segment Report** page, navigate to **Report Dashboard > Segment & User > Real Time tab > User Summary By Segment** on the left pane. Enter relevant search criteria to generate the report. See below for the attribute descriptions of the generated report:

Attribute Name	Description
Segment Id	Unique Id of segment.
Segment Name	Name of segment.
Hierarchical Path	Path of parent segment.
Account Status	User account status: "Activated" or "Disabled".
Total Users	Total number of users associated with the segment.

Group Membership By Segment Report

This report provides information about the group membership by segment in the AccessMatrix system.

To locate the **Group Membership By Segment Report** page, navigate to **Report Dashboard > Segment & User > Real Time tab > Group Membership By Segment** on the left pane. Enter relevant search criteria to generate the report. See below for the attribute descriptions of the generated report:

Attribute Name	Description
Group Name	Name of group.
Segment Id	Id of segment the group belongs to.
Number of Users	Number of users under the group.

Attribute Name	Description
Number of Admin Privilege	Number of Admin Privileges assigned to the group.

Application Summary By Segment Report

This report provides information about the application summary by segment in the AccessMatrix system.

To locate the **Application Summary By Segment Report** page, navigate to **Report Dashboard > Segment & User > Real Time > Application Summary By Segment** on the left pane. Enter relevant search criteria to generate the report. See below for the attribute descriptions of the generated report:

Attribute Name	Description
Segment Id	Unique Id of the segment.
Segment Name	Name of the segment.
Hierarchical Path	Path of the parent segment.
Number of Application Security Agent	Number of Application Security Agents associated with the segment.
Number of Web Security Agent	Number of Web Security Agents associated with the segment.
Number of USO Application	Number of USO Applications associated with the segment.
Total Applications	Total number of applications associated with the segment.

Segment Hierarchy Report

To locate the **Segment Hierarchy Report** page, navigate to **Report Dashboard > Segment & User > Real Time > Segment Hierarchy** on the left pane. Enter relevant search criteria to generate the report. See below for the attribute descriptions of the generated report:

Attribute Name	Description
Segment Id	Unique Id of the segment.
Segment Name	Name of the segment.
Hierarchical Path	Path of the parent segment.

Attribute Name	Description
Total Sub Segments	Number of sub-segments under the parent segment.
Total Users	Number of users under the segment.
Total User Groups	Number of groups under the segment.
Total Token Groups	Number of token groups under the segment.
Total USO Applications	Number of USO applications under the segment.
Total Access Agents	Number of access agents under the segment.

Historical Reports

Historical reports allow the administrator to analyze the past system and user events. Below are the segment and user-related historical reports.

User Activities Report

This report provides information about the events incurred by the user activities during the use of the AccessMatrix system and can be based on certain event sub-categories. The administrator may generate the reports based on the desired report criteria.

To locate the **User Activities Report** page, navigate to **Report Dashboard > Segment & User > Historical > User Activities** on the left pane. Enter relevant search criteria to generate the report. Refer to [User Activities Report Sample](#). See below for the attribute descriptions of the generated report:

Attribute Name	Description
Actor User Id@User Store	Id of the requesting user and the name of the user store the user belongs to.
Audit Id	System-generated Id for audit purposes.
Server Id	Refers to the server Id.
Actor Segment Id@Segment Store	Name of the segment and the segment store the action performing user belongs to.
Agent Segment Id@Segment Store	Name of the agent segment and the segment store the agent segment belongs to. Agent segment is a middle application that connects/access the AccessMatrix server.

Attribute Name	Description
Report Time	Time of user activity.
Event Category	Refers to the type of events.
Actor IP Address	IP address of the user who performed the request.
Target Id	Id of the user/account the activity was done on.
Target Class	The class of the target user.
Target Segment Id@Segment Store	Name of the segment and the segment store the target user belongs to.
Target Field (New Value)	The target field of the user activity and the new value in the target field.
(A = Authenticated, NA = Not Authenticated)	If the event is an authentication event, it indicates what kind of authentication and the status of the authentication result.
Event Result	Indicates whether the event was successful or failed.
Message	Short description of the event.
Result Code	Short description/code of the result.

Inactive User By Date Range Report

This report provides a list of users who were inactive within a specified date range. It can be further filtered by authentication realm Id.

To locate the **Inactive User By Date Range Report** page, navigate to **Report Dashboard > Segment & User > Historical > Inactive User By Date Range** on the left pane. Enter the relevant search criteria to generate the report. See below for the attribute descriptions of the generated report:

Attribute Name	Description
User Id	The user Id of the inactive user.
Last Login Success	The last successful login date and time.
Last Login Failure	The last failed (unsuccessful) login date and time.

Inactive User By Federated Applications Report

This report provides a list of users belonging to the specified federated application(s) who were inactive within a specified date range.

To locate the **Inactive User By Federated Applications Report** page, navigate to **Report Dashboard > Segment & User > Historical > Inactive User By Federated Applications** on the left pane. Enter the relevant search criteria to generate the report. See below for the attribute descriptions of the generated report:

Attribute Name	Description
User Id	The user Id of the inactive user.
User Name	The user name of the inactive user.
Segment Id	The unique Id of the segment.

4.3.3. Authentication Reports

Authentication Reports cover all authentication-related actions in the system.

Real Time Reports

Real-time reports allow the administrator to perform live monitoring of the system and user events.

Below are the authentication-related to real-time reports.

E2EE PIN Mailer Activities Report

This report provides information about the E2EE PIN mailer activities audited in the AccessMatrix system.

To locate the **E2EE PIN Mailer Activities Report** page, navigate to **Report Dashboard >**

Authentication > Real Time > E2EE PIN Mailer Activities on the left pane. Enter the relevant search criteria to generate the report. See below for the attribute descriptions of the generated report:

Attribute Name	Description
PIN Mailer Id	Unique PIN mailer Id.
PIN Mailer Work Flow Module Id	Unique Id of the workflow of the PIN mailer.
Modified Time	Date and time of modification.
PIN Mailer Status	Current status of PIN mailer.
PIN Mailer Template Id	System-generated template Id.
Content Type	PIN mailer content type.
Content Store	PIN mailer user store.
Content Id	PIN mailer user Id.
Content Summary	Summary of PIN mailer message.
Delivery Summary	Summary of delivery details.
Delivery Channel Id	Id of the delivery channel used.
Creator Id@User Store	User Id of PIN mailer creator.

Attribute Name	Description
Created Time	The time when the PIN mailer was created.
Maker Id@User Store	Requested user Id and the user store the requested user belongs to.
Made Time	The time when the PIN mailer request was created.
Checker Id@User Store	Request processor Id and the user store the processor belongs to.
Checked Time	The time when the PIN mailer request was processed.

User Bad Login Report

This report provides information about the bad login attempts.

To locate the **User Bad Login Report** page, navigate to **Report Dashboard > Authentication > Real Time > User Bad Login** on the left pane. Enter the relevant search criteria to generate the report. See below for the attribute descriptions of the generated report:

Attribute Name	Description
User Id	User Id with the bad login.
Segment Id	The name of the segment the user belongs to.
Login Module Id	Id of the login module used.
Bad Login Attempts	Number of failed login attempts.
Login Module Account Status	Status of login module: "Activated" or "Disabled".
Realm Id	Unique Id of the authentication realm.
Last Failed Login	Last failed login date and time.

Historical Reports

Historical reports allow the administrator to analyze the past system and user events. Below are the authentication-related historical reports.

Login Violation Report

This report provides information about the login violation incurred by the user trying to login to the AccessMatrix system and can be based on certain violation types. The administrator may generate the report based on the desired report criteria.

To locate the **Login Violation Report** page, navigate to **Report Dashboard > Authentication > Historical > Login Violation** on the left pane. Enter the relevant search criteria to generate the report. Refer to [Login Violation Report Sample](#). See below for the attribute descriptions of the generated report:

Attribute Name	Description
Actor User Id@User Store	Id of the requesting user and the name of the user store the user belongs to.
Event(Server Id)	System generated event Id.
Event Category	Event category of the login violation.
Type	Subcategory of the login violation.
Violation Message	Description of the login violation.
Violation Type	Type of login violation.
Client Type	Client type
Client Id	Client Id
Login Module Id (A = Authenticated, NA = Not Authenticated)	If the event is an authentication event, it indicates what kind of authentication and the status of authentication result.
Reported Time	The time when the login violation was reported.
Actor IP Address	IP address of requesting user.

User Login Count Report

This report provides information about the number of each user's logins to the AccessMatrix system within a specified date range. The report can be further filtered to only include active users (any user whose login count is not 0), or only include inactive users (any user whose login count is 0). It can also be filtered by specifying a user Id, which limits the results to the login count of that user only.

To locate the **User Login Count** page, navigate to **Report Dashboard > Authentication > Historical > User Login Count** on the left pane. Enter the relevant search criteria to generate the report. See below for the attribute descriptions of the generated report:

Attribute Name	Description
User Id	The user Id of the user whose login count is displayed.
User Name	The user name of the user whose login count is displayed.
Segment Id	The name of the segment the user belongs to.
Login Count	The number of times the user has logged in to the AccessMatrix system. Note: Unsuccessful logins are also counted as logins in this report.

4.3.4. Enterprised SSO & Web SSO

This is a USO and UAM Module-specific reporting function. Refer to the [USO Administration Guide](#) and [UAM Administration Guide](#) for usage and details.

4.3.5. Token & SMS

This is a UAS Module-specific reporting function. Refer to the [UAS Administration Guide](#) for usage and details.

4.3.6. UCM

This is a UCM Module-specific reporting function. Refer to the *UCM Administration Guide* for usage and details.

4.4. Advanced Configurations

Advanced features are available to cater for individual organization's needs. Generally, it is not required to configure these features:

- [Configure Admin Workflow](#)

4.4.1. Configure Admin Workflow

It is a useful feature that allows the administrator to impose certain admin workflow control onto certain user-defined administrative tasks (in terms of resource class and operations) e.g. system modification, user creation, group deletion, and so on. The administrator is also provided with a feature to restrict applying the desired admin workflow control by these available scopes:

- Segment Scope
- Group Scope
- Token Group Scope
- Actor's Group Scope



Notes:

- If any of the above scopes are applied or defined, the admin workflow control will only become effective for the particular administrative operation upon meeting all the scopes defined for the particular administrative operation.
- Apart from the above scopes, the administrator is also provided with an option to specify whether to apply the workflow on administrative tasks performed through Admin Console only.

Currently, only the system-defined "Default Dual-Control" workflow type is supported as the admin workflow control. The administrator is only allowed to view and edit the admin workflow configuration entry.

As an example, when an administrator performs admin privilege assignment for a user or group, while the system-defined "Default Dual-Control" workflow type has been applied onto the "Admin Privilege" resource, a request will be generated when the administrator (known as Maker) submit the above-mentioned administrative operation.

For each request created, the changes will only be effected provided another administrator (known as Checker) approves the request. The Checker must not be the same Administrator Id as the Maker.

Currently, admin workflow control can only be applied to these resources:

- System
 - Policy
 - Admin Role
 - USO Resource
-

-
- Segment
 - Group
 - User
 - Token

The following table shows the various actions that can be performed onto a request by the Maker and Checker respectively:

Maker	Checker
i. Edit: To edit the request. Once edited, the request is resubmitted with same request id but the version number will be increased by one. ii. Withdraw: To withdraw the request.	i. Approve: To approve the request. Once approved, the changes will be effective. ii. Reject: To reject the request. iii. Delete: To delete the request.

The following table shows the approval workflow actions matrix; actions that can be taken on a admin request under various statuses:

Action/Status	Pending	Approved/Withdrawn/Deleted	Rejected
Approve	Yes	No	No
Reject	Yes	No	No
Delete	Yes	No	Yes
Withdraw	Yes	No	Yes
Edit	Yes	No	Yes

For an administrator (maker) to configure an admin workflow:

To locate the **Admin Workflow** page, navigate to **Administration Dashboard > Security Policy > Admin & Audit > Admin Workflow** on the left pane. To edit Admin Workflow configurations, follow the steps below:

1. Click **Edit** on the Admin Workflow's **View Configuration** page.
 2. On the **Edit Configuration** page, click **Edit** under **Attributes Value** to edit the workflow rule.
The **Admin Workflow** dialog box will pop-up.
-

-
3. Edit the existing admin workflow rules as required. Otherwise, to add new workflow rules, click **Add**. Select the desired resource class and operation as well as the workflow scopes, if applicable, and the workflow type.
 4. To add more rules, repeat step 3. As many rules as deemed fit may be added.
 5. To remove a rule, click on the **X** icon on the **Order** column pertaining to the desired rule.
 6. To reposition a rule, click on red up/down arrow icon on the **Order** column pertaining to the desired rule.
 7. Click **OK** when done and click **Save**.

Now that you have configured an admin workflow, for example, configuring an admin role to have the right to create, edit, delete and view a user group.

For an administrator (maker) to create a request :

1. Navigate to **Administration Dashboard > Security Policy > Admin & Audit > Admin Role** on the left pane.
2. Click **Create**.
3. Enter an admin role Id, admin role name and its description in the respective empty fields.
4. On the **Create Admin Role** page, tick the checkboxes for **Create**, **Delete**, **Edit**, and **View** in the **Group** row.
5. Click **Save**. The information alert box with the "Your request has been submitted" message is shown.
6. Add a comment in the empty column as needed. Click **OK**. A request is created.



Note: To view the request(s), refer to [retrieve the request list](#) under the Request Management section.

Admin Workflow Page Attribute Description

Attribute Name	Description
*Configuration Id	System pre-defined Admin Workflow Id.
*Configuration Name	Name of Admin Workflow configuration.
Description	Description of Admin Workflow.

Attribute Name	Description
Version	Admin Workflow entry version.
Implementation	Admin Workflow implementation.
Attributes Tab	
The Attributes tab contains user-defined workflow rules.	

4.5. Utility Tools

AccessMatrix comes with utility tools to assist the user experience:

- [CSV Data Export/Import](#)

4.5.1. CSV Data Export/Import

CSV Data Export/Import allows the administrator to create objects in the Admin Console by using the import feature and export objects by using the export feature. Only certain system-defined objects can be exported/imported via CSV data format.

Locate the **CSV Data Export/Import** page by navigating to **Administration Dashboard > System > CSV Data Export/Import**.

Exporting CSV Data

1. On the **Exporting CSV Data** page, under the **Export** section, select the desired object(s) to export, e.g. **AuthnRealm**.
2. Enter the desired Encryption Password.

 **Note:** CSV data will not be exported if this field is blank.

3. Click the **Export** button to export all data under the selected object, e.g. all Authentication Realms. Otherwise, click the **Preview** button to select the desired object Id to export, e.g. **40151**. Click **Export** to perform data export.
4. Save the CSV file to the desired location.

Importing CSV Data

1. On the **Exporting CSV Data** page, under the **Import** section, enter the desired Encryption Password.

 **Note:** CSV data will not be imported if this field is blank.

2. Untick the **Create new objects only; do not modify existing objects**. attribute to modify the existing objects. By default, this attribute is ticked to only create new objects and will not modify existing objects.
 3. Click **Choose File** to browse for the CSV file to import.
 4. Click the **Import** button and the **Job Completed** table will be displayed. Navigate to the desired object and check it to make sure it is created.
-

5. Appendices

This section contains the following appendices:

- [A. Admin Role and Admin Right Definitions](#)
- [B. Event Category and Types](#)
- [C. Message Template Macros](#)
- [D. Sample Report Templates](#)

5.1. A. Admin Role and Admin Right Definitions

The following table provides a detailed description of the various predefined admin roles. The administrator is allowed to modify the admin rights assigned to the admin role.

Admin Role	Description
InitialAdminRole	By default, this initial admin role allows administrator to perform all operations in the system.
UserAdminRole	By default, this user admin role allows administrator to perform these operations: User Administration (except login account time editing), Group Administration, USO Resource and Entitlement, Administration, Token Viewing, UAM Administration, Admin Privilege Assignment, System Viewing, Policy Viewing and Segment Administration.
PolicyAdminRole	By default, this policy admin role allows administrator to perform these operations: System Administration, Policy Administration, NAS Client Administration and Encryption Key Administration (except deletion).

Apart from the above predefined admin roles, administrator is also able to define other admin roles containing desired admin rights. The following table provides detailed description for the various system-defined admin rights in terms of the menu option accessible and actions performable:

Resource	Operation	Menu Option Accessible	Actions Performable
User Admin Rights			
User	Create, Edit, Delete, View	Operation Dashboard > User & Segment option > User Tab > Search, Create	Admin Right Dependency: User - Create, Edit, Delete i) Able to search, view, create, edit, and delete user information ii) Only able to view login account
	Assign Group Membership	-	Admin Right Dependency: User - Create, Edit Group - View i) Able to assign/unassign group membership

Resource	Operation	Menu Option Accessible	Actions Performable
	Edit Login Account Status, Edit Login Account Time	-	Admin Right Dependency: User - View Policy – View i) Able to edit login account start time and end time
	Reset Password	-	Admin Right Dependency: User - View User - Reset Password i) Able to reset password
	Able to assign/unassign admin privilege	-	Admin Right Dependency: User - View Admin Privilege - Assign i) Able to assign/unassign admin privilege
Group	Create, Edit, Delete, View	Operation Dashboard > User & Segment option > Group Tab > Search, Create	Admin Right Dependency: Group - Create, Edit, Delete i) Able to search, view, create, edit, and delete group information ii) Only able to view membership
	Able to assign/unassign admin privilege	-	Admin Right Dependency: Group - View Admin Privilege - Assign Able to assign/unassign admin privilege
System Configuration Rights			
System	Create, Edit, Delete, View	Administration Dashboard > Configuration >	Admin Right Dependency: System - Create, Edit, Delete, View

Resource	Operation	Menu Option Accessible	Actions Performable
		Authentication > User Store > Search, Create; - Administration Dashboard > Configuration > Authentication > User Lookup Module > Search, Create; - Administration Dashboard > Configuration > Authentication > Login Module > Search, Create; - Administration Dashboard > Configuration > Authentication > Authentication Realm > Search, Create; - Administration Dashboard > System > Housekeeping > Housekeeping Task; - Administration Dashboard > System > Server & Cache > Server > Search, Create	i) Able to search, view, create, edit, delete user store information ii) Able to search, view, create, edit and delete user lookup module information iii) Able to search, view, create, edit and delete login module information iv) Able to search, view, create, edit and delete authentication realm information v) Able to view, edit housekeeping task information vi) Able to search, view, create, edit and delete server information
Policy	Create, Edit, Delete, View	- Administration Dashboard > Security Policy > Admin & Audit > Admin Role > Search, Create - Administration Dashboard > Security Policy > Authentication & USO > Login Policy > Search, Create - Administration Dashboard > Security Policy > Authentication & USO > Password Expiry Policy > Search, Create - Administration Dashboard > Security Policy > Authentication &	Admin Right Dependency: Policy – View, Create, Edit, Delete i) Able to search, view, create, edit and delete Admin Role information ii) Able to Search, View, Create, Edit and Delete Login Policy iii) Able to Search, View, Create, Edit and Delete Password Expiry Policy iv) Able to Search, View, Create, Edit and Delete Password Quality Policy

Resource	Operation	Menu Option Accessible	Actions Performable
		USO > Password Quality Policy > Search, Create - Administration Dashboard > Security Policy > Authentication & USO > Housekeeping > Housekeeping Policy - Administration Dashboard > System > Server Monitoring	v) Able to View, and Edit Housekeeping Policy Configuration vi) Able to View, and Edit Sever Monitoring Configuration
Report Rights			
System & Admin related reports	View	Real Time Report: Monitoring Report Dashboard > System & Admin > Real Time > Monitoring	Admin Right Dependency: Report - View, System- View
		Real Time Report: Admin Privilege Report Dashboard > System & Admin > Real Time > Admin Privilege	Admin Right Dependency: Report - View, User - View, Role - View
		Historical Report: Events Report Dashboard > System & Admin > Historical > Events	Admin Right Dependency: Report - View, System- View
		Historical Report: Modification History Report Dashboard > System & Admin > Historical > Modification History	Admin Right Dependency: Report - View, System- View
		Historical Report: Admin Workflow Request Report Dashboard > System & Admin > Historical > Admin Workflow Request	Admin Right Dependency: Report - View, Request - View
Segment & User related reports	View	Real Time Report: Inactive User Report Dashboard > Segment & User > Real Time > Inactive User	Admin Right Dependency: Report - View, User -View
		Real Time Report: User Login Account Status Report Dashboard > Segment & User > Real Time > User Login Account Status	Admin Right Dependency: Report - View, User -View

Resource	Operation	Menu Option Accessible	Actions Performable
		Real Time Report: User Group By Segment Report Dashboard > Segment & User > Real Time > User Grouped By Segment	Admin Right Dependency: Report - View, Segment-View, User - View
		Real Time Report: User Summary By Segment Report Dashboard > Segment & User > Real Time > User Summary By Segment	Admin Right Dependency: Report - View, Segment-View
		Real Time Report: Group Membership By Segment Report Dashboard > Segment & User > Real Time > Group Membership By Segment	Admin Right Dependency: Report - View, Segment-View, Group- View
		Real Time Report: Application Summary By Segment Report Dashboard > Segment & User > Real Time > Application Summary By Segment	Admin Right Dependency: Report - View, Segment-View
		Real Time Report: Segment Hierarchy Report Dashboard > Segment & User > Real Time > Segment Hierarchy	Admin Right Dependency: Report - View, Segment-View
		Historical Report: User Activities Report Dashboard > Segment & User > Historical > User Activities	Admin Right Dependency: Report - View, User-View
Authentication related reports	View	Real Time Report: E2EE PIN Mailer Activities Report Dashboard > Authentication > Real Time > E2EE PIN Mailer Activities	Admin Right Dependency: Report - View, Pin Mailer-View
		Real Time Report: User Bad Login Report Dashboard > Authentication > Real Time > User Bad Login	Admin Right Dependency: Report - View, User-View

Resource	Operation	Menu Option Accessible	Actions Performable
		Real Time Report: Login Violation Report Dashboard > Authentication > Historical > Login Violation	Admin Right Dependency: Report - View, User - View
Others			
Segment	Create, Edit, Delete, View	Segment/OU Store	Admin Right Dependency: Segment - View, Create, Edit, Delete Able to search, view, create, edit and delete segment/sub-segment information

5.2. B. Event Category and Types

Locate the **Create Event Notification Policy** page by navigating to **Administration Dashboard > Configuration > Messaging & Response > Event Notification Policy**. Select **Create**. In the **Attributes** tab, under the **Event filter** section, select **Add** to view the **Event Category** drop-down list.

Commonly used event category and types:

Event Category	Event Type	Description
AdminRequest	All	Handles all event types related to the administrator's request.
	Add	The event when the administrator's request is added.
	Process	The event when the administrator's request is in process.
Authentication	All	Handles all event types related to authentication.
	GenerateOtp	The event when OTP is generated.
	Login	The event when user login.
	Logout	The event when user logout.
	PreAuthenticate	The event when a user is pre-authenticated.
	Reauthentication	The event when a user is re-authenticated.
AuthnConfig	All	Handles all event types related to authentication configuration.
	AddLoginModule	The event when the login module is added.
	AddLookupModule	The event when the lookup module is added.
	AddRealm	The event when an authentication realm is added.

Event Category	Event Type	Description
	DeleteLoginModule	The event when the login module is deleted.
	DeleteLookupModule	The event when the lookup module is deleted.
	DeleteRealm	The event when an authentication realm is deleted.
	ModifyLoginModule	The event when the login module is modified.
	ModifyLookupModule	The event when the lookup module is modified.
	ModifyRealm	The event when an authentication realm is modified.
	RegisterLoginModuleHandler	The event when a new login module handler is registered.
Event	All	Handles all event types related to an event.
	Notification	The event when the notification is added.
Housekeeping	All	Handles all event types related to housekeeping.
	Completed	The event when housekeeping task is completed.
	DeleteInactiveUsers	The event when inactive users are deleted after task execution.
	Started	The event when housekeeping task is started.
	SuspendInactiveUsers	The event when inactive users are suspended after task execution.
Licence	LicenceWarningEvent	The event when a license restriction is violated and a license violation warning is issued.

Event Category	Event Type	Description
Modification	All	Handles all event types related to the modification.
	Add	The event when the modification is added.
	Delete	The event when the modification is deleted.
	Modify	The event when the modification is changed.
Password	All	Handles all event types related to password.
	ChangePassword	The event when a password is changed.
	ResetPassword	The event when a password is reset.
	VerifyPassword	The event when a password is verified.
Principal	All	Handles all event types related to principal/users.
	AddAuthnAccount	The event when the authentication account is assigned to a user.
	AddGroup	The event when a user group is created.
	AddLoginAccount	The event when a login account is assigned to a user.
	AddOAuthClientPrivilege	The event when a user/user group is granted an OAuth client privilege.
	AddSAMLPrivilege	The event when a user/user group is granted a SAML application privilege.
	AddUser	The event when a user is created.
	AddUserStore	The event when a user store is created.
	DeleteAuthnAccount	The event when the user's authentication account is deleted.
	DeleteGroup	The event when a user group is deleted.

Event Category	Event Type	Description
	DeleteLoginAccount	The event when the user's login account is deleted.
	DeleteUser	The event when the user is deleted.
	DeleteUserStore	The event when the user store is deleted.
	LinkExternalIdentity	The event when the user is linked to external application identity.
	ModifyAuthnAccount	The event when the user's authentication account is modified.
	ModifyGroup	The event when the user group is modified.
	ModifyLoginAccount	The event when the user's login account is modified.
	ModifyUser	The event when the user is modified.
	ModifyUserStore	The event when the user store is modified.
	RemoveOAuthClientPrivilege	The event when a user's/user group's OAuth client privilege is removed.
	RemoveSAMLSPPrivilege	The event when a user's/user group's SAML Service Provider application privilege is removed.
	SaveSecondaryAccount	The event when saving the secondary account.
	UnlinkExternalIdentity	The event when the user is unlinked from external application identity.
Privilege	All	Handles all event types related to the privilege.
	AddPrivilege	The event when a privilege is added to a user.
	DeletePrivilege	The event when a user's privilege is deleted.

Event Category	Event Type	Description
	ModifyPrivilege	The event when a user's privilege is modified.
Segment	All	Handles all event types related to the segment.
	Add	The event when a segment is added.
	Delete	The event when a segment is deleted.
	Modify	The event when a segment is modified.
Server	All	Handles all event types related to the server.
	Add	The event when the server is added.
	Delete	The event when the server is deleted.
	Modify	The event when the server is modified.
	MonitoringCritical	The event when the server's critical state is monitored.
	MonitoringInfo	The event when the server's information is monitored.
	MonitoringWarn	The event when the server's warning is monitored.
	Shutdown	The event when the server is shut down.
	Started	The event when the server is started.
Session	All	Handles all event types related to sessions.
	Expired	The event when the user's session is already expired.
	IdleTimeout	The event when the session reaches the set idle time.

5.3. C. Message Template Macros

Coordinate-based Template Macros

Macro	Description
<code>\${user.name}</code>	Refers to the user name
<code>\${user.id}</code>	Refers to the user Id
<code>\${userAttr.customAttributeName}</code>	Refers to the user custom attribute value based on the provided custom attribute name.
<code>\$pinmailer</code>	Refers to the pin mailer object.

Example:

Row	Column	Text
7	25	<code>\$user.name</code>
8	25	<code>\$userAttr.AddressLine1, \$userAttr.AddressLine2</code>
9	25	<code>\$userAttr.City, \$userAttr.State - \$userAttr.ZipCode</code>
10	25	<code>\$userAttr.Country</code>
4	1	<code>\$pinmailer.id</code>

To display current date, use `${dateTool.format('hh:mm a dd-MMM-yyyy', $currentDate)}`. The following table provides the details for each of the valid dateTool symbols:

Symbol	Meaning [Example]
y	year [e.g. yyyy for 2012]
M	month in a year [e.g. MMMM for September, MMM for Sep, MM for 09 and M for 9]
d	day in a month [e.g. dd for 10]
h	hour in am/pm (1 to 12) [e.g. hh for 11]
H	hour in a day (0 to 23) [e.g. HH for 23]

Symbol	Meaning [Example]
m	minute in an hour [e.g. mm for 30]
s	second in a minute [e.g. ss for 50]
a	am/pm marker [e.g. a for PM]
z	time zone [e.g. z for PST]

Default Template Engine Macros

Macro	Description
User	
<code>\${userId}</code>	Refers to session's user Id.
<code>\${userAttr.get("<name>")}</code>	Refers to the session's user attribute value based on the provided attribute name.
<code>\${userName}</code>	Refers to the session's user name.
<code>\${userDN}</code>	Refers to the session's user distinguished name.
<code>\${userEmail}</code>	Refers to the session's user e-mail.
<code>\${userMobile}</code>	Refers to the session's user mobile.
Generic Event Macro	
<code>\${targetId}</code>	Refers to the event's target Id.
<code>\${event.<name>}</code>	Refers to the special attribute passed by client API.
<code>\${targetUserEmail}</code>	Refers to the email attribute of target of some events (e.g. token assignment with its category-event type as "Token-Assign") where the object class of the event's target is User.
<code>\${targetUserMobile}</code>	Refers to the mobile attribute of target of some events.(e.g. token assignment with its category-event type as "Token-Assign") where the object class of the event's target is User.
<code>\${event.newPassword.clearValue}</code>	Refers to the new password in change and reset password event. To use this macro, it is required to enable in Basic Login Policy or allowed reset password by system in Login Module.
<code>\${actorId}</code>	Refers to the event's actor Id.

Macro	Description
\${targetUserName}	Refers to the event's user name.
\${targetUserDN}	Refers to the event's distinguished name.
SMS OTP Macro	
\${otp}	Refers to the One Time Password for SMS-OTP and other OTP related event.
\${pac}	Refers to the Personal Assurance Code for SMS-OTP and other OTP related event.
\${issuedate}	Refers to the issue date of OTP.
\${issuetime}	Refers to the issue time of OTP.
\${issueDateObj}	Refers to the issue date object.
\${expiredDateObj}	Refers to the expired date object.
\${timeout}	Refers to the timeout value of OTP.
Sender and Recipients	
\${user}	Refers to the session's user (the actual address will be resolved automatically based on delivery channel).
\${userEmail}	Refers to the session's user e-mail.
\${userMobile}	Refers to the session's user mobile.
\${target}	Refers to the event's target (the actual address will be resolved automatically based on delivery channel).
\${actor}	Refers to the event's actor (the actual address will be resolved automatically based on delivery channel).
\${targetUserEmail}	Refers to the email attribute of target of some events(e.g. token assignment with its category-event type as "Token-Assign") where the object class of the event's target is User.
\${targetUserMobile}	Refers to the mobile attribute of target of some events(e.g. token assignment with its category-event type as "Token-Assign") where the object class of the event's target is User.
\${targetUserAttr.get("<name>")}	Refers to the event's user attribute value based on the provided attribute name.
\${eventMsg}	Refers to the event's message.

Macro	Description
<code>\${eventResultCode}</code>	Refers to the event's result code.
<code>\${eventExtCode}</code>	Refers to the event's extended result code.
<code>\${eventResult}</code>	Refers to the event's result.
<code>\${eventObj}</code>	Refers to the event's object.
Processing	
<code>\${dateTool}</code>	Refers to the Velocity Template DateTool to be used with the Java Date object. Example: The SMS-OTP is <code>\${pac}-\${otp}</code> for Account Activation. Enter by <code>\${dateTool.format('hh:mm a dd-MMM-yyyy', \$expiredDateObj)}</code> .
<code>\${b64Tool}</code>	Refers to the Velocity Template Base64Tool to be used with the Java Date object. Example: To convert <code>\$event.pdf</code> from a byte array to a Base64 encoded string, enter <code>\${b64Tool.toBase64(\$event.pdf)}</code> .
Token Inventory Monitoring	
<code>\${event.status}</code>	Refers to the statuses that a token must have to be assignable. Tokens that do not have any of these statuses cannot be assigned to users.
<code>\${event.tokenGroup}</code>	Refers to the token group.
<code>\${event.vendorId}</code>	Refers to the token vendor.
<code>\${event.threshold}</code>	Refers to the threshold set in the Token Inventory Monitoring configurations.
<code>\${event.tokenCount}</code>	Refers to the number of assignable tokens in the specified token group that have not yet been assigned to any user.
<code>\${event.totalToken}</code>	Refers to the total number tokens that are in the specified token group.
<code>\${event.usedToken}</code>	Refers to the number of assigned tokens in the specified token group that do not have the status "PreActivation".
<code>\${event.activatedToken}</code>	Refers to the number of assigned tokens in the specified token group with the status "Activated".
<code>\${event.deactivatedToken}</code>	Refers to the number of assigned tokens in the specified token group with the status "Deactivated".

Macro	Description
\${event.lockedToken}	Refers to the number of assigned tokens in the specified token group with the status "Locked".
License Warnings	
\${event.message}	Refers to warnings from license violations (e.g. the maximum number of users assigned to a login module has been exceeded).
i	Note: Refer to Coordinate-based Template Macros for each date symbol and its meaning.

HTML Content Template Engine Macros

i	Note: Refer to Default Template Engine Macros for the macros.
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Token PIN Mailer Coordinate-based Template Macros

MACRO	Description
\${tokenSerialNumber}	Refers to the token serial number.
\${tokenBatchId}	Refers to the token batch Id.
\${tokenModel}	Refers to the token model.
\${tokenVendor}	Refers to the token vendor.
\${tokenPIN}	Refers to the token PIN.
\${date}	Refers to the print date in "dd.MM.yy" format.

i	Note: Refer to Coordinate-based Template Macros for the date symbols & its meaning.
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Macro availability matrix for Subject and Body fields based on event types (O: Available, X: Not available):

Macros	Description	Credential-based events	Request-based events	Multiple - requests-based events
\${toEmails}	List containing emails of	O	O	O

Macros	Description	Credential-based events	Request-based events	Multiple - requests-based events
	recipients in the To list.			
<code>\${toMobiles}</code>	List containing mobiles of recipients in the To list.	O	O	O
<code>\${ccEmails}</code>	List containing emails of recipients in the Cc list.	O	O	O
<code>\${ccMobiles}</code>	List containing mobiles of recipients in the Cc list.	O	O	O
<code>\${eventType}</code>	Event's type	O	O	O
<code>\${dateTool.format('yyyy-MM-dd HH:mm:ss', \${eventTime})}</code>	Event's time	O	O	O
<code>\${eventResult}</code>	Event's result	O	O	O
<code>\${actorId}</code>	Event's actor Id	O	O	O
<code>\${userId}</code>	Session's user Id	O	O	O
<code>\${userName}</code>	Session's user name	O	O	O
<code>\${userDN}</code>	Session's user distinguished name.	O	O	O
<code>\${userEmail}</code>	Session's user e-mail	O	O	O
<code>\${userMobile}</code>	Session's user mobile	O	O	O

Macros	Description	Credential-based events	Request-based events	Multiple - requests-based events
<code>\${userAttr.get("<attribute name>")}</code>	Session's user attribute value based on the provided attribute name.	O	O	O
<code>\${credential.<field>}</code>	UCM credential, e.g.: "\$credential.id" to get the credential Id.	O	O	X
<code>\${targetResourceId}</code>	UCM connector's Id	O	O	X
<code>\${requester.<name>}</code>	UCM requester's name	X	O	X
<code>\${requestId}</code>	UCM request's title	X	O	X
<code>\${requestAction}</code>	UCM request's action	X	O	X
<code>\${requestReason}</code>	UCM request's reason	X	O	X
<code>\${requestComment}</code>	UCM request's comment	X	O	X
<code>\${approverMobilePhoneNumbers}</code>	UCM approver(s) mobile numbers (in array).	X	O	X
<code>\${requests.size()}</code>	Number of UCM requests.	X	X	O
<code>\${requests.get(n).title}</code>	Title of the nth UCM request.	X	X	O

Macros	Description	Credential-based events	Request-based events	Multiple - requests-based events
<code>\${requests.get(n).ExternalReferenceId}</code>	External reference Id of the nth UCM request.	X	X	O
<code>\${requests.get(n).attributes.get("CheckOutDuration").intValue}</code>	Checkout duration of the nth UCM request.	X	X	O
<code>\${dateTool.format('yyyy-MM-dd HH:mm:ss', \${requests.get(n).checkoutStartTime})}</code>	Reservation time of the nth UCM request	X	X	O
<code>\${requests.get(n).changeHistory.get(0).attr5Char1024}</code>	Reason of the nth UCM request.	X	X	O
<code>\${requests.get(n).changeHistory.get(0).attr4Char128}</code>	Comment of the nth UCM request.	X	X	O
<code>\${requests.get(n).credential.connector.id}</code>	Connector's Id of the nth UCM request.	X	X	O
<code>\${requests.get(n).credential.id}</code>	Credential's Id of the nth UCM request.	X	X	O
<code>\${requests.get(n).credential.userId}</code>	Credential's user Id of the nth UCM request.	X	X	O

How to use the macros:

Just enter the macro value into the message body.

Example:

"A \$requestAction action was submitted."

To use a macro with a custom attribute, just enter the macro and the custom attribute name such as "\$userAttr.get("<attr_name>")" into the message body.

Example:

"Email was sent to \$userAttr.get("myOtherEmail")."

5.4. D. Sample Report Templates

Admin Privilege Report Sample

AccessMatrix

User Admin Privilege Report

Report Filtering Criteria

User Store : Default Store

User or Group : User

User Id : amadmin1, amadmin, User02, User04

Role Id : All

User Id : User02

Segment Id : -

User Name : Test User 02

Description : Test User

Account Status : Activated

Role Id

Role Name

Entitlement

Delegation Type

Restricted Segment Id(s)

Restricted Token Group Id(s)

InitialAdminRole

Initial Admin Role

Inherited (InitialAdminGroup)

OwnPlusDelegate

-

-

User Id : User04

Segment Id : -

User Name : Test User 04

Description : Test User

Account Status : Activated

Role Id

Role Name

Entitlement

Delegation Type

Restricted Segment Id(s)

Restricted Token Group Id(s)

AuditorRole

AuditorRole

Inherited (AuditAdminRole)

OwnOnly

-

-

Own

OwnPlusDelegateOneLevel

-

-

User Id : amadmin

Segment Id : -

User Name : amadmin

Description : Initial Administrator

Account Status : Activated

Role Id

Role Name

Entitlement

Delegation Type

Restricted Segment Id(s)

Restricted Token Group Id(s)

InitialAdminRole

Initial Admin Role

Inherited (InitialAdminGroup)

OwnPlusDelegate

-

-

User Id : amadmin1

Segment Id : -

User Name : amadmin1

Description : Initial Administrator

Account Status : Activated

Role Id

Role Name

Entitlement

Delegation Type

Restricted Segment Id(s)

Restricted Token Group Id(s)

InitialAdminRole

Initial Admin Role

Inherited (InitialAdminGroup)

OwnPlusDelegate

-

-

Events Report Sample

Printed Date : 2017-08-08 16:46:00

Time zone : SGT

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 Events Report

Report Filtering Criteria

User Store	: Default Store	IP Address	: -
Actor User Id	: All		
Event Start	: 2016-08-01 00:00:00	Event End	: 2017-08-08 16:45:01
Event Result	: All	Include actions done by background tasks	: Yes
Target User Store	: All	Target Id	: -
Agent User Id	: All	Agent User Store	: All
Target Segment Id	: -	Target Segment Store	: All
Actor Segment Id	: -	Actor Segment Store	: All
Agent Segment Id	: -	Agent Segment Store	: All
Query Limit	: 1000	Is Query Result Exceeding Query Limit	: false
Event Category	: All		

Audit Id (Server Id)/ Actor User Id@User Store/ Actor IP Address/ Actor Segment Id@ Segment Store/ Agent Segment Id@ Segment Store	Reported Time/ Event Category	Target Id (Target Class) (Target Segment Id@ Segment Store)	Target Field (New Value)	(A = Authenticated, NA = Not Authenticated)	Event Result / Message/ Result Code
2454 (Jami-PC)/ amadmin@DefaultStore/ 0.0.0.0: Authentication-Login 0.0.1/-	2017-06-20 13:51:34/ 0.0.1/-	amadmin (User)(-)	Client Type (USO) Client Id (0.0.0.0.0.0.1)	-	Failed (Realm has not been assigned for realmId=40151) Authentication Realm is not assigned
2453 (Jami-PC)/ amadmin@DefaultStore/ 0.0.0.0: Authentication-Login 0.0.1/-	2017-06-20 13:34:13/ 0.0.1/-	amadmin (User)(-)	Client Type (Admin Console) Client Id (0.0.0.0.0.0.1)	Login Module Id (SystemPasswordBasicA (ForceChgPwd: / -))	Successful (ForceChgPwd: / -)
2450 (Jami-PC)/ amadmin@DefaultStore/ 0.0.0.0: Principal-ModifyUserStore 0.0.1/-	2017-06-20 12:34:26/ 0.0.1/-	testUSO (User Store)(-)	Client Type (Admin Console) Client Id (0.0.0.0.0.0.1)	Discriminator2 (APP-SG-450) Units ("Idap://192.168.1.8:389")	Successful (-)
2449 (Jami-PC)/ amadmin@DefaultStore/ 0.0.0.0: AuthnConfig-ModifyLookupModule 0.0.1/-	2017-06-20 12:33:53/ 0.0.1/-	DefaultActiveDirectoryClient Type (Admin Console) (User Lookup Module) Client Id (-)	-	_UserStoreUUID (testUSO)	Successful (-)
2448 (Jami-PC)/ amadmin@DefaultStore/ 0.0.0.0: AuthnConfig-ModifyLookupModule 0.0.1/-	2017-06-20 12:33:45/ 0.0.1/-	DefaultActiveDirectoryClient Type (Admin Console) (User Lookup Module) Client Id (-)	-	_UserStoreUUID (-)	Successful (-)
2447 (Jami-PC)/ amadmin@DefaultStore/ 0.0.0.0: AuthnConfig-ModifyLookupModule 0.0.1/-	2017-06-20 12:33:25/ 0.0.1/-	DefaultActiveDirectoryClient Type (Admin Console) (User Lookup Module) Client Id (-)	-	TranslateDomainToStore (-) _UserStoreUUID (testUSO)	Successful (-)
2446 (Jami-PC)/ amadmin@DefaultStore/ 0.0.0.0: AuthnConfig-ModifyLookupModule 0.0.1/-	2017-06-20 12:32:30/ 0.0.1/-	DefaultActiveDirectoryClient Type (Admin Console) (User Lookup Module) Client Id (-)	-	Find UserStore based on NT Domain value (true)	Successful (-)

Modification History Report Sample

Printed Date : 2017-08-08 16:41:39

Time zone : SGT

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 **User Modification History Report**

Report Filtering Criteria			
Target Id	: *	Event Result	: All
Event Start	: 2016-08-01 00:00:00	Event End	: 2017-08-08 16:40:38
Query Limit	: 1000	Is Query Result Exceeding Query Limit	: false
Event Category	: All		

Target Id : jamie				
Audit Id (Server Id)/ Actor User Id@User Store/ Actor IP Address	Reported Time/ Event Category	Target Id (Target Class)	Target Field (Old Value) (New Value)	Event Result (Message)
1356 (Jamie-PC)/ amadmin@DefaultStore/ 0: 0:0:0:0:0:1/	2017-05-31 16:00:36/ Password- ResetPassword	jamie (User)	-	Client Type Login Module Id Failed(LDAP error code 53 (Admin Console) (TstLogin) Client Id - 0000001F: SvcErr: DSID- (0:0:0:0:0:0:1) 031A0FC0, problem 6003 (WILL_NOT_PERFORM), data 0 J)

Target Id : otpstuser				
Audit Id (Server Id)/ Actor User Id@User Store/ Actor IP Address	Reported Time/ Event Category	Target Id (Target Class)	Target Field (Old Value) (New Value)	Event Result (Message)
2055 (Jamie-PC)/ amadmin@DefaultStore/ 0: 0:0:0:0:0:1/	2017-06-13 17:09:30/ Principal- AddAuthnAccount	otpstuser (User)	Resim Id (SMSOTP)	Status (Activated) Client Type (Admin Console) Client Id (0:0:0:0:0:0:1)
2066 (Jamie-PC)/ amadmin@DefaultStore/ 0: 0:0:0:0:0:1/	2017-06-13 17:09:30/ Privilege-AddPrivilege	otpstuser (User)	Privilege (InitialAdminRole)	Client Type (Admin Console) Client Id (0:0:0:0:0:0:1)
2067 (Jamie-PC)/ amadmin@DefaultStore/ 0: 0:0:0:0:0:1/	2017-06-13 17:09:30/ Principal-AddUser	otpstuser (User)	Id (-)(otpstuser) Name (-)(OTP Test User) Status (-)(Activated) Mobile (-)(92473507) Email (-)(jamie.zun@sprint.com) addGroup (-)(InitialAdminGroup)	Client Type (Admin Console) Client Id (0:0:0:0:0:0:1)
2068 (Jamie-PC)/ amadmin@DefaultStore/ 0: 0:0:0:0:0:1/	2017-06-13 17:09:50/ Principal-ModifyUser	otpstuser (User)	-	Client Type (Admin Console) Client Id (0:0:0:0:0:0:1)

Admin Workflow Request Report Sample

Printed Date : 2017-08-11 17:06:50 Time zone : SGT Page : 1 / 3

AccessMatrix Admin Workflow Request Report

Report Filtering Criteria			
User Store	: All		
Actor User Id	: All		
Event Start	: 2016-08-02 00:00:00	Event End	: 2017-08-11 17:06:20
Event type	: All	Event Result	: All
Query Limit	: 1000	Is Query Result Exceeding Query Limit	: false
Request Id	: -		
Request Action	: All		
Request Status	: All		
Request Status	: All		

Audit Id (Server Id)/ Actor User Id@User Store/ Actor IP Address/ Actor Segment Id@ Segment Store/ Agent Segment Id@ Segment Store	Reported Time/ Event type	Request Id/ Request Content Type	Content Summary/ Workflow Type	Request Action/ Request Status/ Changes Version	Field Scope Field Name Old Value New Value	Event Result / Message/ Result Code
2814 (Jamie-PC)/ User08@DefaultStore/ 0:0:0:0:0: 0:1/ -/-	2017-08-11 11:22:52/ Process	00000003/ Role	Role - SuperUserRole/ DefaultDualControl	Withdraw/ Withdrawn/ -	-	Successful
2824 (Jamie-PC)/ User01@DefaultStore/ 0:0:0:0:0: 0:1/ -/-	2017-08-11 11:35:46/ Add	00000004/ Role	Role - AdminRoleABC/ DefaultDualControl	Submit/ Pending/ -	-	Successful
2831 (Jamie-PC)/ amadmin@DefaultStore/ 0:0:0:0:0: 0:0:1/ -/-	2017-08-11 11:38:14/ Process	00000002/ Role	Role - NewAdminRole/ DefaultDualControl	Approve/ Approved/ -	-	Successful
2833 (Jamie-PC)/ amadmin@DefaultStore/ 0:0:0:0:0: 0:0:1/ -/-	2017-08-11 11:39:25/ Process	00000004/ Role	Role - AdminRoleABC/ DefaultDualControl	Reject/ Rejected/ -	-	Successful
2842 (Jamie-PC)/ amadmin@DefaultStore/ 0:0:0:0:0: 0:0:1/ -/-	2017-08-11 15:07:02/ Process	00000001/ Role	Role - testingrole/ DefaultDualControl	Edit/ Pending/ -	-	Successful
2846 (Jamie-PC)/ amadmin@DefaultStore/ 0:0:0:0:0: 0:0:1/ -/-	2017-08-11 16:07:33/ Add	00000005/ User	User06/ DefaultDualControl	Submit/ Pending/ 0	Object Fields Mobile - 98765432 Object Fields addGroup - UCMAdminRole	Successful
2849 (Jamie-PC)/ amadmin@DefaultStore/ 0:0:0:0:0: 0:0:0:1/ -/-	2017-08-11 16:11:10/ Add	00000006/ User	User10/ DefaultDualControl	Submit/ Pending/ 0	Object Fields Email - abc@def.com Object Fields addGroup - TestAdminGroup	Successful
2854 (Jamie-PC)/ amadmin@DefaultStore/ 0:0:0:0:0: 0:0:0:1/ -/-	2017-08-11 16:14:16/ Process	00000005/ User	User06/ DefaultDualControl	Approve/ Approved/ -	-	Successful

User Activities Report Sample

Printed Date : 2017-08-22 11:21:01

Time zone : SGT

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 User Activities Report

Report Filtering Criteria						
User Store	:	All		IP Address	:	-
Actor User Id	:	All				
Event Start	:	2016-08-02 00:00:00		Event End	:	2017-08-22 11:18:05
Event Result	:	All		Include actions done by background tasks	:	No
Target User Store	:	All		Target Id	:	-
Agent User Id	:	All		Agent User Store	:	All
Target Segment Id	:	-		Target Segment Store	:	All
Actor Segment Id	:	-		Actor Segment Store	:	All
Agent Segment Id	:	-		Agent Segment Store	:	All
Query Limit	:	1000		Is Query Result Exceeding Query Limit	:	false
Event Category	:	All				

Actor User Id@User Store		: amadmin@DefaultStore				
Audit Id (Server Id) / Actor Segment Id@ Segment Store / Agent Segment Id@ Segment Store	Reported Time / Event Category	Actor IP Address	Target Id (Target Class) (Target Segment Id@ Segment Store)	Target Field (New Value)	(A = Authenticated, NA = Not Authenticated)	Event Result / Message/ Result Code
2443 (Jamie-PC) / - / -	2017-06-20 12:18:37 / Modification-Modify	0:0:0:0:0:0:1	ESSOServerConfiguration (Admin Console) Client Id (0:0:0:0:0:0:1)	-	Default authentication for USO users (40171)	Successful
2442 (Jamie-PC) / - / -	2017-06-20 12:16:10 / Modification-Modify	0:0:0:0:0:0:1	ESSOServerConfiguration (Admin Console) Client Id (0:0:0:0:0:0:1)	-	-	Successful
2441 (Jamie-PC) / - / -	2017-06-20 12:04:05 / Authentication-Login	0:0:0:0:0:0:1	amadmin (User) Client Id (0:0:0:0:0:0:1)	Realm Id (SystemRealm)	Login Module Id (SystemPasswordBasic:A)	Successful
2438 (Jamie-PC) / - / -	2017-06-20 11:44:48 / Authentication-Login	0:0:0:0:0:0:1	amadmin (User) Client Id (0:0:0:0:0:0:1)	Realm Id (40151)	-	Failed/ Realm has not been assigned for realmId=40151 Authentication Realm is not assigned Successful
2384 (Jamie-PC) / - / -	2017-06-19 10:40:08 / Session-Expired	0:0:0:0:0:0:1	amadmin (User) Client Type (Admin Console) Client Id (0:0:0:0:0:0:1)	Reported By (CurrentThread: 110) Last Refreshed Time (2017-06-16 11:46:16)	SystemPasswordBasic:A	

Login Violation Report Sample

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Time zone : SGT

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 Login Violation Report

Report Filtering Criteria

User Store	: All	Actor User Id	: All
Include actions done by background tasks	: No	IP Address	: -
From Date	: 2016-08-01 00:00:00	To Date	: 2017-08-22 11:53:31
Agent User Id	: All	Agent User Store	: All
Is Query Result Exceeding Query Limit	: false	Query Limit	: 1000
Violation Type	: All		

Actor User Id@User Store : amadmin1@DefaultStore

Event(Server Id)	Event Category	Type	Violation Message / Violation Type	Client Type	Client Id	Login Module Id (A = Authenticated, NA = Not Authenticated)	Reported Time	Actor IP Address
2072(Jamie-PC)	Authentication	Login	Wrong password / OTP/signature/challenge response/password verification failure	Admin Console	0:0:0:0:0:0:1	SystemPasswordBasic:NA (Realm Id) SystemRealm	2017-06-13 17:17:02	0:0:0:0:0:0:1
2704(Jamie-PC)	Authentication	Login	Wrong password / OTP/signature/challenge response/password verification failure	Admin Console	0:0:0:0:0:0:1	SystemPasswordBasic:NA (Realm Id) SystemRealm	2017-08-08 15:34:03	0:0:0:0:0:0:1
Subtotal Violations : 2								

Actor User Id@User Store : jamie@TestAD

Event(Server Id)	Event Category	Type	Violation Message / Violation Type	Client Type	Client Id	Login Module Id (A = Authenticated, NA = Not Authenticated)	Reported Time	Actor IP Address
1351(Jamie-PC)	Authentication	Login	invalid credentials/ OTP/signature/challenge response/password verification failure	Admin Console	0:0:0:0:0:0:1	TstLogin:NA (Realm Id) AD-OTP	2017-05-31 15:58:45	0:0:0:0:0:0:1
1358(Jamie-PC)	Authentication	Login	invalid credentials/ OTP/signature/challenge response/password verification failure	Admin Console	0:0:0:0:0:0:1	TstLogin:NA (Realm Id) AD-OTP	2017-05-31 16:01:29	0:0:0:0:0:0:1
1359(Jamie-PC)	Authentication	Login	ForceChgPwd: / 18888	Admin Console	0:0:0:0:0:0:1	TstLogin:A SMSOTPLgin:NA (Realm Id) AD-OTP	2017-05-31 16:01:40	0:0:0:0:0:0:1
Subtotal Violations : 3								

System Summary Report Sample

Printed Date : 2021-10-29 12:00:17

Time zone : ICT

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System Summary Report

Name	Value
AccessMatrix Version	5.7.2.7200-SIT19
JVM Version	11.0.12+7
OS Version	Windows 7
Computer RAM	2.00 GB
JAVA Memory	0.38 GB
Number of processor	2
Web Container	AccessMatrix
Number of Servers	1
User count per Login Module	Last checked at 2021-10-29 23:26:59
Pseudo E2EEA Login Module	4
System Password Basic	7
Count of assigned Tokens	Last checked at 2021-10-29 23:26:59
VASCO (DP270)	1
OATH (YeeKey)	2
Number of users in Default userstore	12

5.5. E. Common Issues

The following section provides solutions to common issues when using AccessMatrix.

Online Report Generation

Fontconfig Error

When generating a report using the Admin Console, a **Fontconfig** package that has been incorrectly installed will cause the process to fail. The exception "Could not initialize class net.sf.jasperreports.engine.util.JRStyledTextParser" will be prompted.

i **Note:** This is currently a known issue for Linux environments only.

To resolve this issue, try performing the following steps:

1. Execute the following command to install the **Fontconfig** package: `yum install fontconfig libfontconfig.so.1`
2. Download and unzip the [java-font-test-20200 815T104410Z-001.zip](#) file to */tmp*.

i **Note:** You can obtain this file by contacting i-Sprint's Professional Services personnel.

3. Execute the following commands to test that the fonts have been loaded correctly:
 - o `which java`
 - o `/usr/lib/jvm/jdk-11.0.8+10-jre/bin/java -cp /tmp FontTest`

i **Note:** The resulting output should not display any errors and instead display the installed fonts.
